

Modelling the dynamics of decision-making

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13 Aug 2026



Perceptual decision making (under uncertainty)

Integrate sensory information

Select an action/choice among alternative competing options

Respond quickly and/or accurately



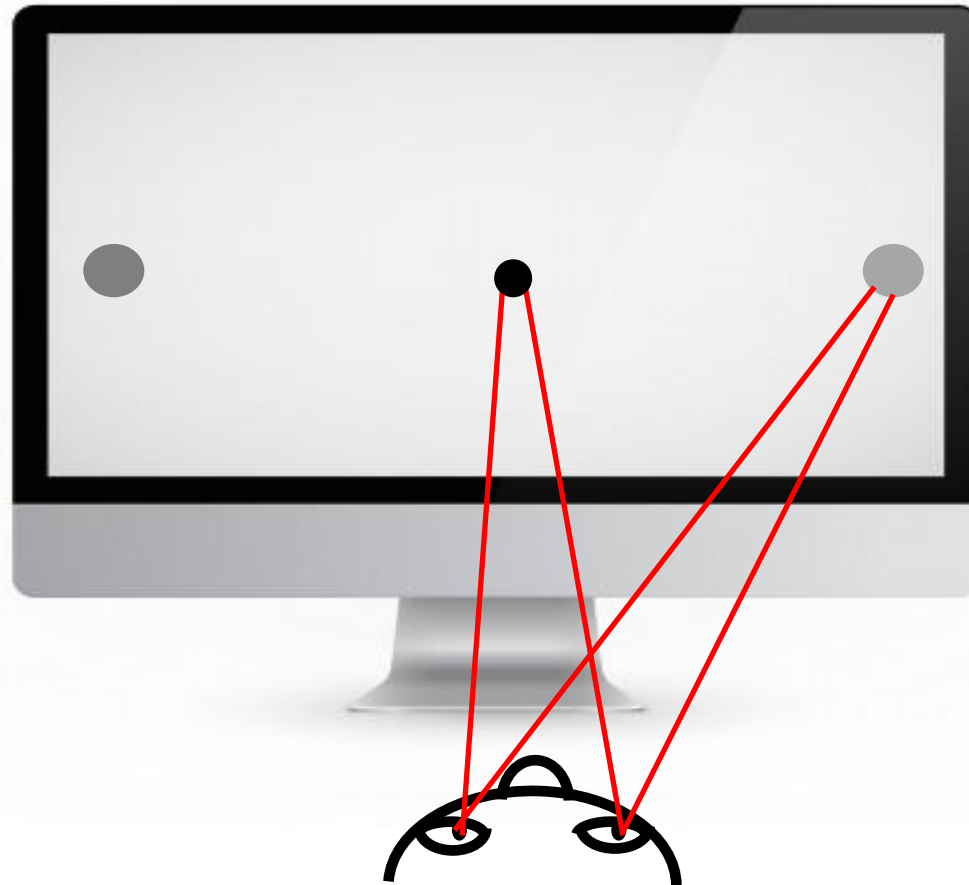


Sensory perception as a disturbance that passed along nerve fibres to the brain – a **mechanistic** machine. But **deterministic** reflex!

- René Descartes (1629) *Treatise of Man*.
- Sir Charles Sherrington (1906) *The Integrative Action of the Nervous System*

A colour/brightness discrimination task

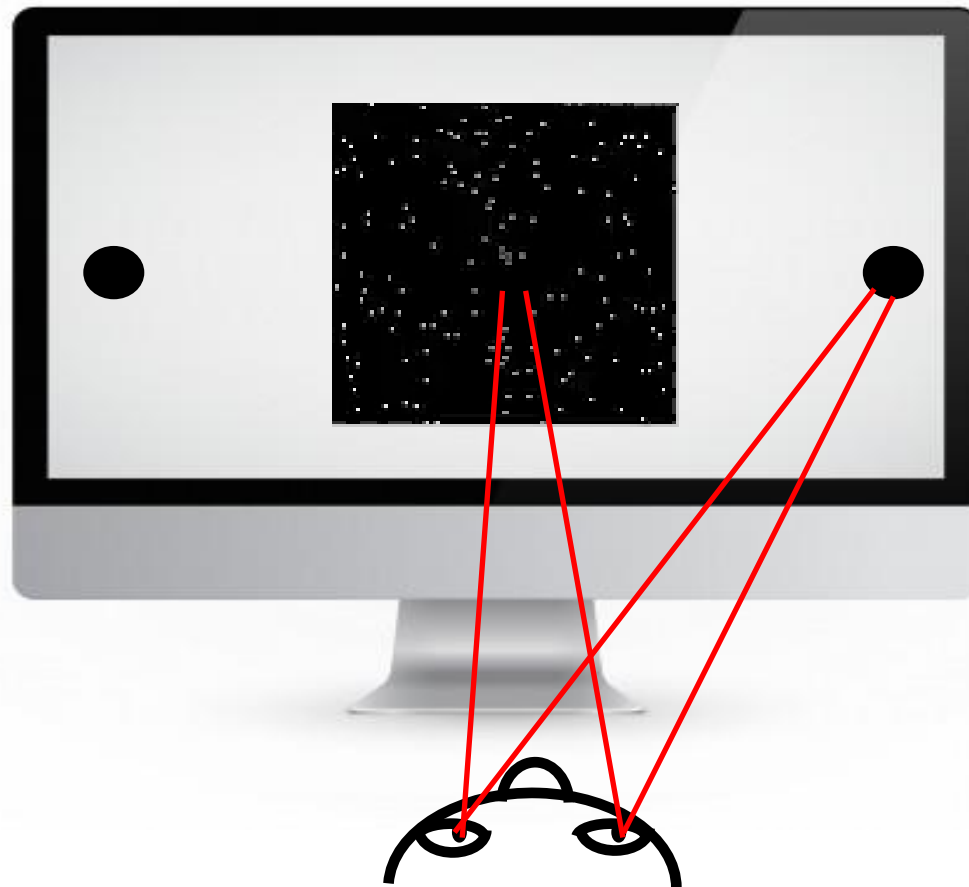
Researchers control difference in colours/brightness



“Correct”

Lab: A visual motion direction discrimination task

Experimentalist controls % of dots moving in the same direction (e.g. more rightward than leftward)

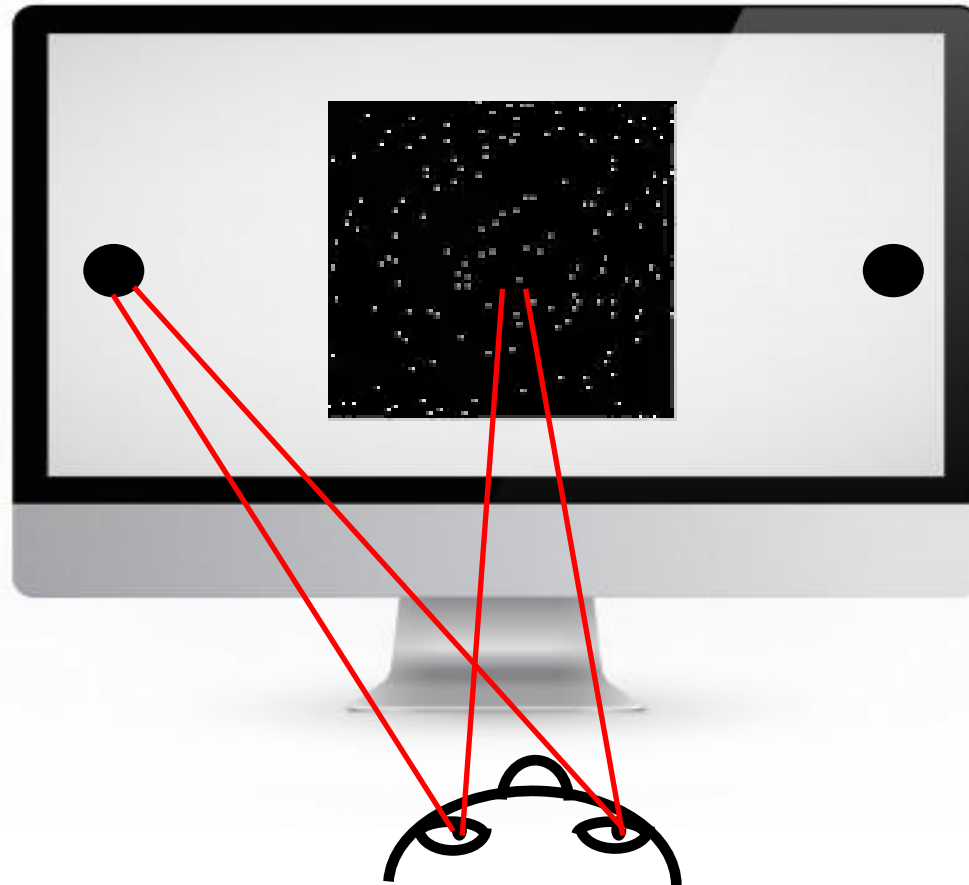


“Correct”

Lab: A visual motion direction discrimination task

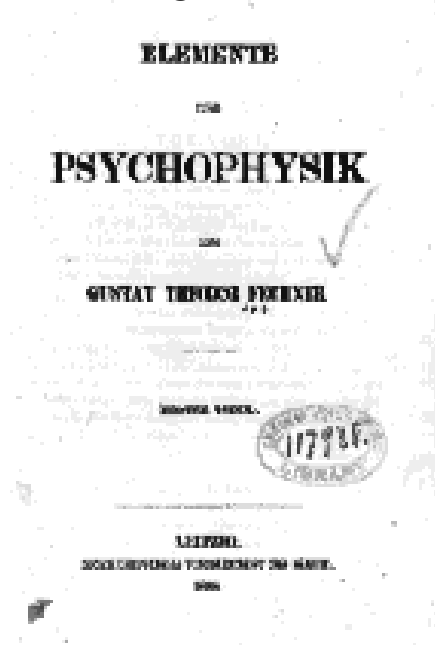
Experimentalist controls % of dots moving in the same direction (e.g. more rightward than leftward)

“Error”

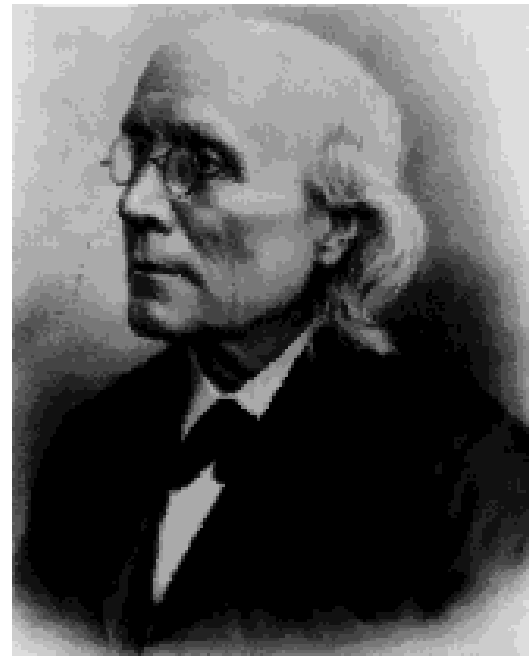


Psychophysics

... the analysis of perceptual processes by studying the effect on a subject's experience or behaviour of systematically varying the properties of a stimulus along one or more physical dimensions.



Publication of Fechner's *Elemente der Psychophysik* (1860) is generally considered to mark the formal beginning of experimental psychology.



Gustav Theodor Fechner (1801–1887).

→ Mathematical/quantitative formulation of perception

Golden age of Mathematical Psychology

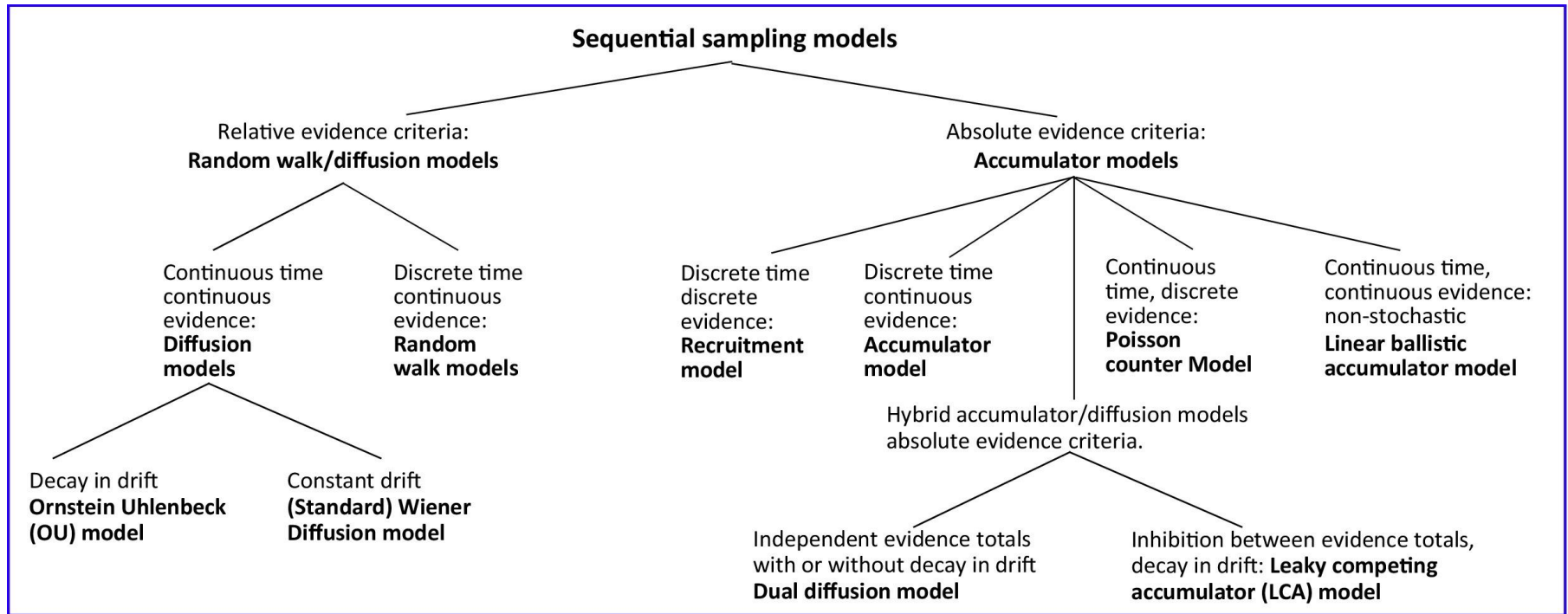
(1940's) ...

Coinciding with the maturation of some mathematical (information-theoretic and probabilistic) frameworks to describe individual's choice behaviour and response times.

... & birth of Decision Sciences

- *Abraham Wald (1947) Sequential analysis.*
- *Jacob Wolfowitz (1949) Sequential decision.*
- *R. Duncan Luce (1959) Individual choice behavior: a theoretical analysis.*
- *Mervyn Stone (1960) Models for choice reaction time. Psychometrika.*
- *Donald R. J. Laming (1968) Information theory of choice-reaction times.*
- *Roger Ratcliff (1978) A theory of memory retrieval. Psychol. Rev.*
- *R. Duncan Luce (1986) Response times: their role in inferring elementary mental organization.*

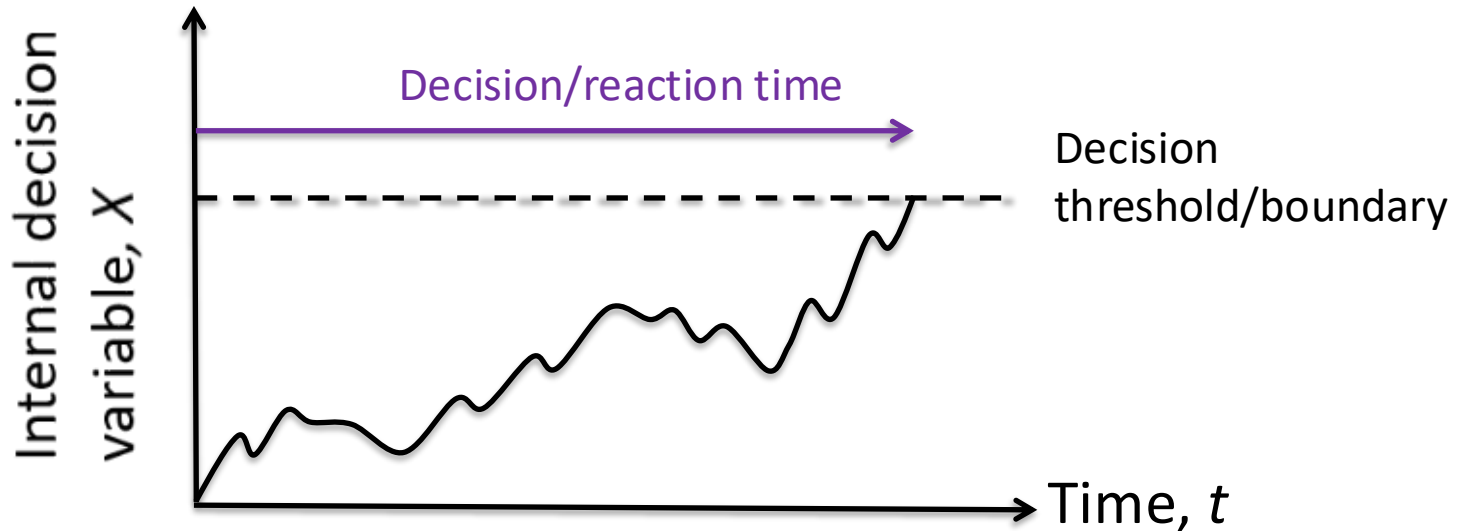
Models in Mathematical Psychology of decision-making



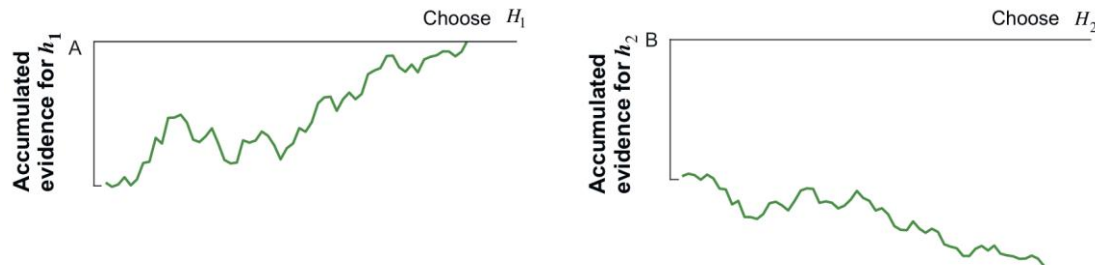
Common feature in cognitive models:

- Evidence accumulation -

Sequential sampling model framework



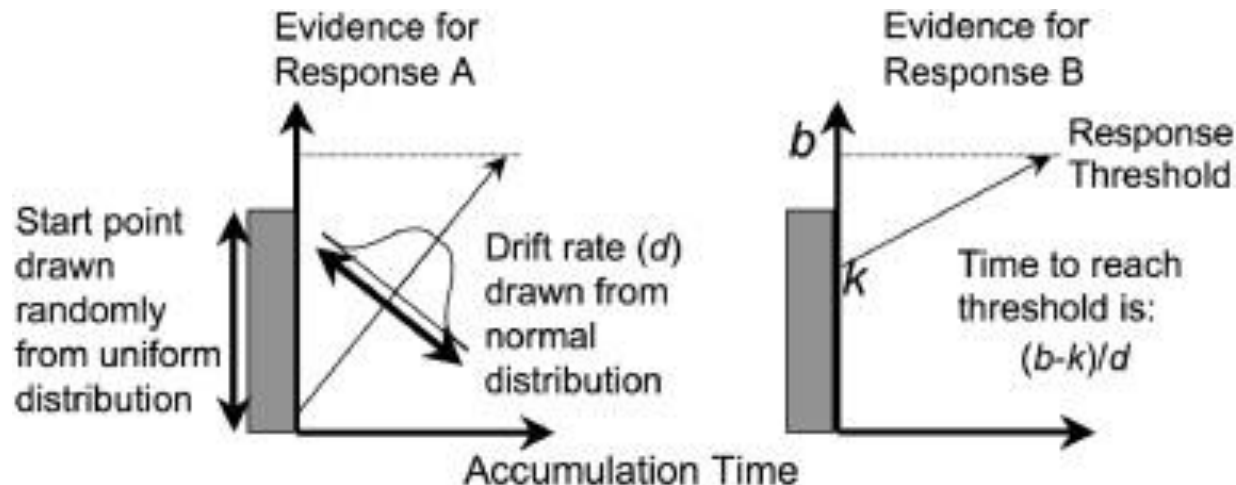
Race model



Common feature in cognitive models:

- Evidence accumulation -

Linear Ballistic Accumulator (LBA) Model



Brown & Heathcote (2008)

<https://itsdfish.github.io/SequentialSamplingModels.jl/dev/lba/>

Elements of a decision: a probabilistic formalism

- Often, perceptual decisions are uncertain or comes with various forms of noise.
- Choice behaviour becomes non-deterministic and probabilistic.
- What is the (unknown) state of the world given noisy data from sensory systems?
- Select among competing hypotheses $h_1, h_2, h_3, \dots h_n$. WLOG, focus on $n = 2$.

Definitions:

- Prior probability, $P(h_i)$: Probability before obtaining any evidence e .
- Likelihood, $P(e|h_i)$: Values that e can attain when h_i is true.
- Decision variable (DV): Accrual of all sources of priors, evidence, and values into a quantity interpreted by decision rule to produce a choice H_i (associated with hypothesis h_i).

Goal in decisions: achieve desired outcomes (correct) and avoid undesired ones (incorrect).

Signal detection theory (SDT)

SDT prescribes a process to **convert a single observation of noisy evidence into a categorical choice** (Green & Swets, 1966).

For binary decisions, the **decision variable DV** is related to the **ratio of the likelihoods** of h_1 and h_2 given e :

$$l_{12}(e) = \frac{P(e|h_1)}{P(e|h_2)}$$

Then impose a decision rule (criterion or threshold) to make a choice.

E.g. Choose h_1 if and only if $l_{12}(e) \geq \beta$, where β is some constant. Hence, variety of goals can be implemented.

Sequential analysis (SA) framework: extension to SDT

Accommodates multiple pieces of evidence observed over time ($t_1, \dots, t_n$).

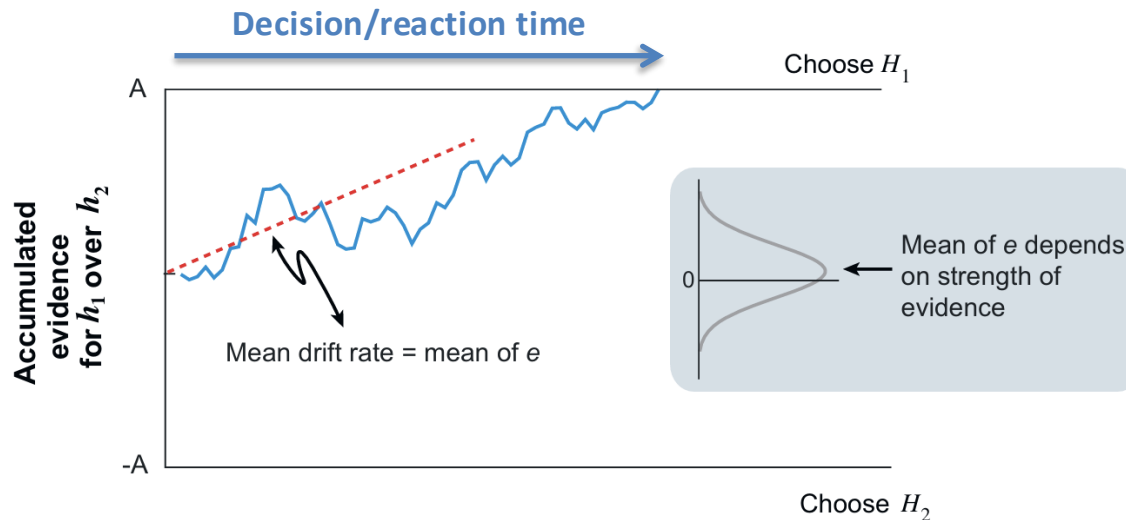
A simple DV is the logarithm of the likelihood, i.e. $\log LR$, from multiple, independent pieces of evidences $e_1, e_2, \dots, e_n$

$$\begin{aligned} \log LR_{12} &= \log \frac{P(e_1, e_2, \dots, e_n | h_1)}{P(e_1, e_2, \dots, e_n | h_2)} \\ &= \sum_{i=1}^n \log \frac{P(e_i | h_1)}{P(e_i | h_2)} \end{aligned}$$

A simple stopping rule: Update DV with new evidences till reaching some upper or lower threshold.

Sequential analysis framework

$$\begin{aligned} e_0 &\rightarrow f_0(e_0) \Rightarrow \text{Stop or} \\ &\quad \downarrow \\ e_1 &\rightarrow f_1(e_0, e_1) \Rightarrow \text{Stop or} \\ &\quad \downarrow \\ e_2 &\rightarrow f_1(e_0, e_1, e_2) \Rightarrow \text{Stop or} \\ &\quad \vdots \\ e_n &\rightarrow f_1(e_0, e_1, \dots, e_n) \Rightarrow \dots \end{aligned}$$

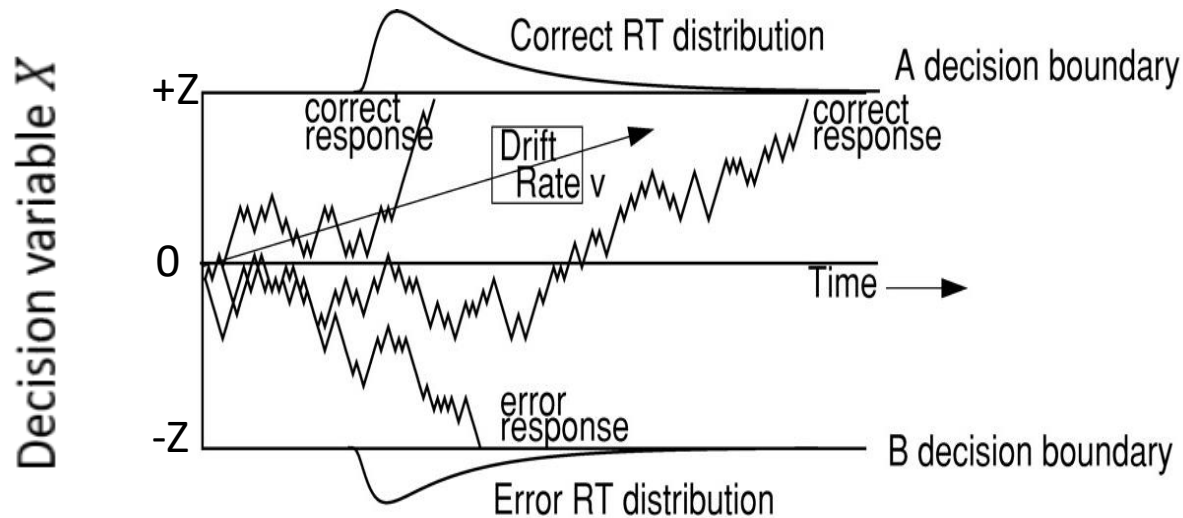


Gold & Shadlen (2007)

This process comprises the **Sequential Probability Ratio Test (SPRT)** – **most efficient test by achieving a desired error rate (accuracy) with smallest number of samples** (Wald & Wolfowitz, 1947) – speed-accuracy trade-off. Marian Adam Rejewski, Alan Turing et al. used this approach to break the German enigma cipher in WWII.

A cognitive model: Drift-diffusion model (DDM)

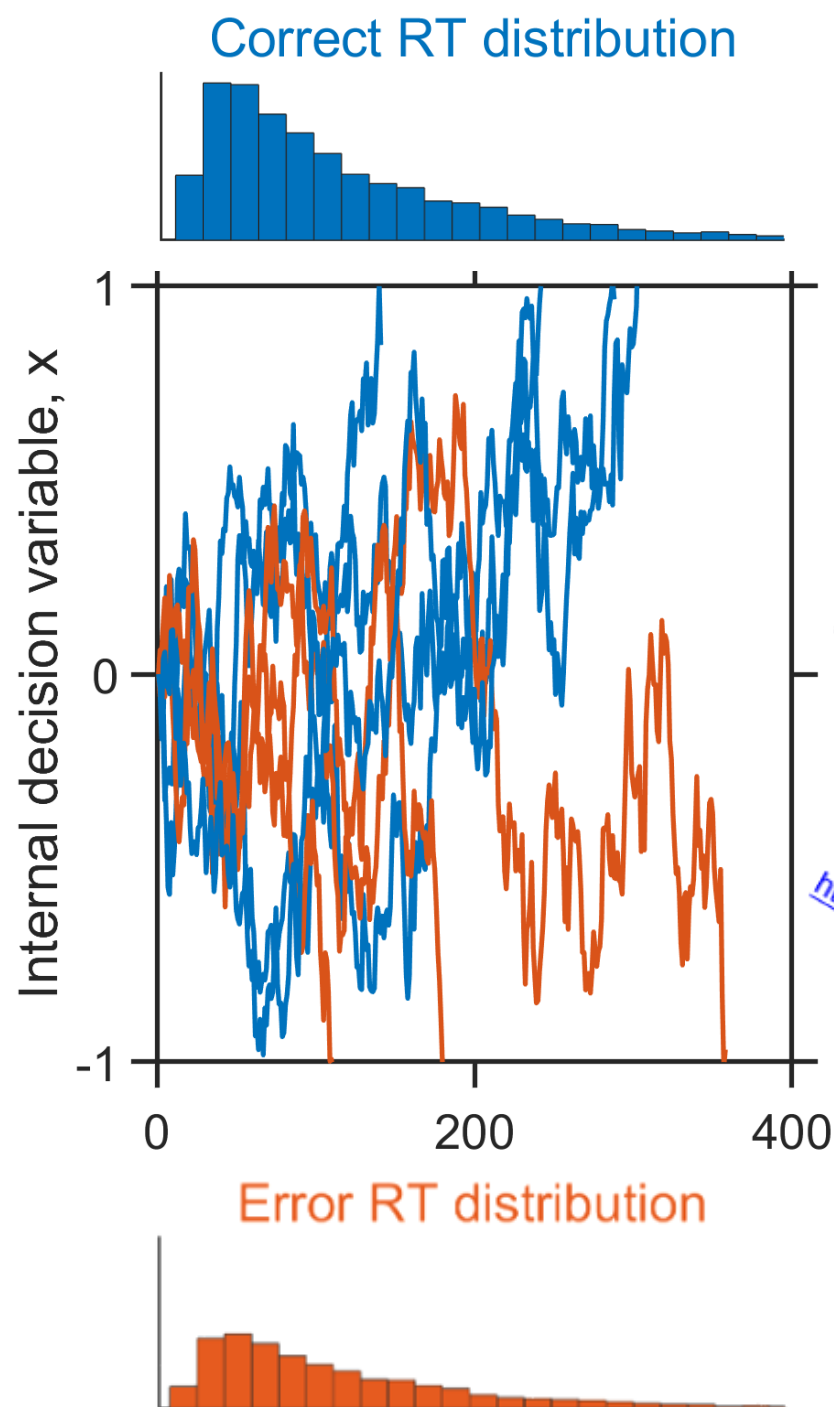
1D stochastic dynamics for 2-choice tasks



$$\begin{aligned} X &= \int dX = \int v \, dt + \sigma \, dW \\ &= \text{signal difference} + \text{noise} \end{aligned}$$

Computer simulations over several trials

With upward drift rate (signal)



Reaction time (RT)

Accuracy =
(No. corrects) / (Total no. choices)

<https://www.ejwagenmakers.com/EZ.html>

<https://github.com/murraylab/PyDDM>

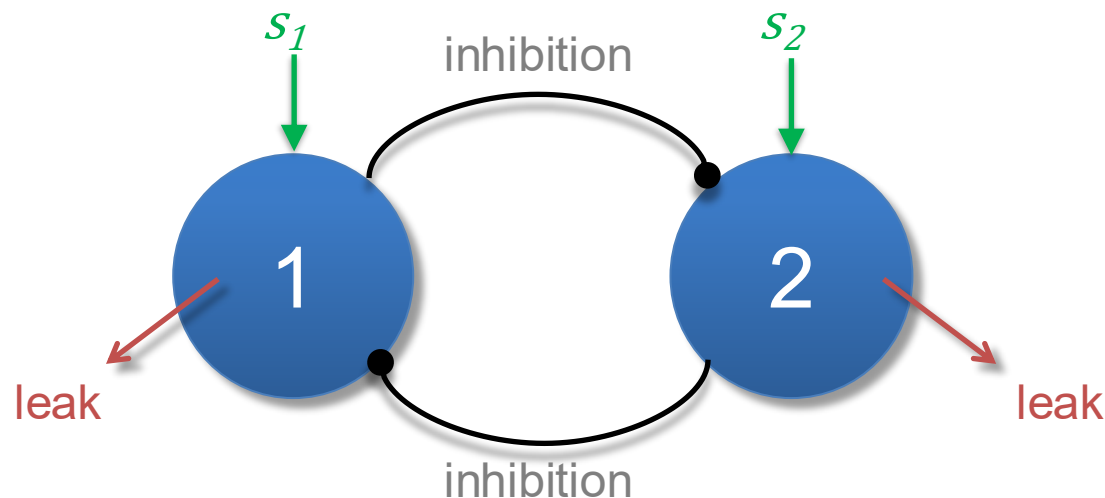
A simple neural network (connectionist) model

Leaky Competing Accumulator (LCA) model

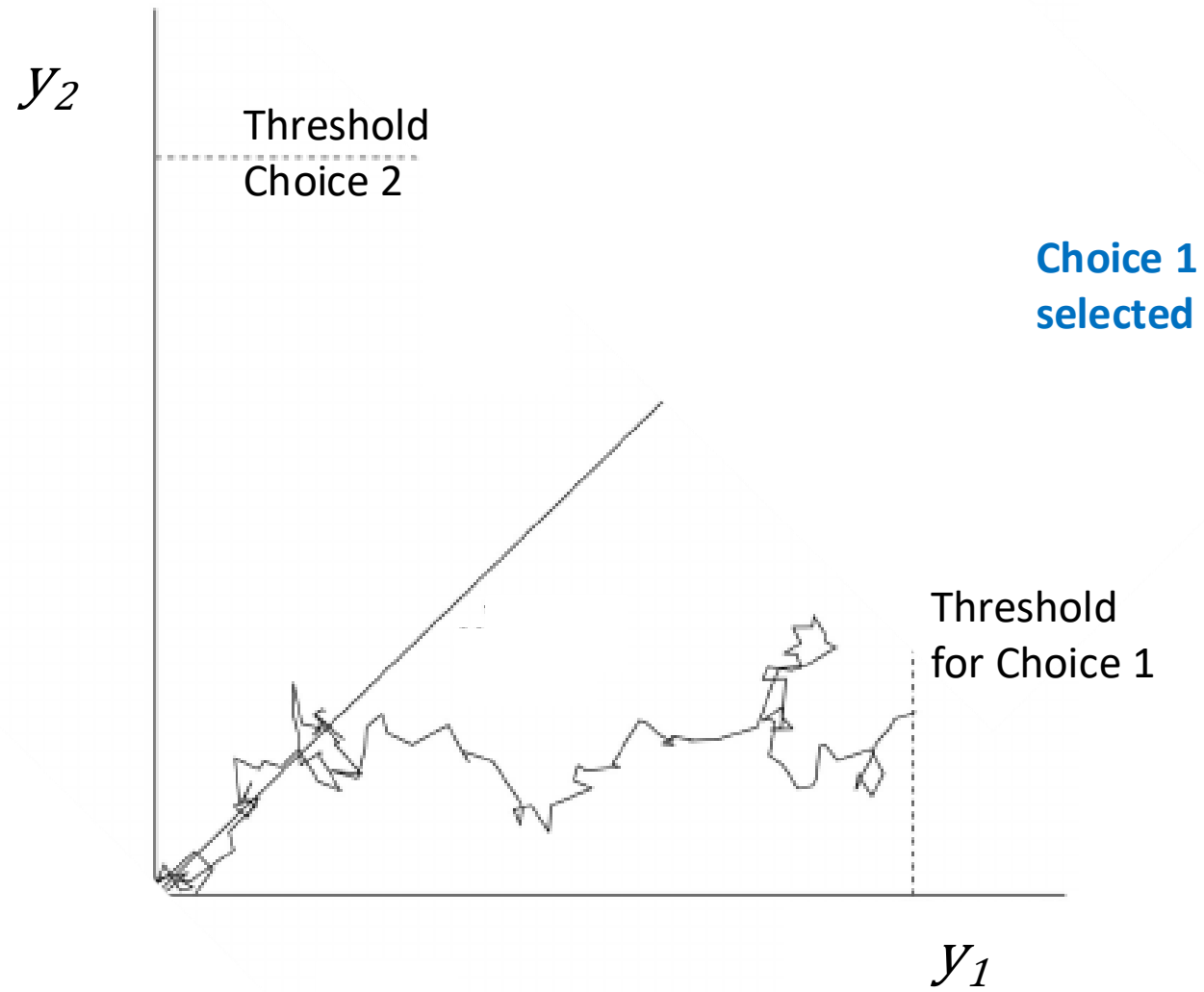
Temporal integration of evidence by competing accumulators – *winner-take-all* behaviour to facilitate process, or there will be conflict and indecisiveness.

$$\begin{aligned} dy_1 &= [-\gamma y_1 + f(-\beta y_2) + s_1] dt + \sqrt{D} dW_1 \\ dy_2 &= [\underbrace{-\gamma y_2}_{\text{leak}} + f(\underbrace{-\beta y_1}_{\text{inhibn}}) + \underbrace{s_2}_{\text{stim}}] dt + \underbrace{\sqrt{D} dW_2}_{\text{noise}} \end{aligned}$$

Usher & McClelland (1995, 2001)

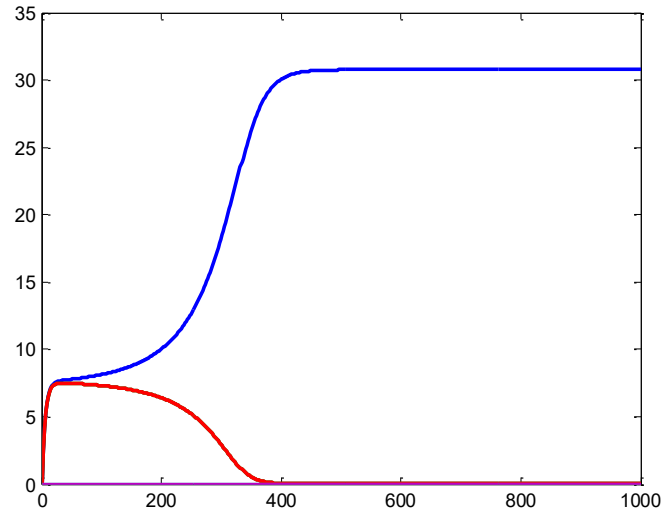


Sample simulated trajectory of nonlinear LCA model in activity (phase/state) space

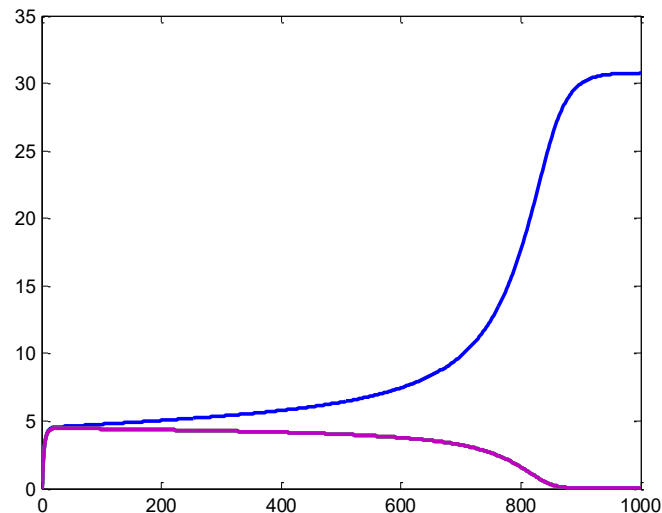


A multi-unit winner-take-all LCA model

Hick's or Hick–Hyman law: Describes the time it takes for a person to make a decision as a result of the possible choices: increasing choice options will increase the decision time ~logarithmically



With 2 choices
(short latency or
response time)

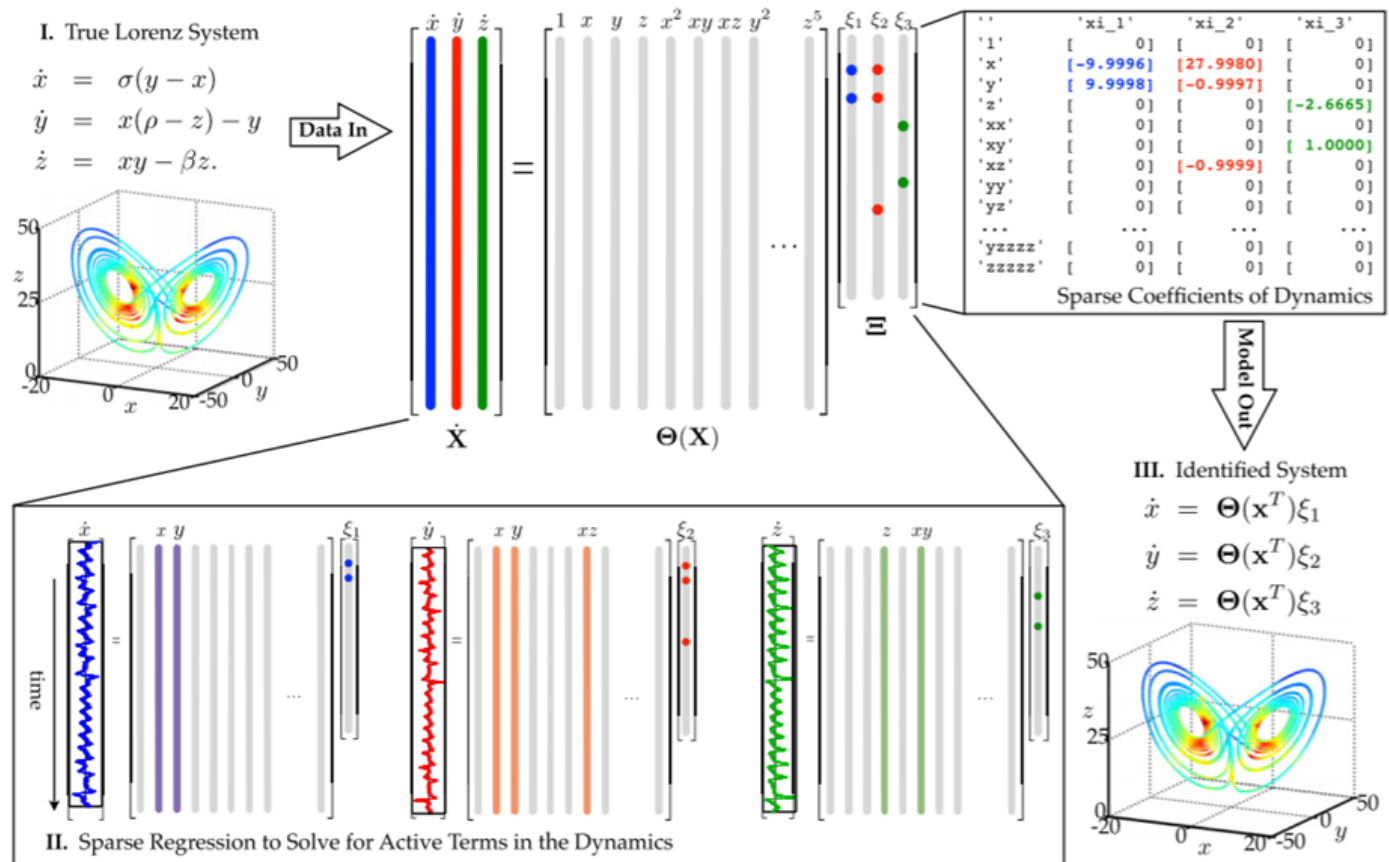


With 5 choices
(long latency or
response time)

Note: The other 3 neural
populations follow the
same dynamics as purple

All model parameter values remain the same in both cases.

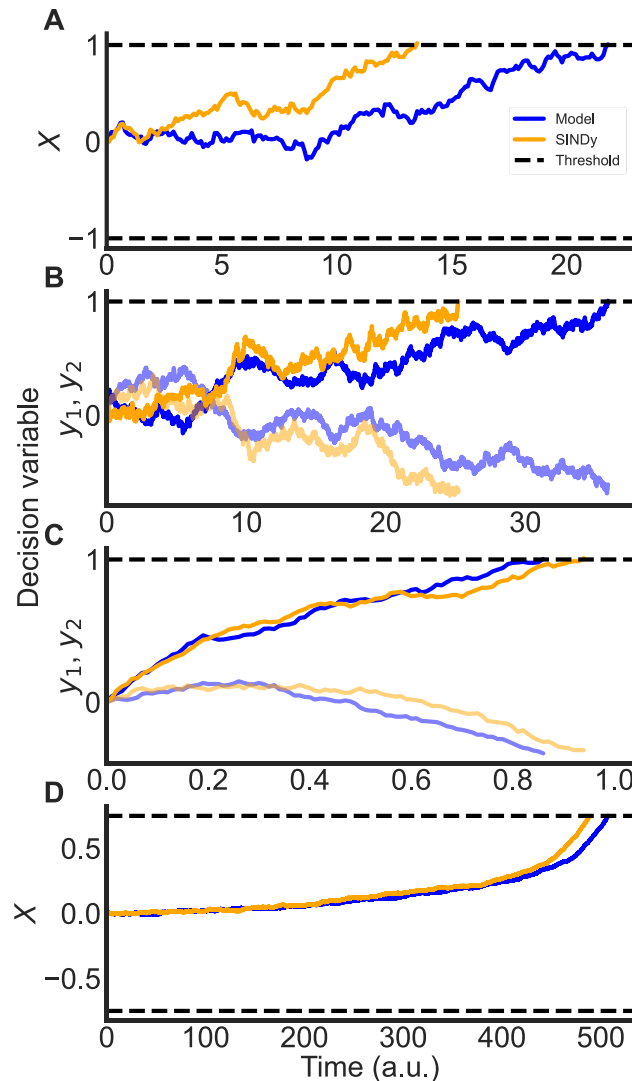
Using machine learning (symbolic regression) to uncover dynamical equations of low-dimensional state trajectory



Sparse identification of nonlinear dynamics (SINDy) to rediscover governing equations

Using machine learning (symbolic regression) to uncover dynamical equations of low-dimensional neural trajectory

Single-trial examples



Decision models

Drift-diffusion model (DDM)

Leaky competing accumulator (LCA) model approx. DDM with fine-tuning (LCA-DDM)

LCA model with saddle fixed point

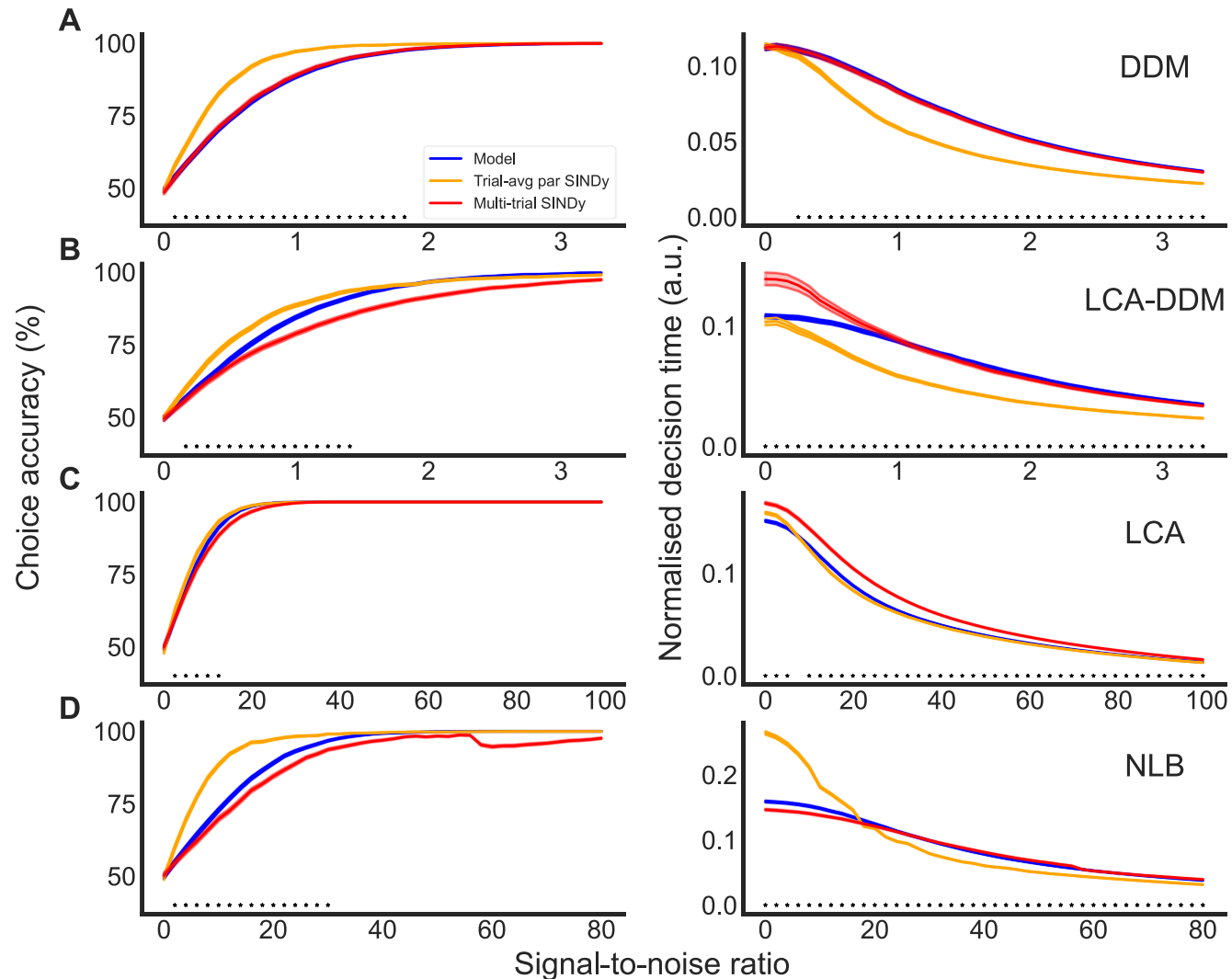
Nonlinear bistable model (NLB)

Brendan Lenfesty



Using machine learning (symbolic regression) to uncover dynamical equations of low-dimensional neural trajectory

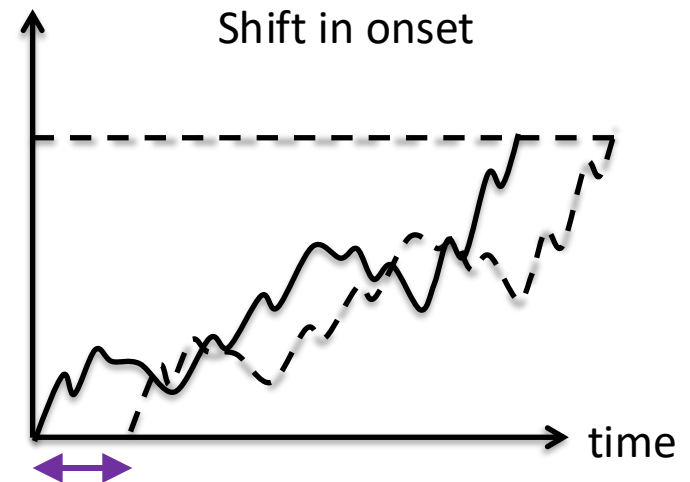
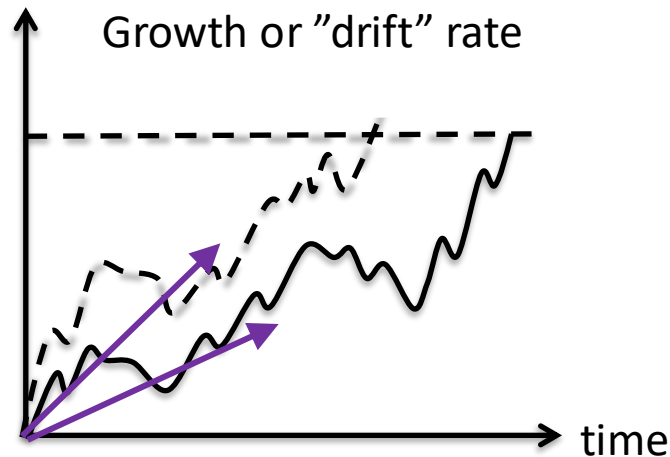
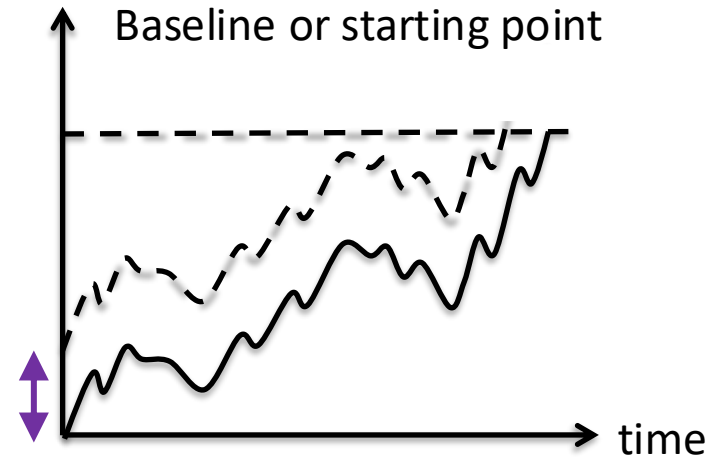
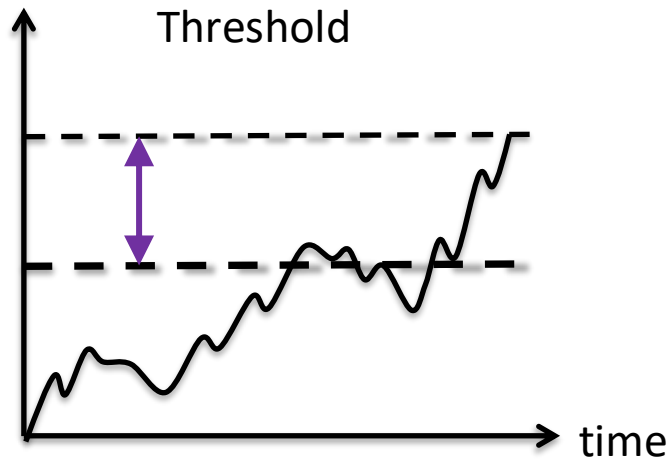
Trial-averaged approaches



Brendan Lenfesty



“Biases” or “controls” in decision making

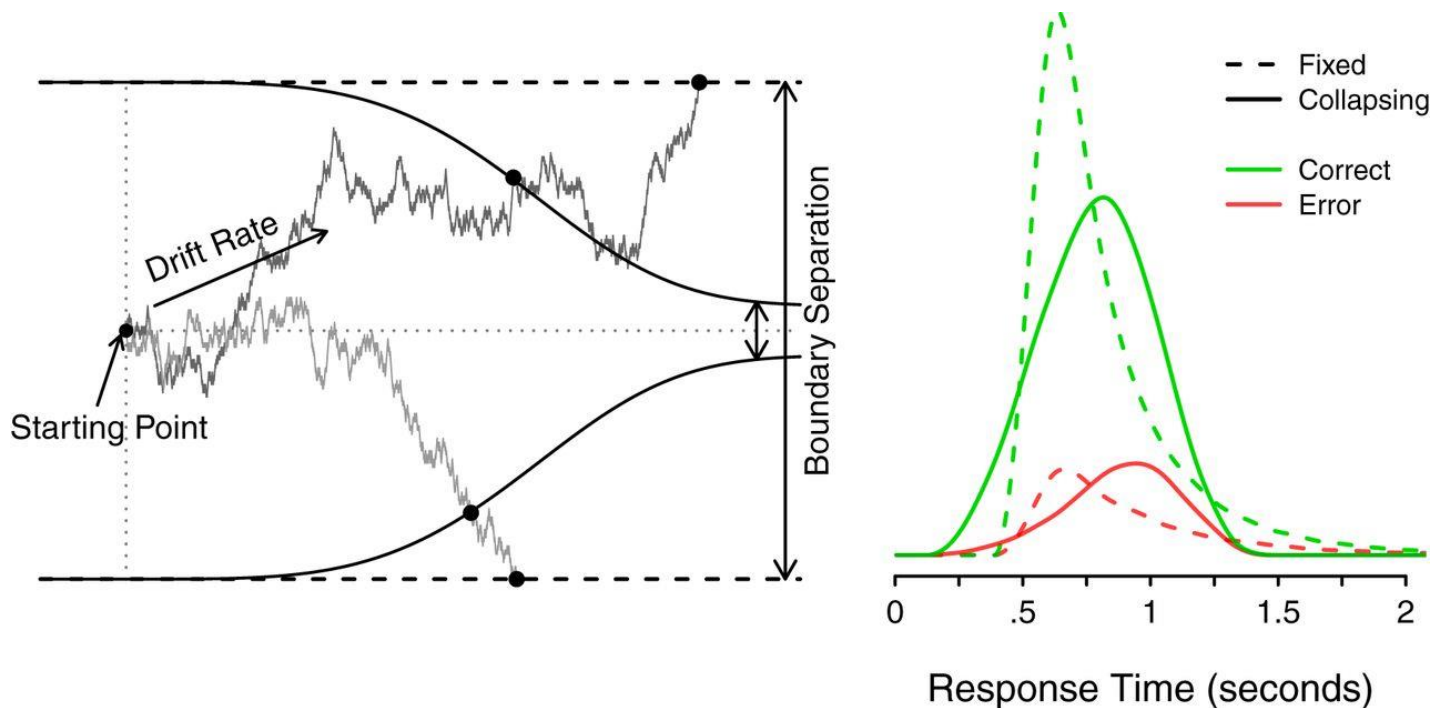


Speed-accuracy tradeoff (SAT)

“Biases” or “controls” in decision making

Threshold, drift rate, noise can be time-dependent, e.g. urgency

Collapsing bound over time can lead to faster correct than error decisions in DDM

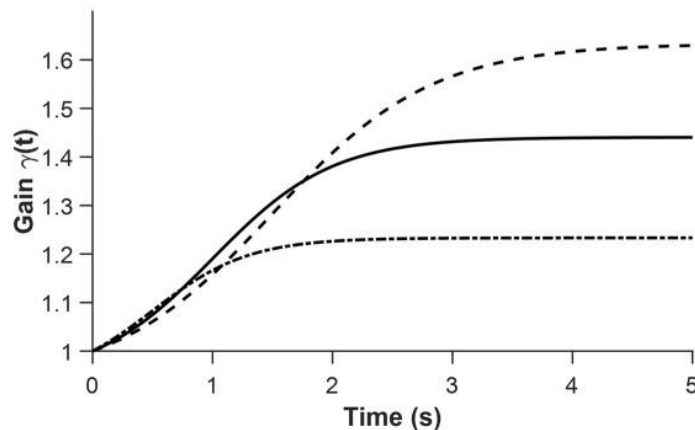
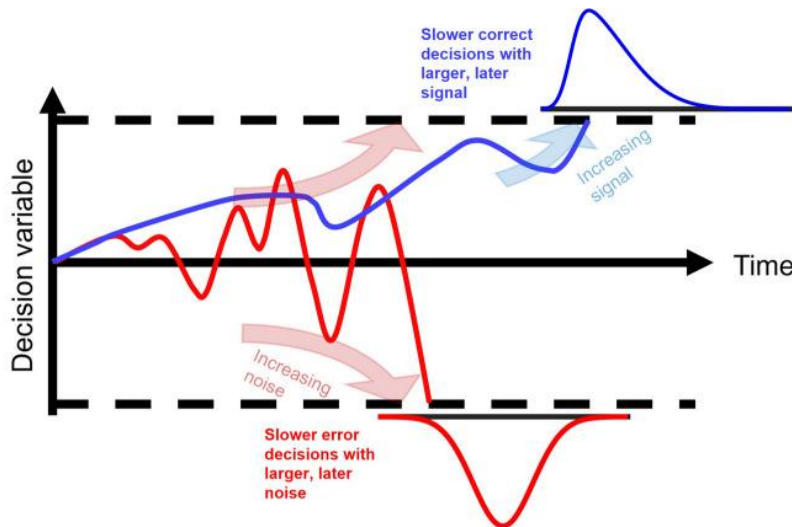


Churchland et al. (2008); Hawkins et al. (2015)

“Biases” or “controls” in decision making

Threshold, drift rate, noise can be time-dependent, e.g. urgency

Time-variant gain in noise but not signal leads to slower error decisions in DDM

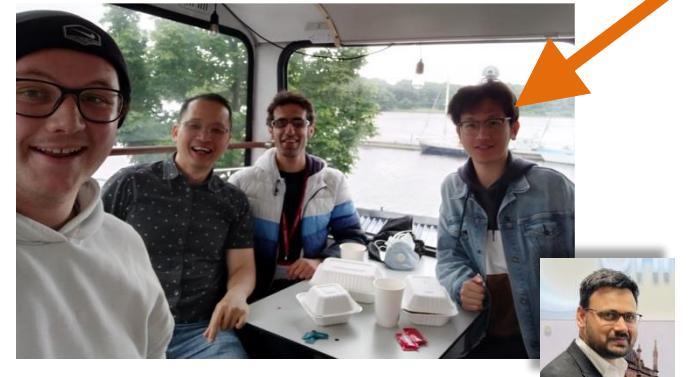


Type I

$$dX = v \gamma(t) dt + \sigma dW$$

Type II

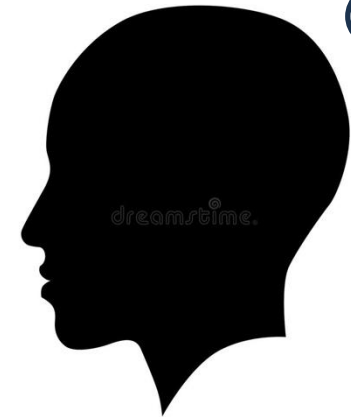
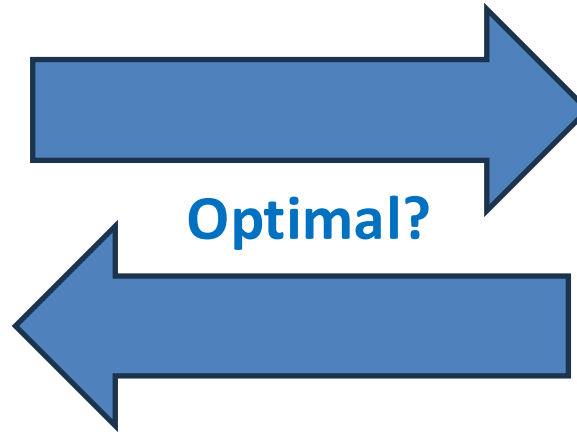
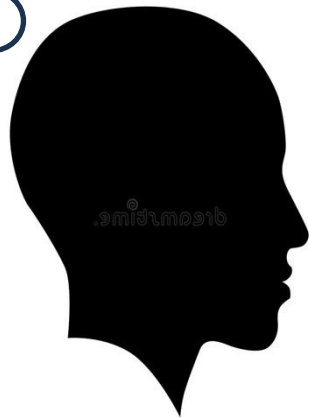
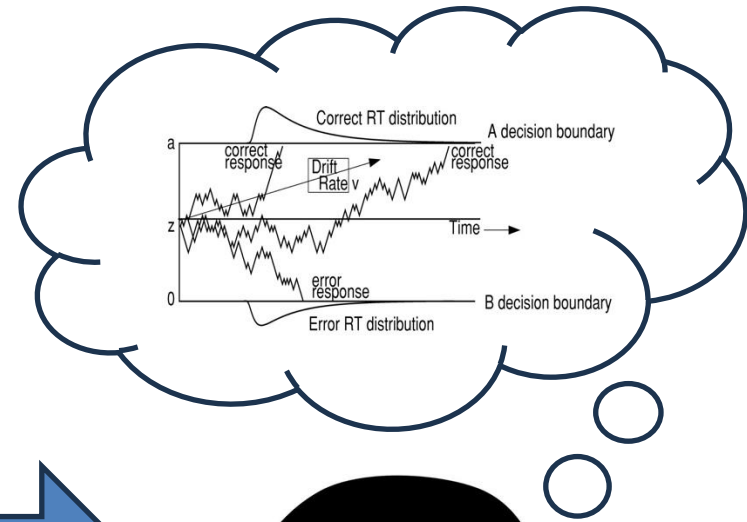
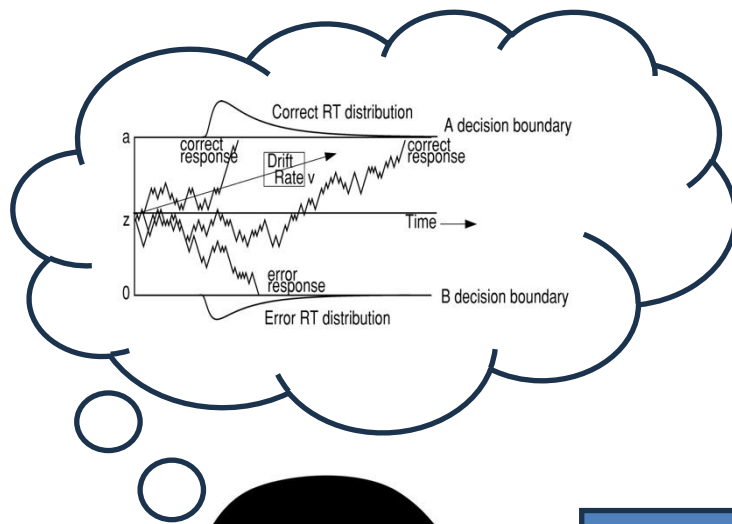
$$dX = v dt + \sigma \gamma(t) dW$$



Tan, Faraz, Lenfesty, Asadpour & Wong-Lin (2023);

Asadpour, Tan, Lenfesty & Wong-Lin (2024)

Coupling DDMs (“digital brains”) for group decision-making



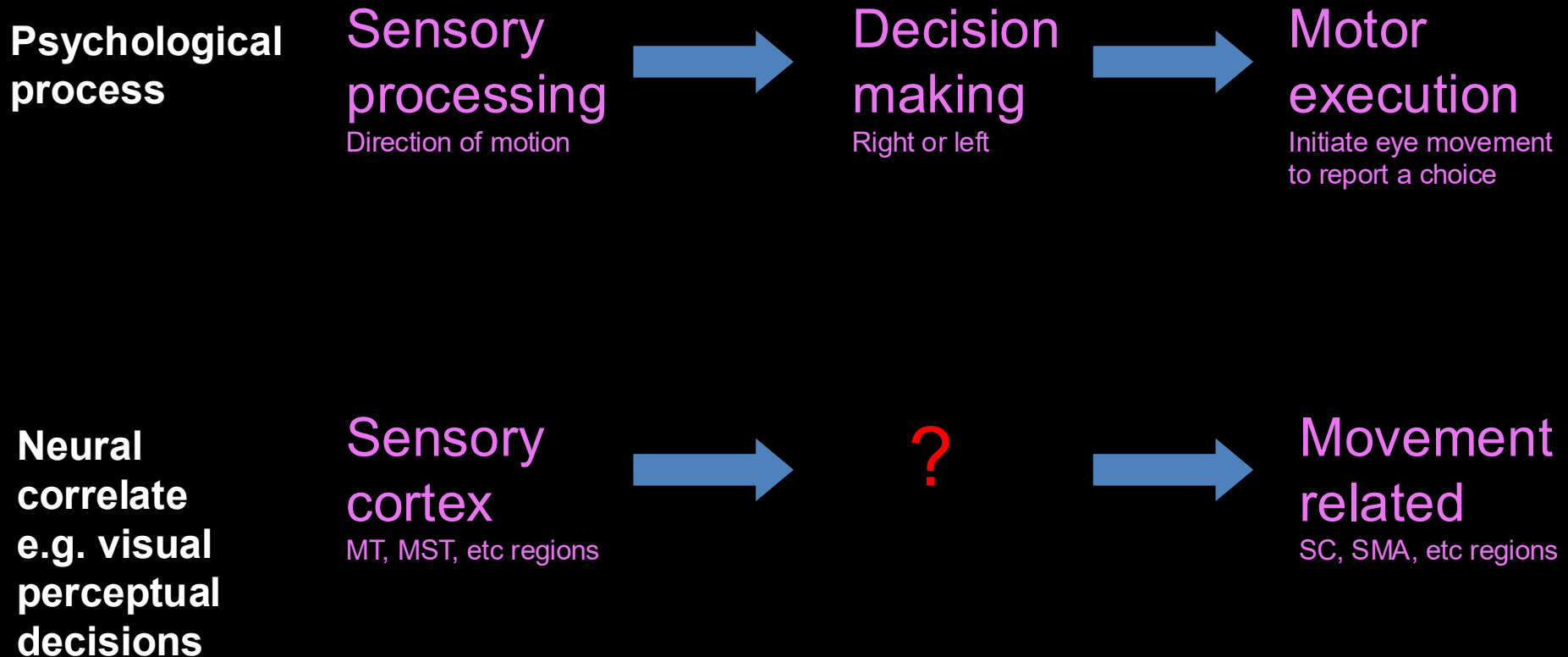
Toby Newey



Srivastava & Leonard (2015)
Newey et al. (2025; in prep)

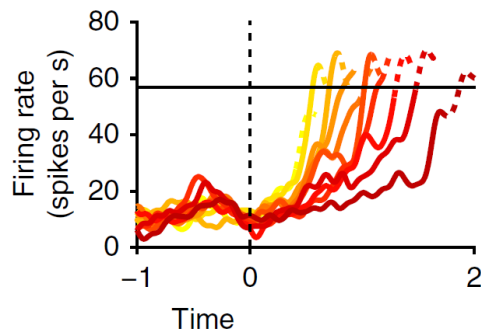
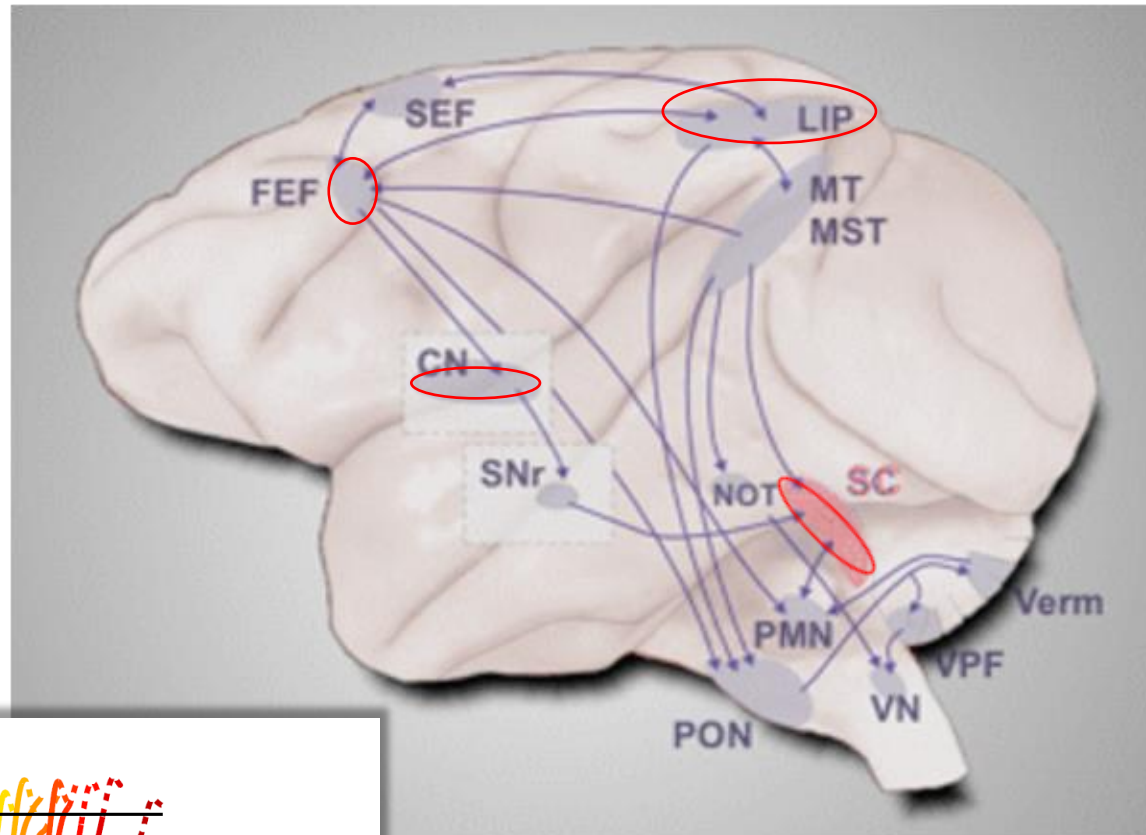
Is there evidence in brain physiology of such processes in perceptual decision-making?

Neural correlates of perceptual decision making



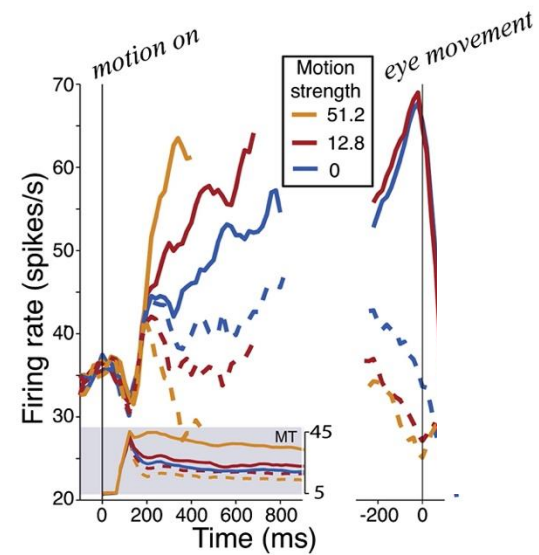
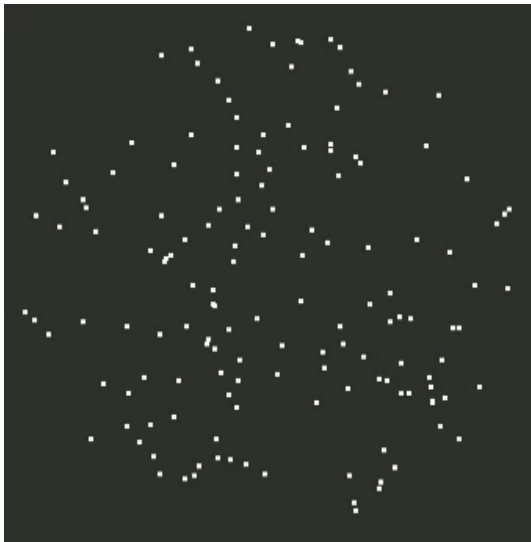
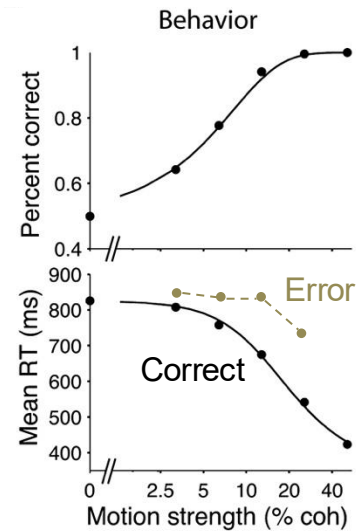
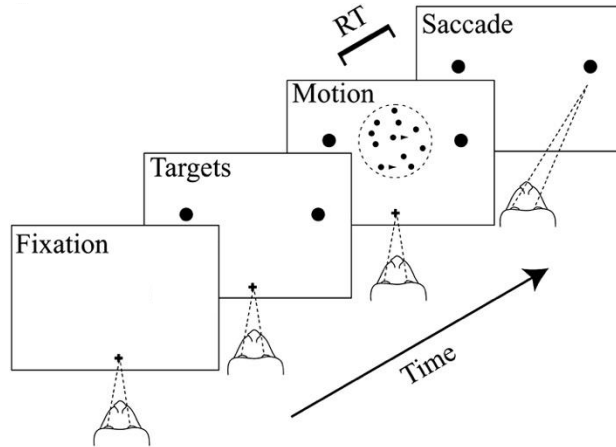
Brain circuitry controlling (saccadic) eye movements

The search for “neural integrators” in the brain



$$\int \dots dt$$

Classic experiment

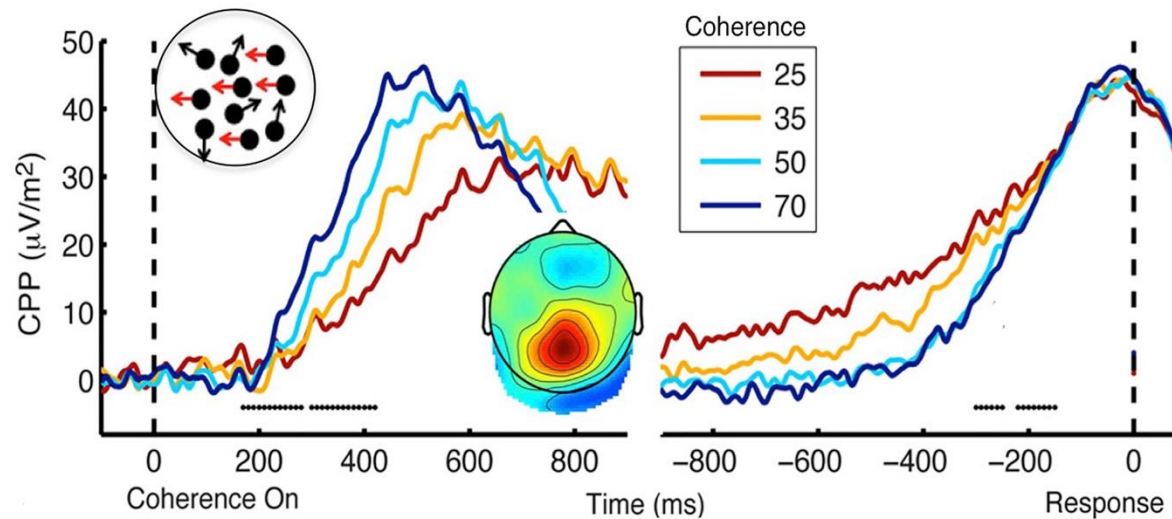


Neuronal population in area LIP (lateral intraparietal area)

Recording of a particular (LIP) brain cell in a non-human primate performing motion discrimination task



Similar macroscale brain activity dynamics in humans



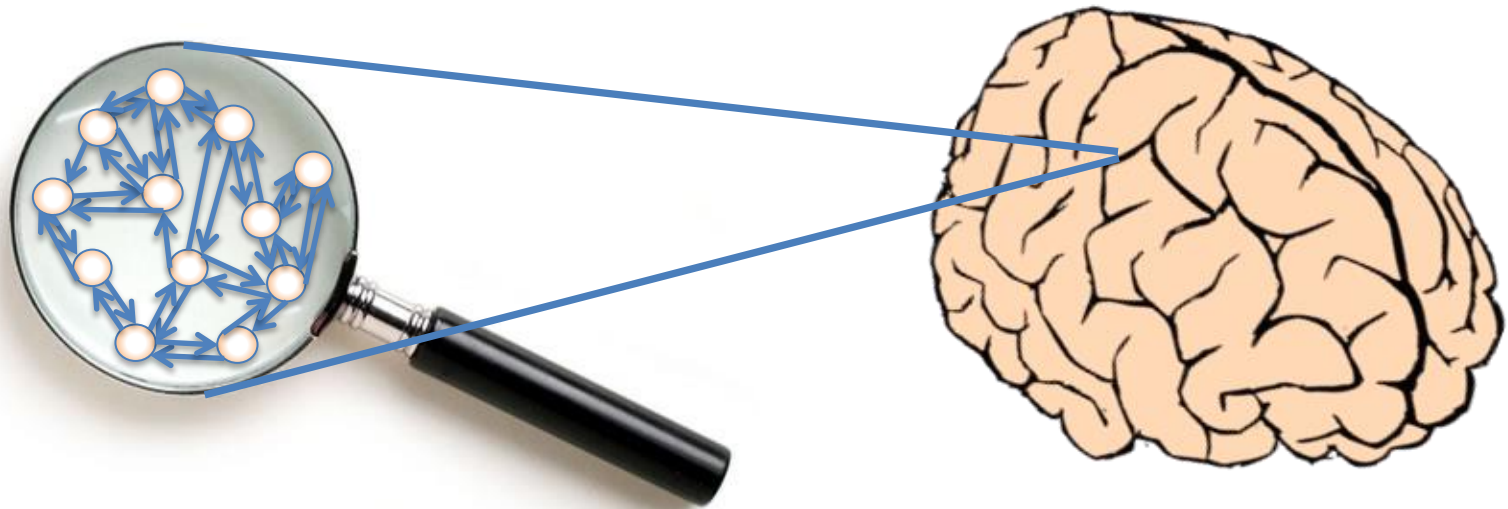
Kelly & O'Connell (2013)



Electroencephalography (EEG) measures ensembles of neural activity with high temporal resolution

How to link cognitive models to elements of the brain, i.e. brain cells (neurons) and their connections (synapses)?

A network of neurons and emergent behaviour

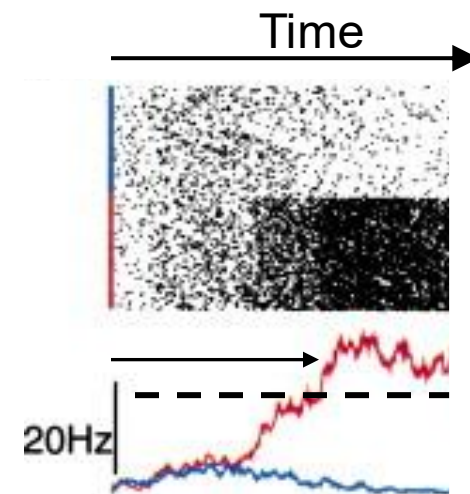
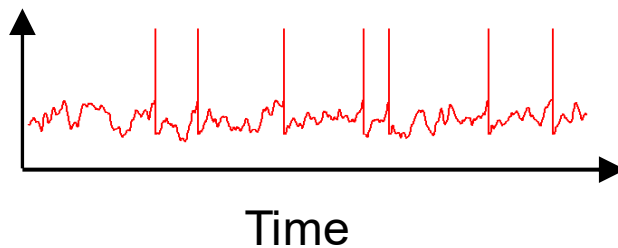


Spiking activity from a single model neuron, and network of connected neurons

Leaky Integrate-and-fire neuronal model
with realistic synaptic dynamics

$$dV/dt = -g_L(V-E_L) + I_e - \sum g_s s(t) (V-E_s)$$

Neuronal
membrane
Potential, V



X-J Wang
(2002)

Key mechanisms in this model?

Simulate this model more efficiently?

What is the link to cognitive models?



Model reduction

A “mean-field” (neural population) approach

Neural population dynamics described by

$$\tau \frac{dr}{dt} = -r + \varphi(I)$$

$$\varphi(I) = \frac{c I - I_0}{1 - \exp(-g (c I - I_0))}$$

an approximation of a first passage time formula for LIF neurons (via Fokker-Planck approach), and I is averaged total input to a neuron,

$$I = W S + I_{stimulus} + I_{noise}$$

where W is the synaptic (connectivity) strength, and

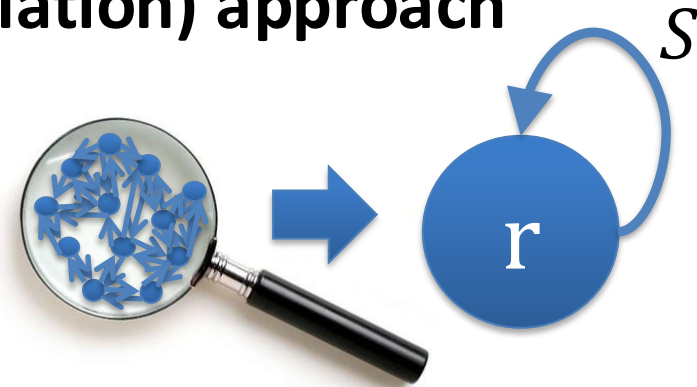
$$\tau_{noise} \frac{dI_{noise}}{dt} = -I_{noise} + \eta \sqrt{\tau_{noise} \sigma_{noise}^2}$$

(Ornstein-Uhlenbeck process)

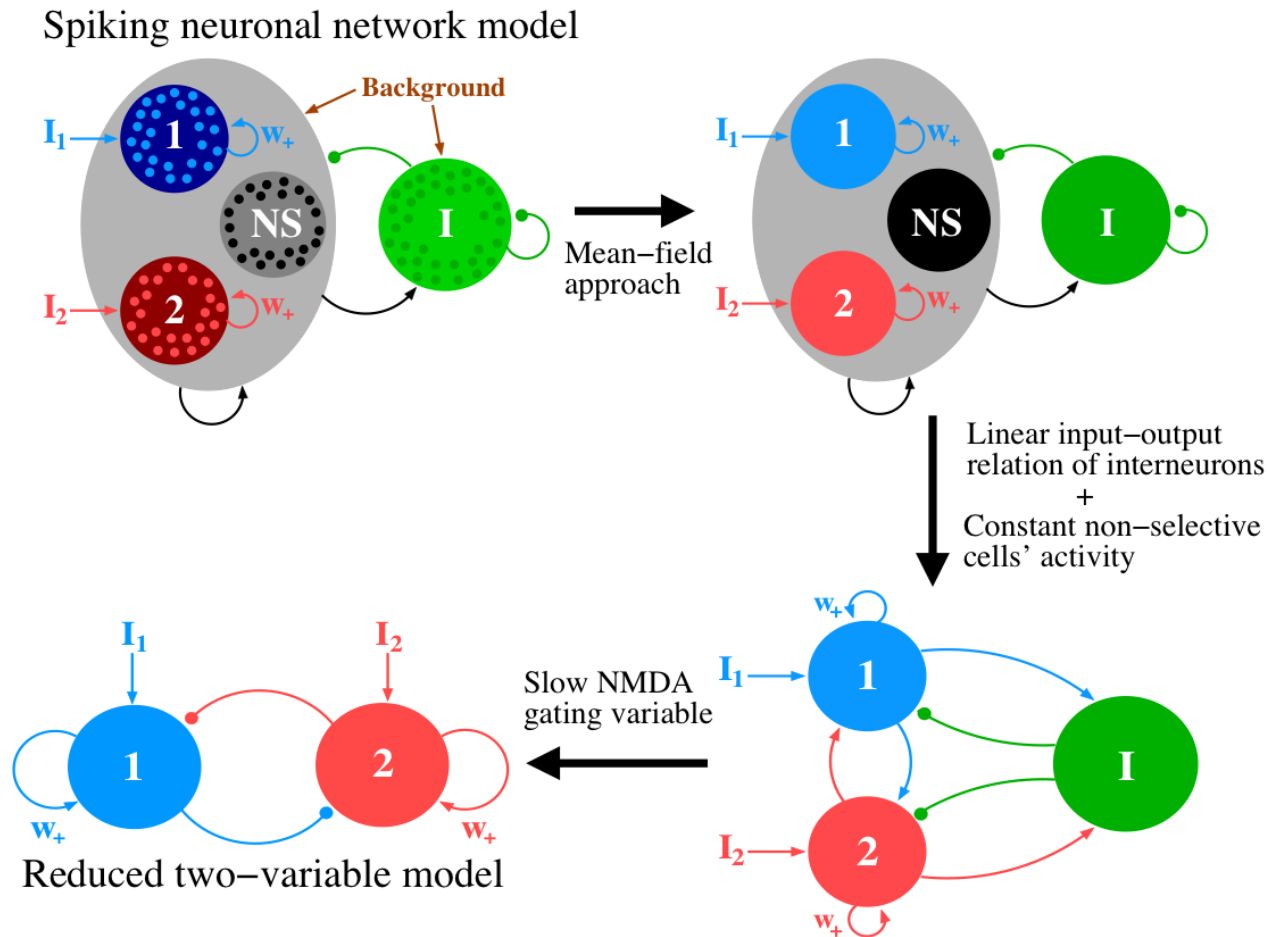
Dynamics describing the averaged (slow) synaptic dynamics for the neural population

$$\frac{dS}{dt} = -\frac{S}{\tau_s} + (1 - S) \gamma r$$

(with interspike interval distribution $\sim$ Poisson)



Reducing a biophysical model of decision-making

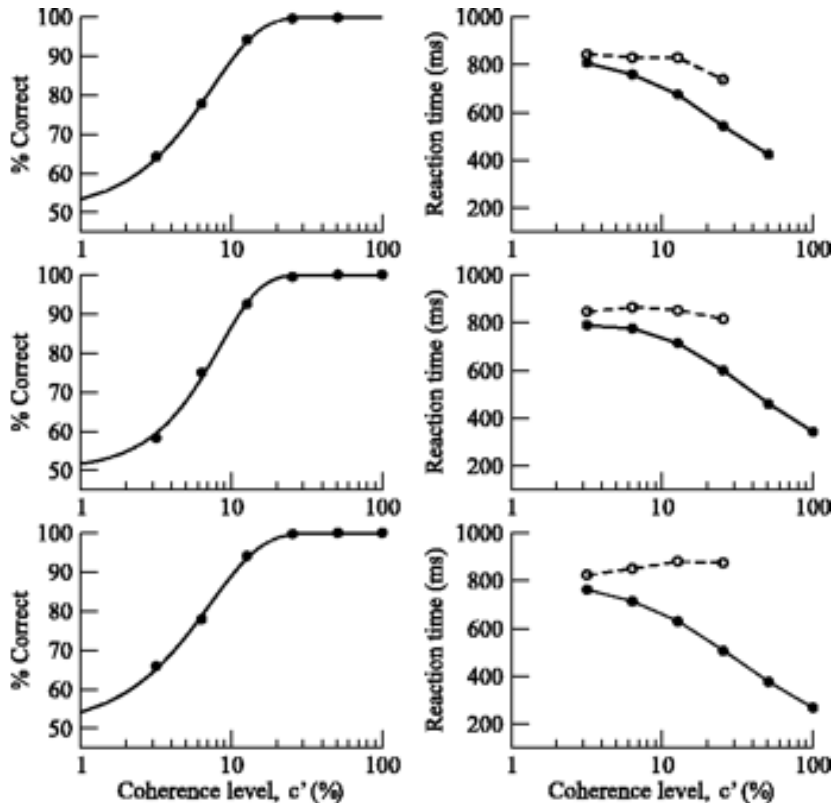


Dynamical equations described by population-averaged of slow NMDA-mediated synaptic dynamics:

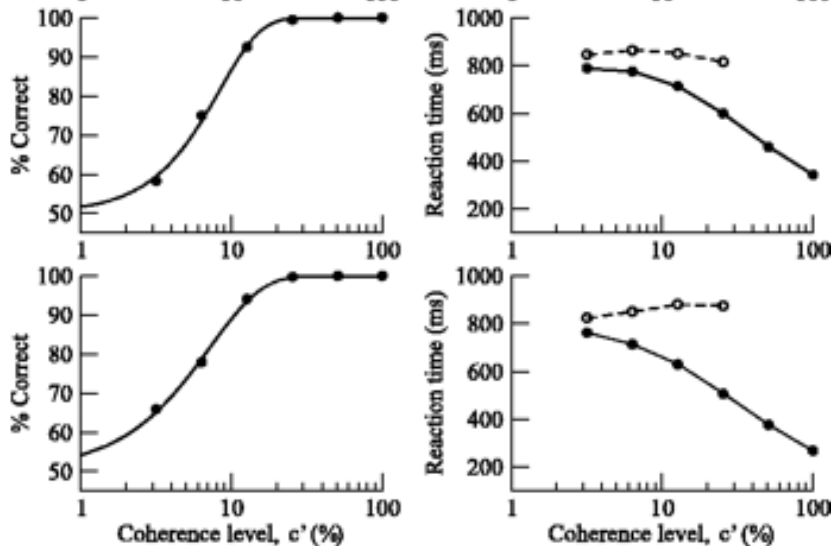
$$\begin{aligned} \frac{dS_1}{dt} &= -S_1/\tau_S + (1 - S_1) F(W_+ S_1 - W_- S_2 + I_1 + I_{\text{noise}}) \\ \frac{dS_2}{dt} &= -S_2/\tau_S + (1 - S_2) F(W_+ S_2 - W_- S_1 + I_2 + I_{\text{noise}}) \end{aligned}$$

A reduced (attractor) neural network model with strong positive feedback

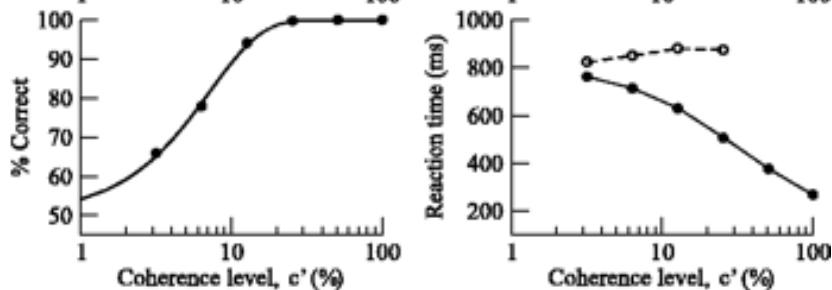
Experimental data



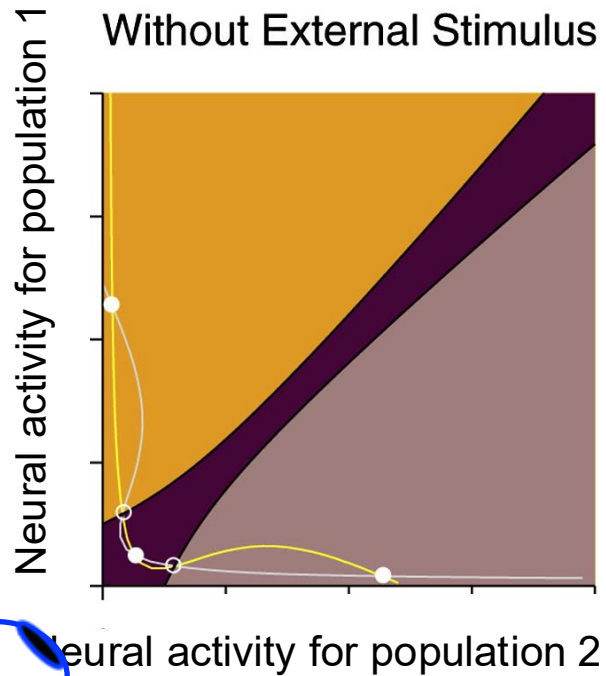
Spiking neuronal network model



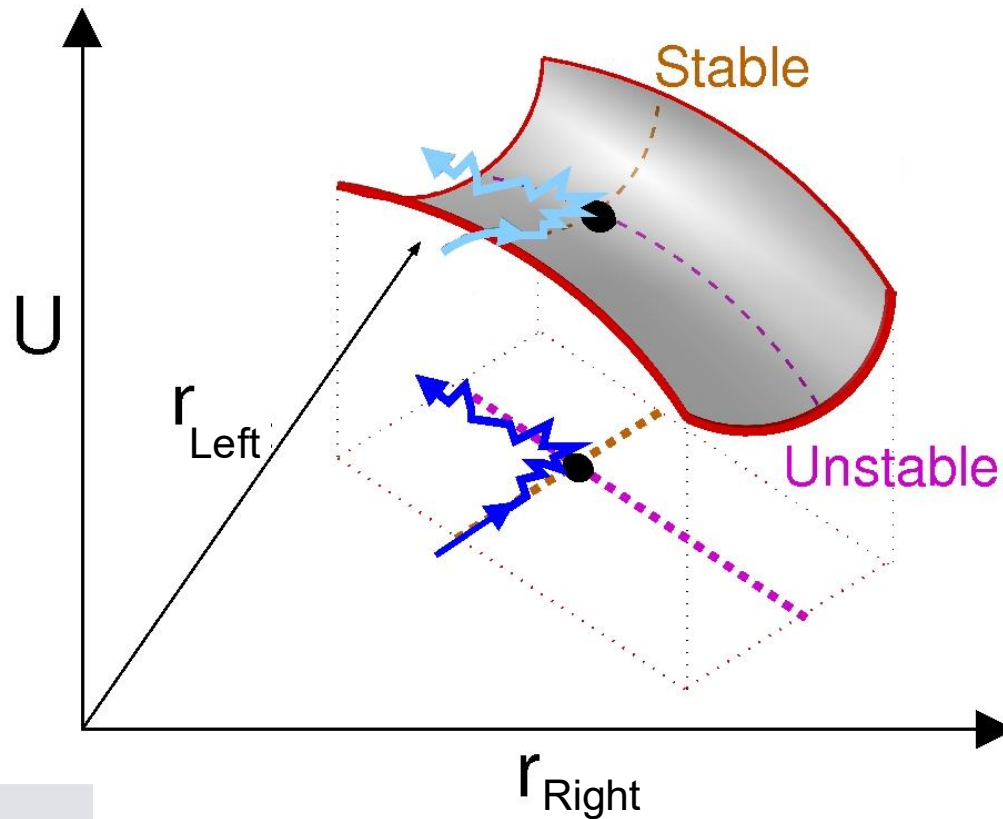
Reduced two-variable model



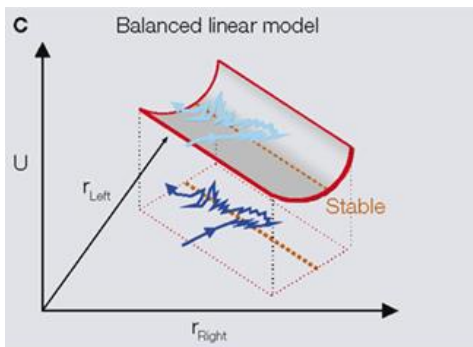
Phase-plane analysis of attractor network model



“Potential energy” landscape in phase space



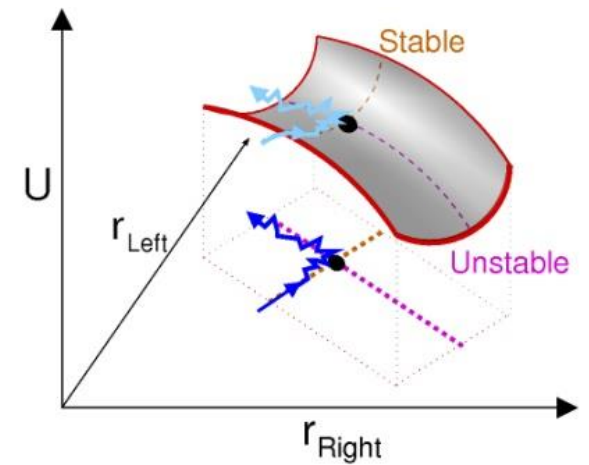
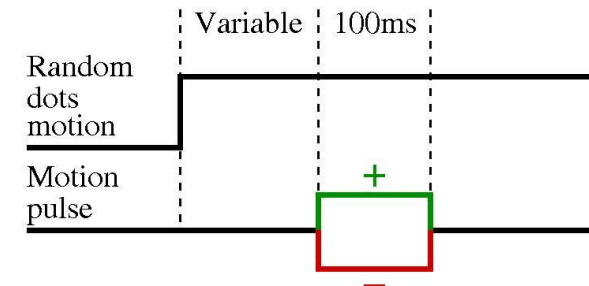
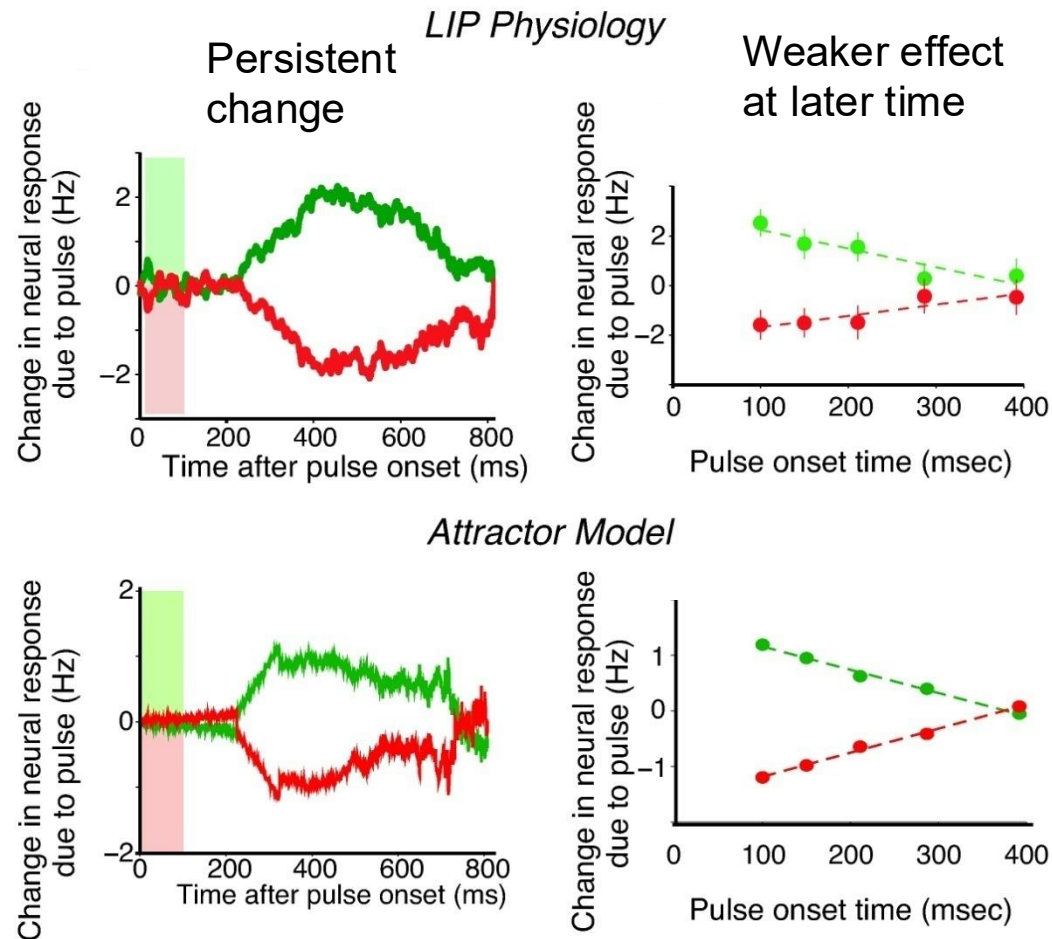
In the vicinity of the unstable “**saddle**” steady state



Wong & Huk (2008)

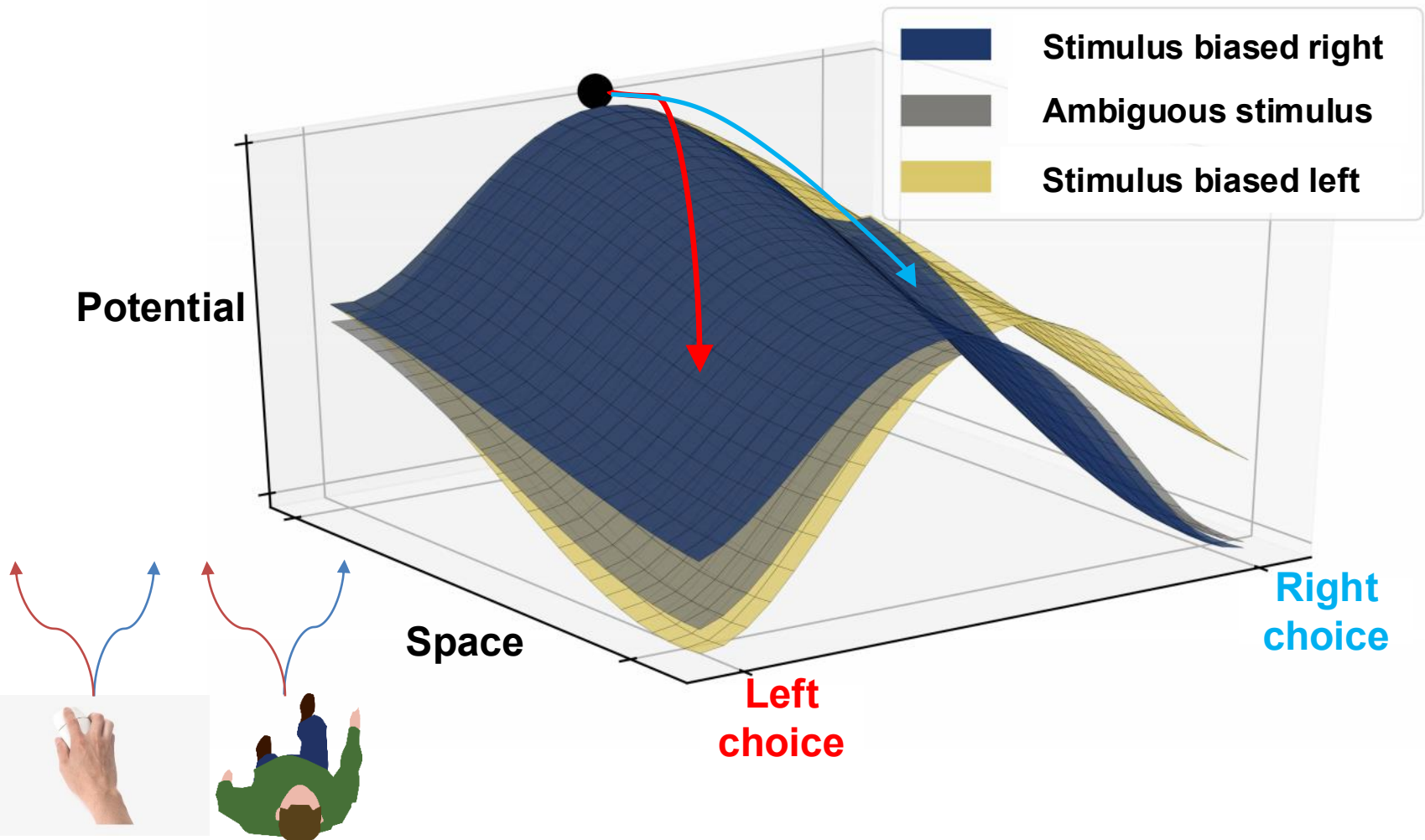
Bogacz et al. (2006)

Model prediction: Attractor network model with “runaway” temporal integration can account for weaker perturbative effects at later times



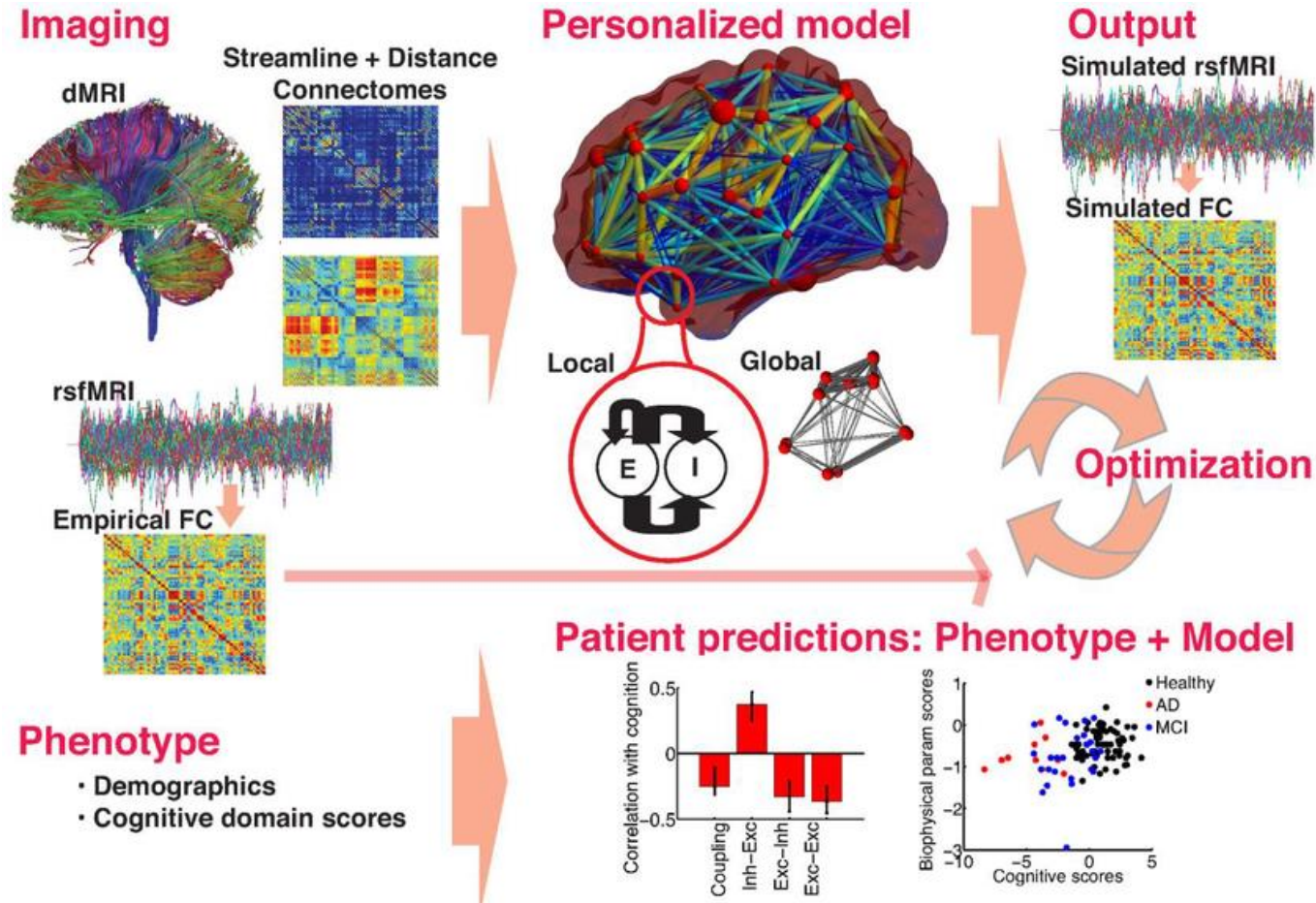
“Potential energy” landscape in space

Reconstructed from walking or computer mouse trajectories



Adapted from Zgonnikov, Atiya, O’Hora, Rano & Wong-Lin (2019)

Off-track: Mean-field model in The Virtual Brain (TVB)

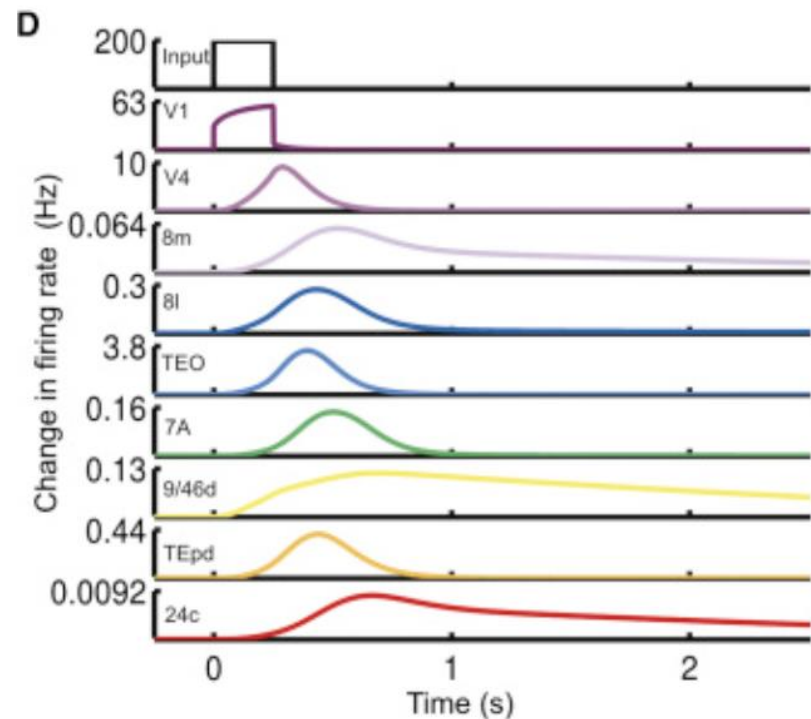
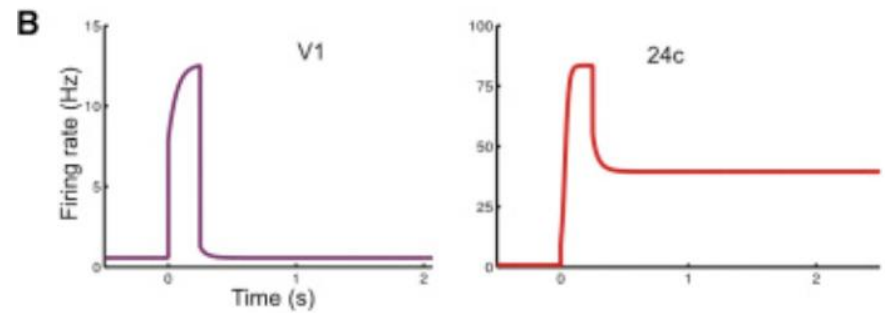
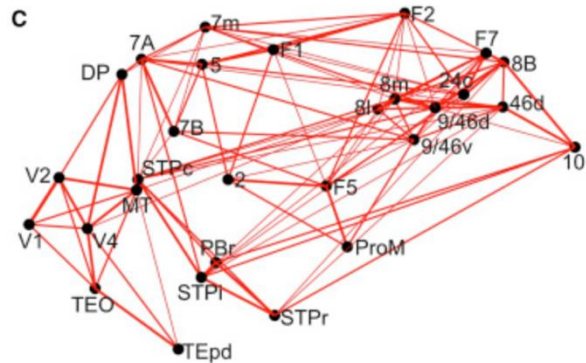
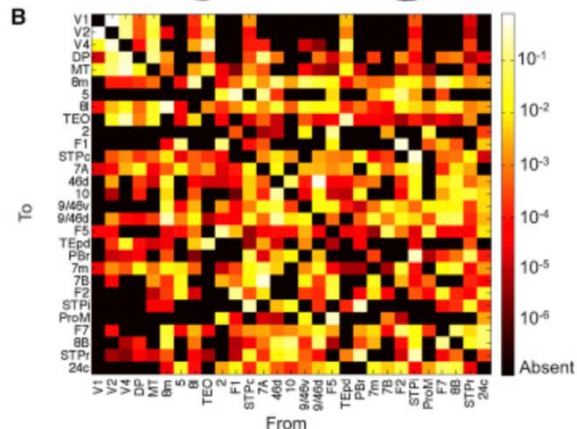
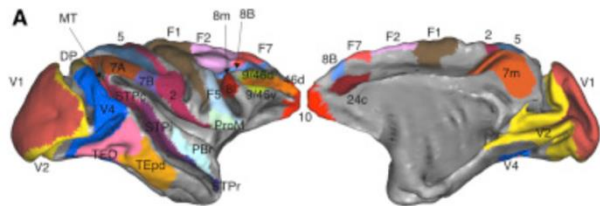


Sanz-Leon et al. (2013; 2015)



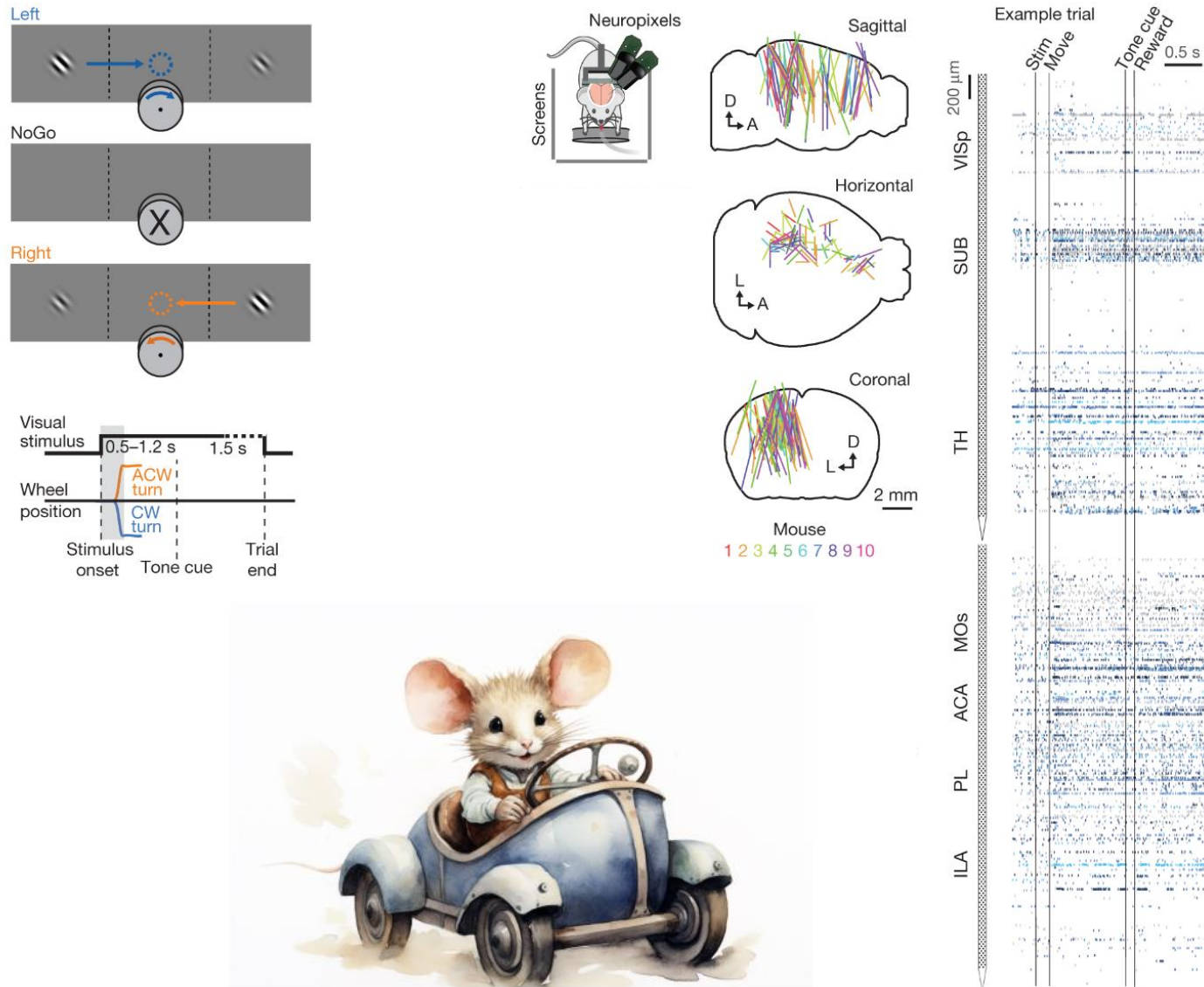
THEVIRTUALBRAIN.

Off-track: Mean-field model in monkey brains



Chaudhuri et al. (2015)

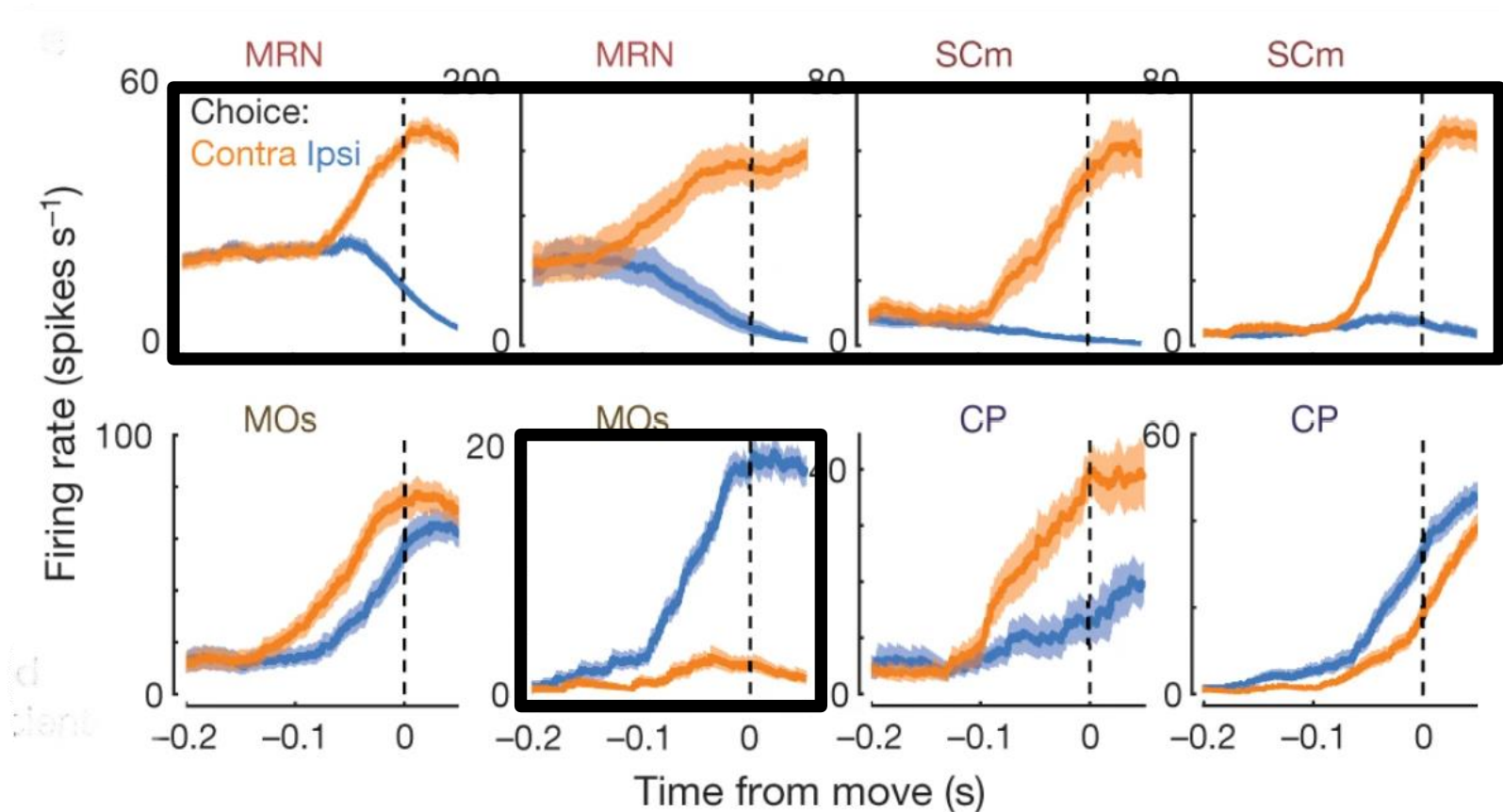
Neural dynamics in higher dimensions from simultaneously recorded neurons



Steinmetz et al.
(2019)

Neural dynamics in higher dimensions from simultaneously recorded neurons

Some sample neuronal activities encoding choice information exhibit winner-take-all behaviour

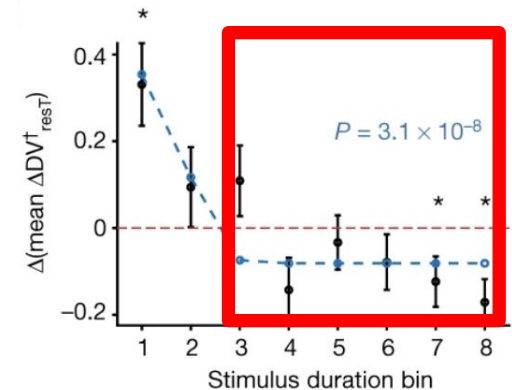
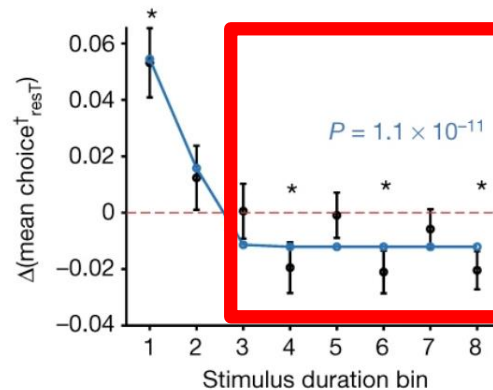
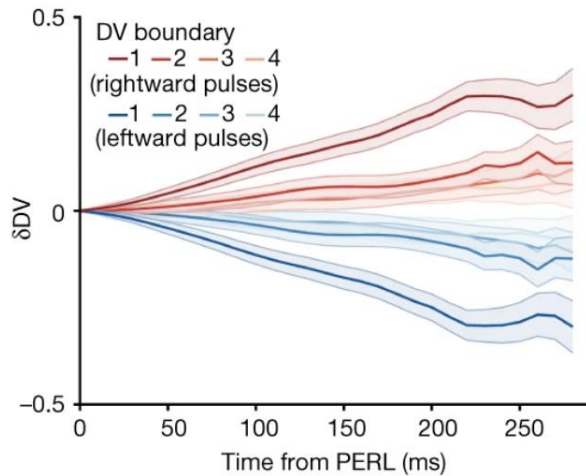
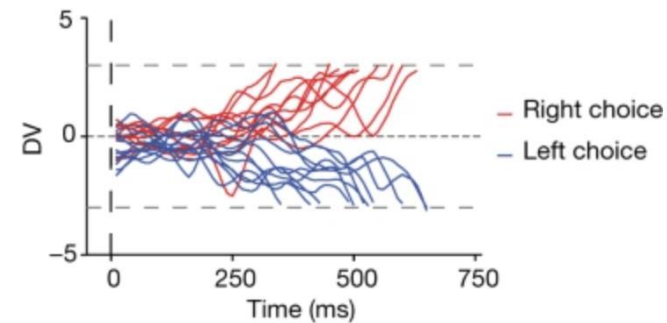
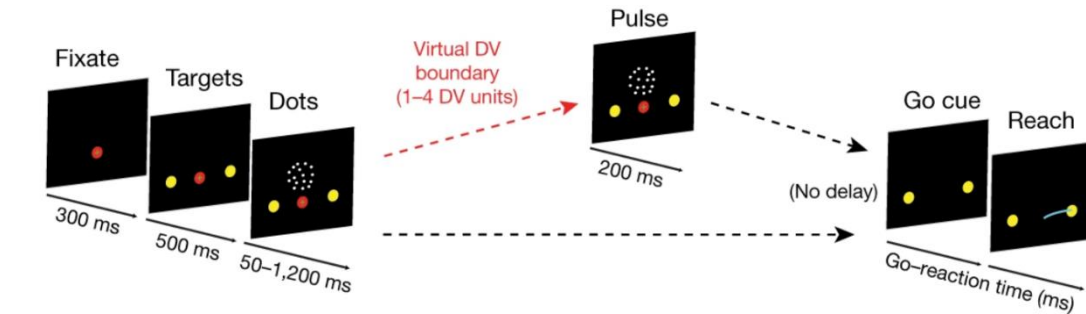


Activities averaged across experimental trials and aligned from movement onset

*Steinmetz et al.
(2019)*

Neural dynamics in higher dimensions from simultaneously recorded neurons

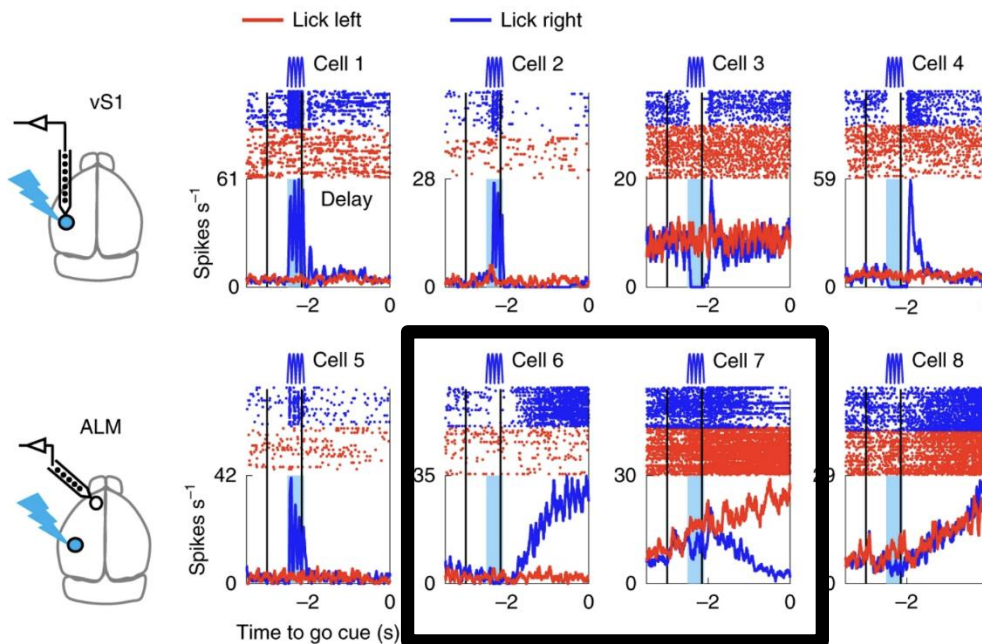
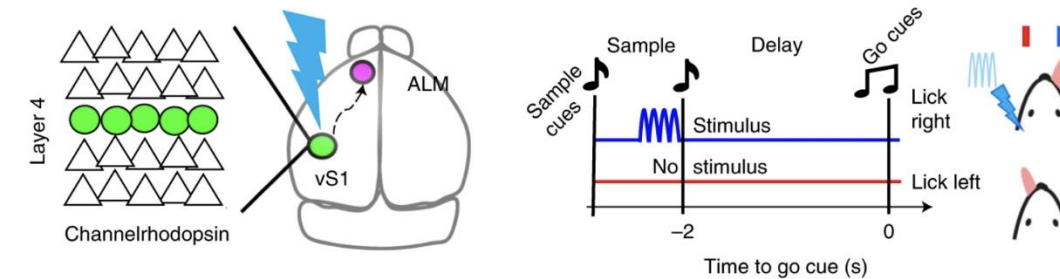
Peixoto et al. (2021)



Neural dynamics in higher dimensions from simultaneously recorded neurons

Finkelstein et al. (2021)

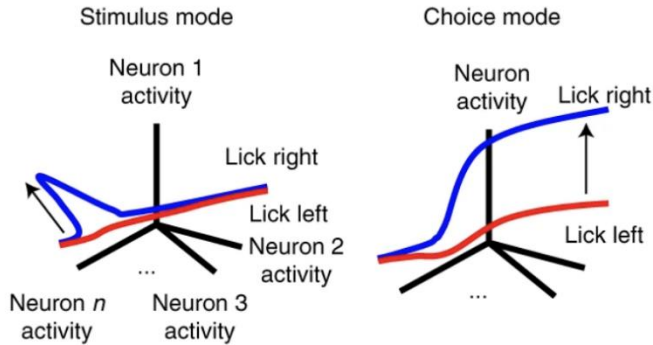
Direct cortical photostimulation in tactile decisions



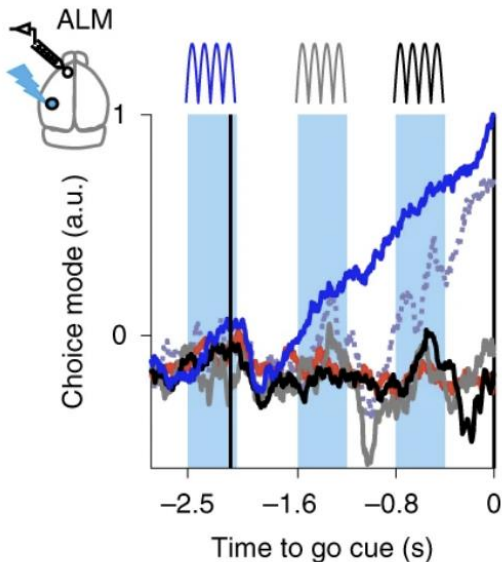
Neural dynamics in higher dimensions from simultaneously recorded neurons

Finkelstein et al. (2021)

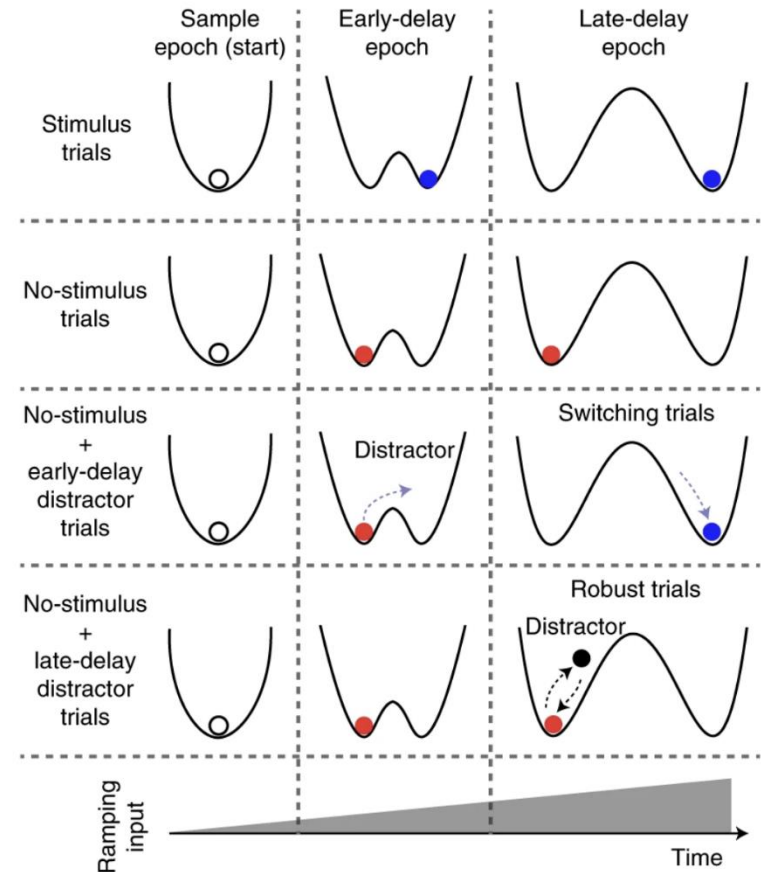
Dimensionality reduction
of population activity in the ALM



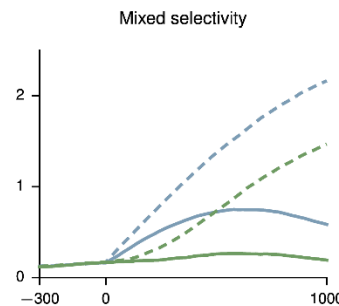
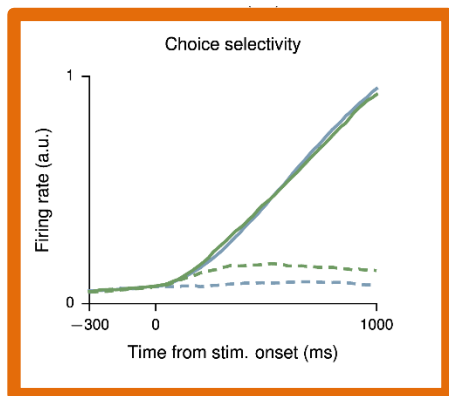
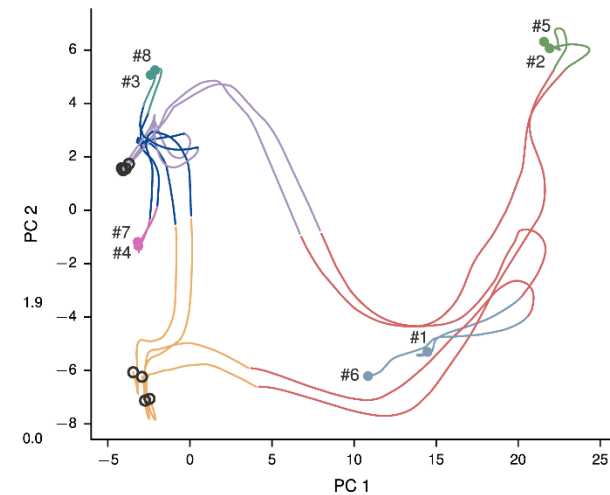
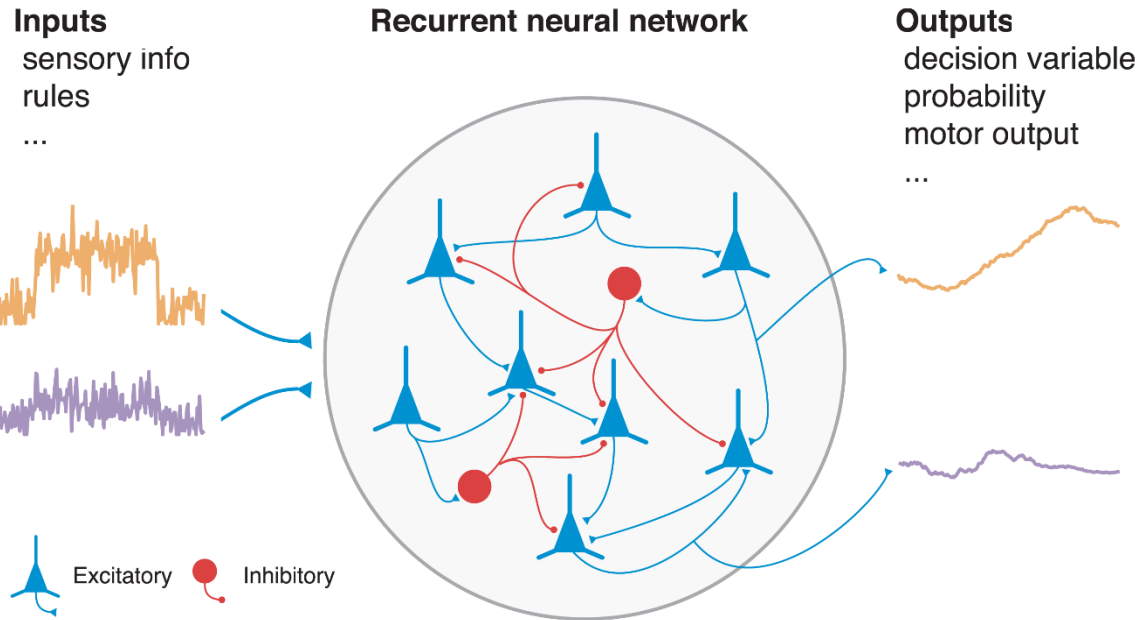
Gating of distractors
in the ALM choice mode



Ramping results in
temporal gating of distractors

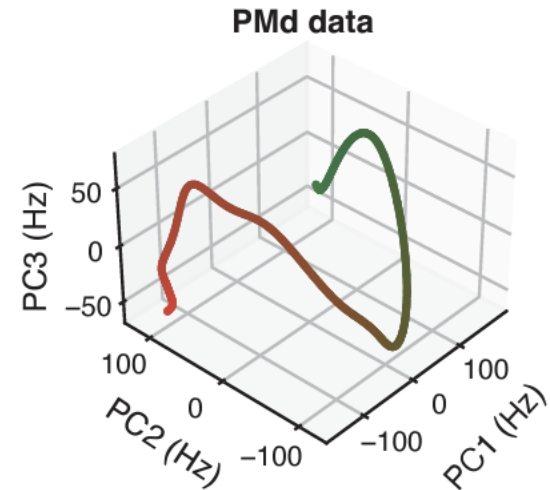
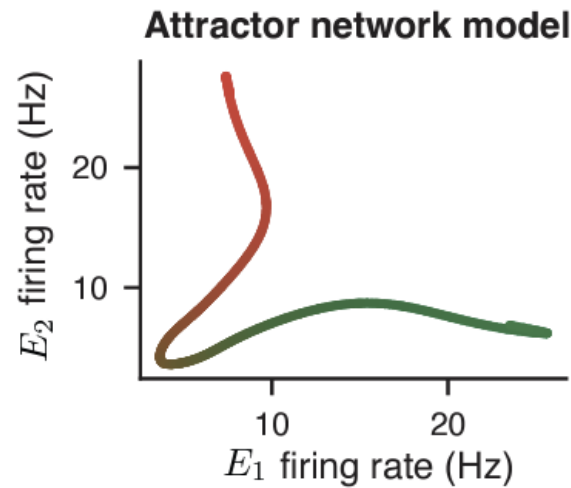
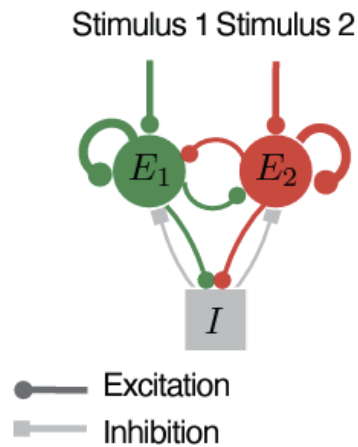


Training excitatory-inhibitory recurrent neural network (RNN) model in decision-making tasks



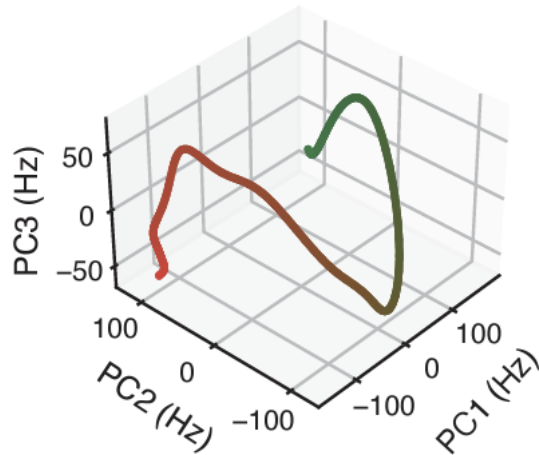
*Neural units embedded in trained RNN with **time integration & winner-take-all behaviour** (amongst others)*

Additional non-selective neuron causes neural trajectories to traverse into additional dimension

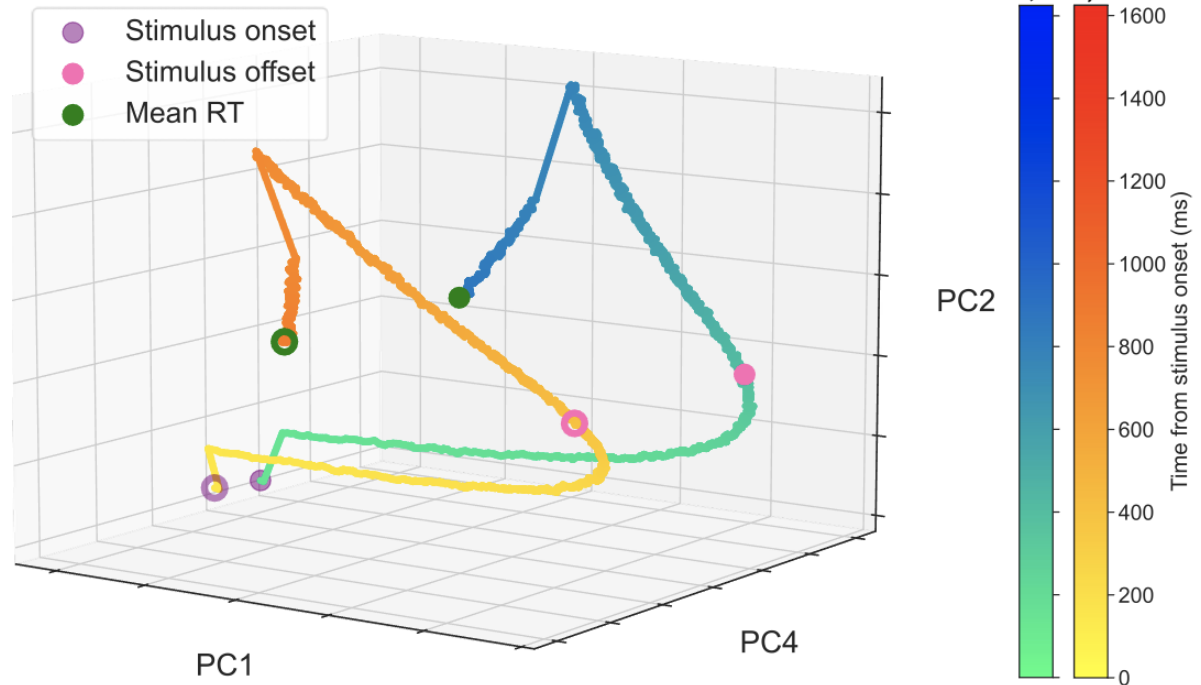
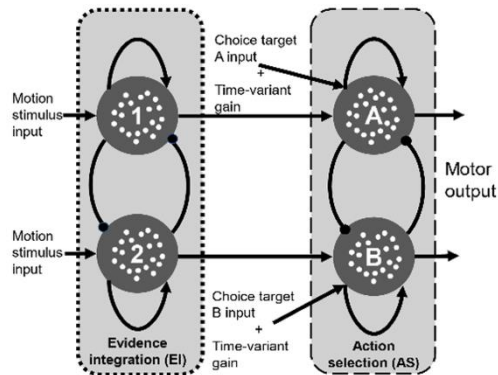


Genkin, Shenoy, Chandrasekaran & Engel (2025)





Neural trajectory with 4 neural units (2 evidence integration + 2 action selection)



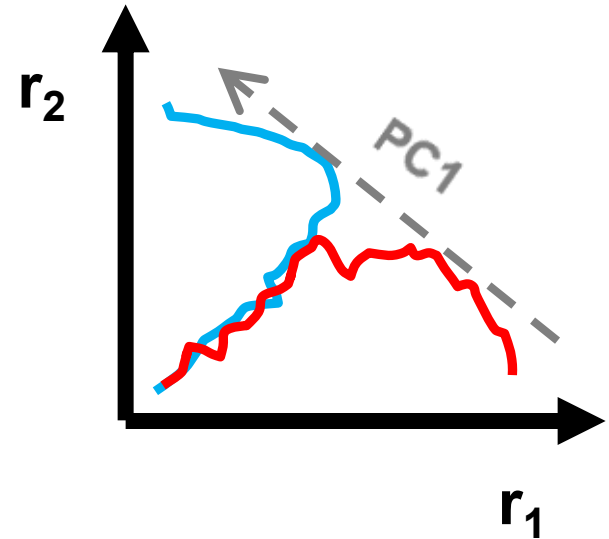
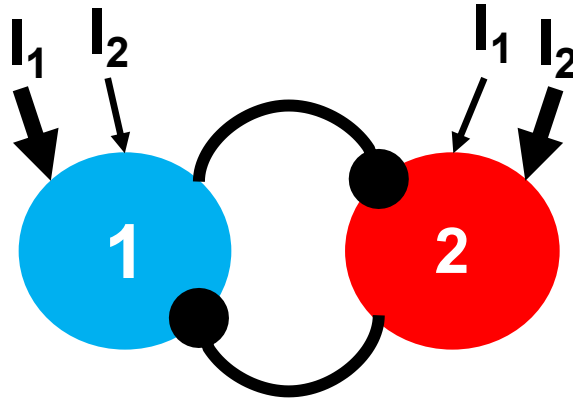
Brendan Lenfesty



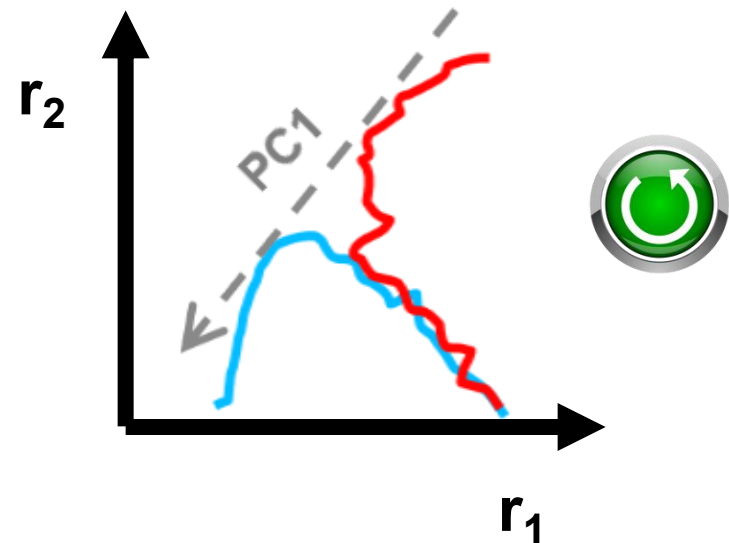
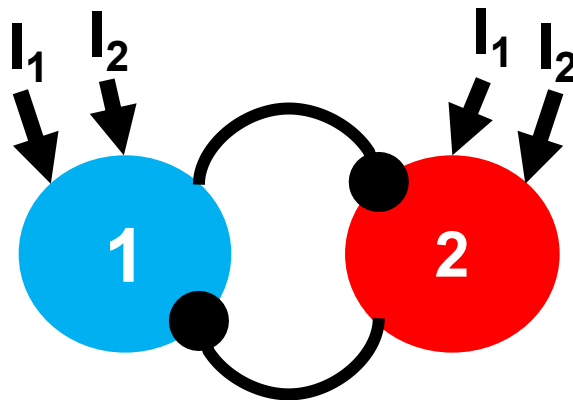
Lenfesty et al. (2025)

Rotation in neural latent space due to mixed selectivity of neurons – robust encoding

More pure selectivity

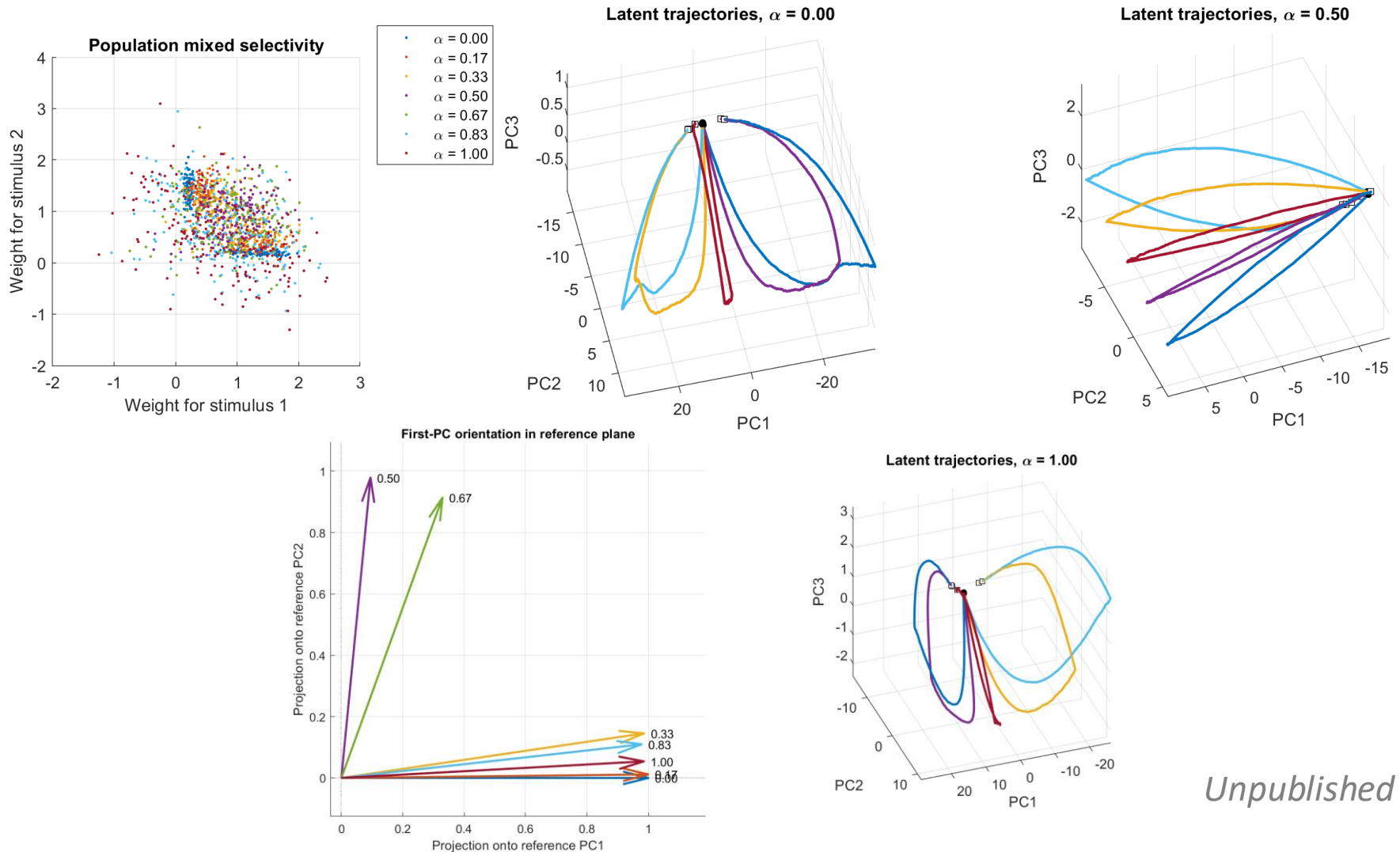


More mixed selectivity



Rotation in neural latent space due to mixed selectivity of neurons – robust encoding

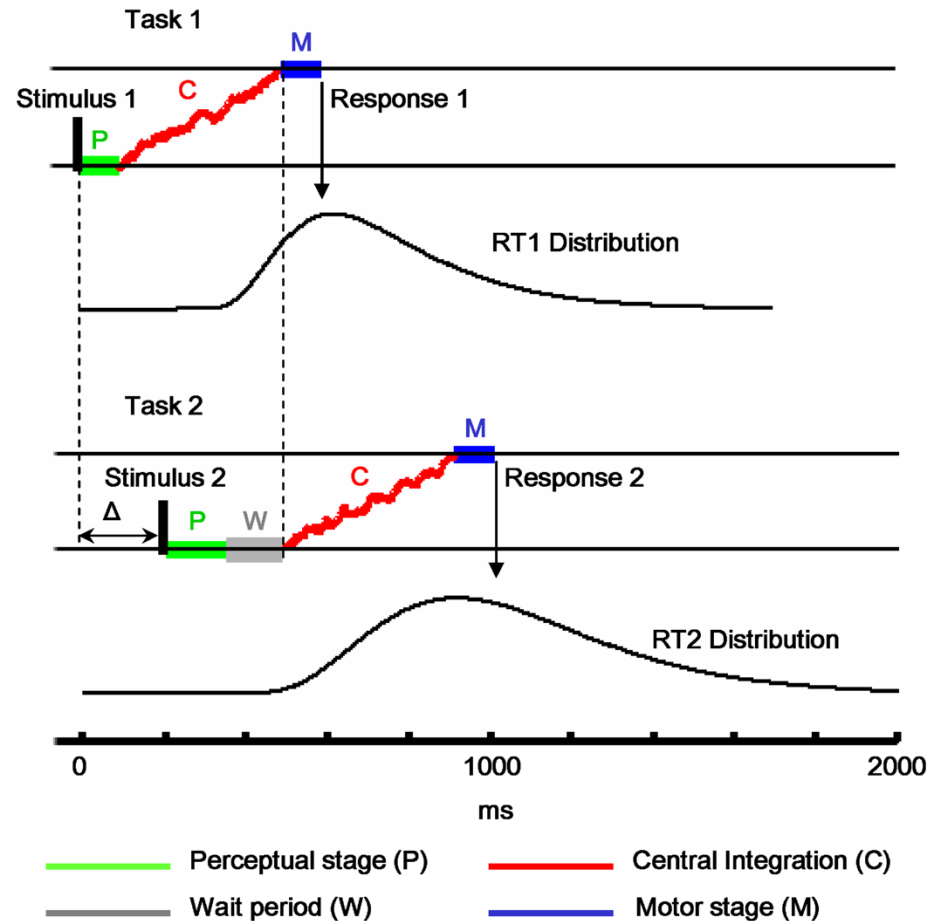
Validation with 200 neurons with mixed selectivity factor α for 2-choice task



Unpublished

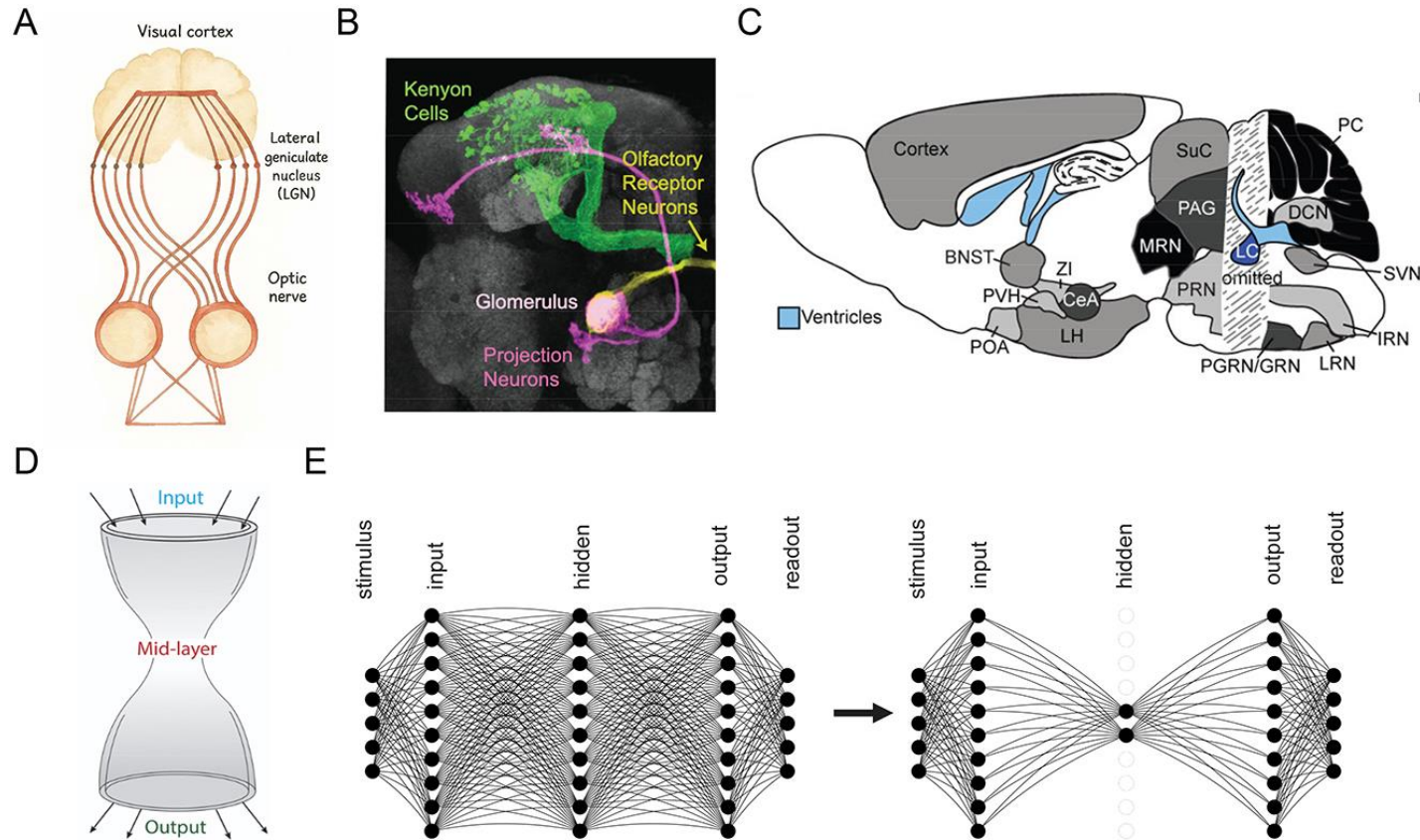
Decision-making as a central bottleneck process

Behavioural evidence



Decision-making as a central bottleneck process

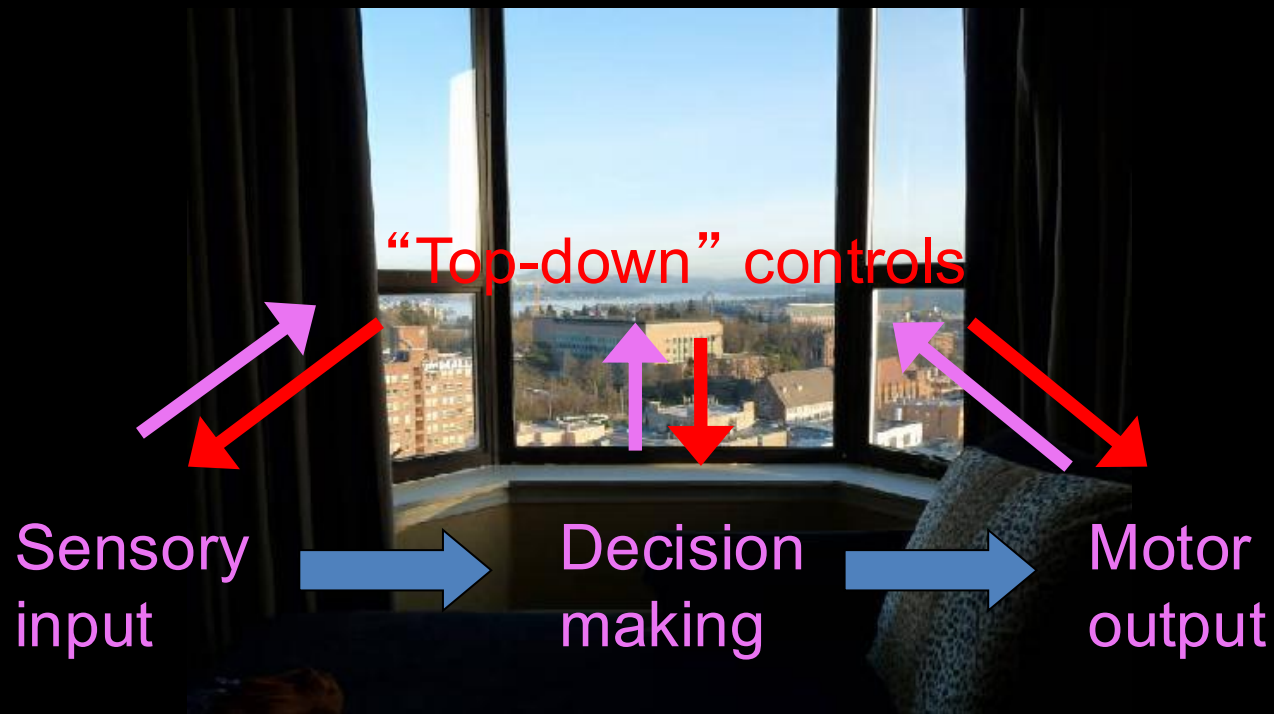
ANN evidence (with non-negative connectivity)



More conducive for controls?

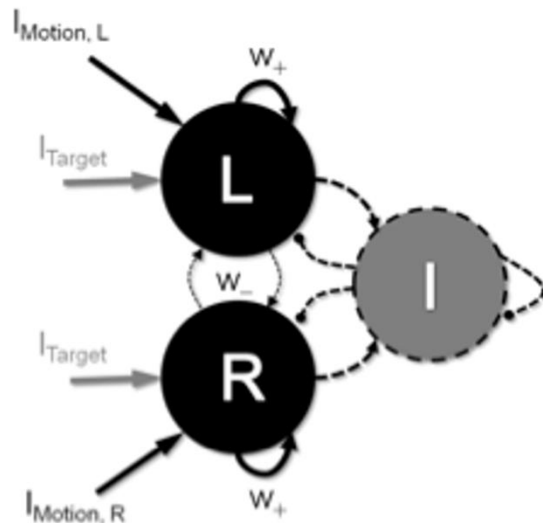
Liu, Du, Wong-Lin & Wang (2025)

Perceptual decision making: a window to understanding higher cognition

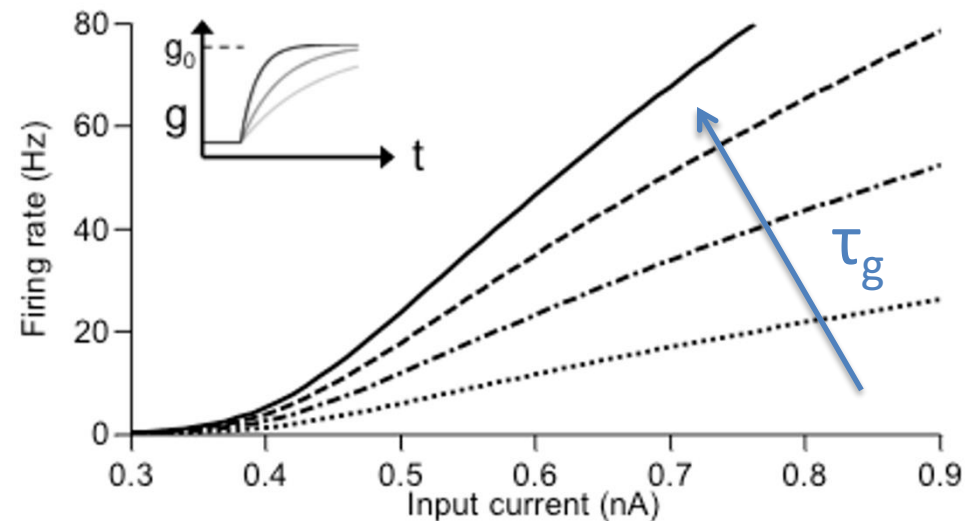


Dynamic gain modulation on excitatory and inhibitory neurons

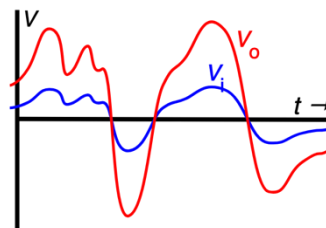
Motivated by attention-induced and urgency studies



Single-cell input-output (response) function with different gains

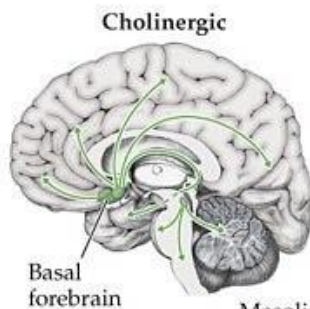
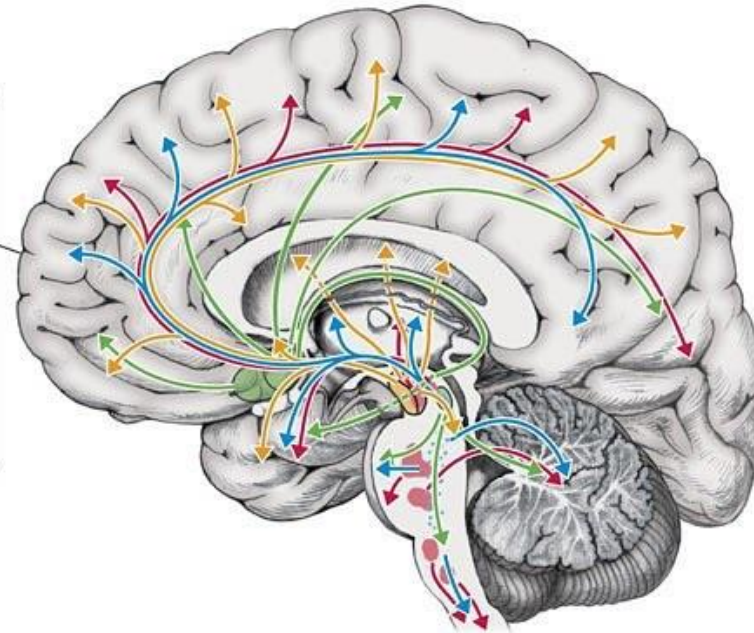


Equivalence in electronics:

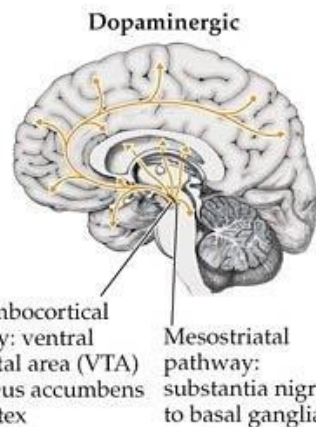


Chemical neuromodulation – Expanding computational complexity

In this midline view, the brain nuclei containing cell bodies of neurons that release four of the major transmitters are shown in different colors, as are the projections of their axons. Although the projections may overlap, each neurotransmitter projects to a distinct set of brain targets.

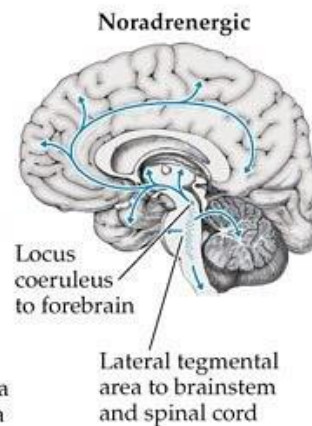


Basal forebrain



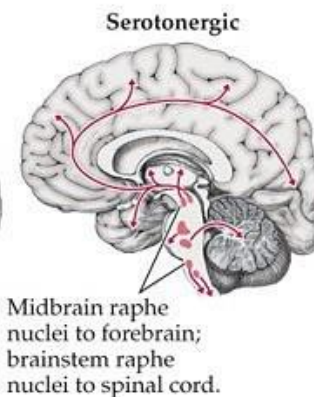
Mesolimbocortical pathway: ventral tegmental area (VTA) to nucleus accumbens

Mesostriatal pathway: substantia nigra to basal ganglia



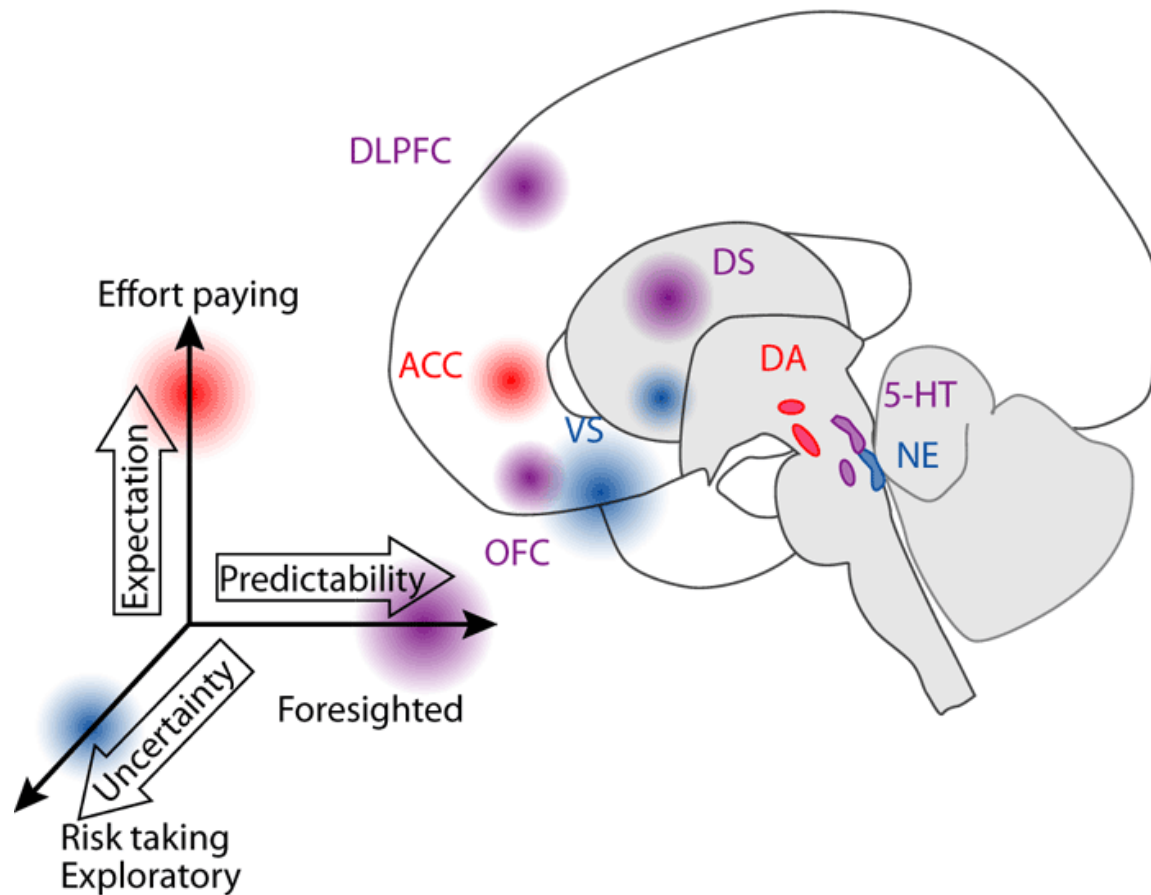
Locus coeruleus to forebrain

Lateral tegmental area to brainstem and spinal cord



Midbrain raphe nuclei to forebrain; brainstem raphe nuclei to spinal cord.

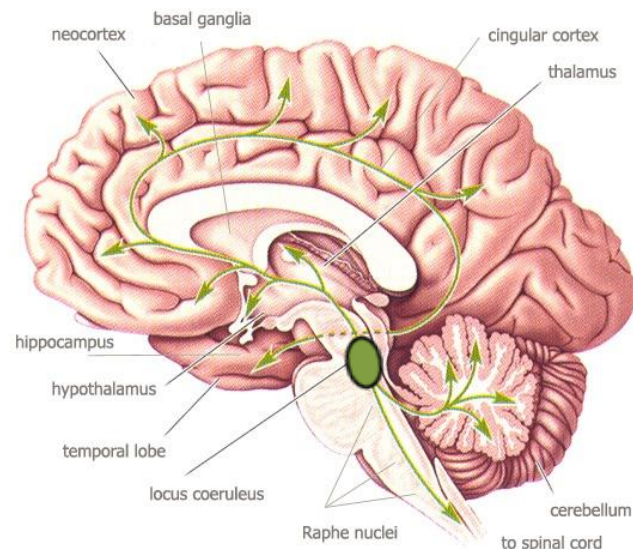
Theories of chemical neuromodulation on decision-making



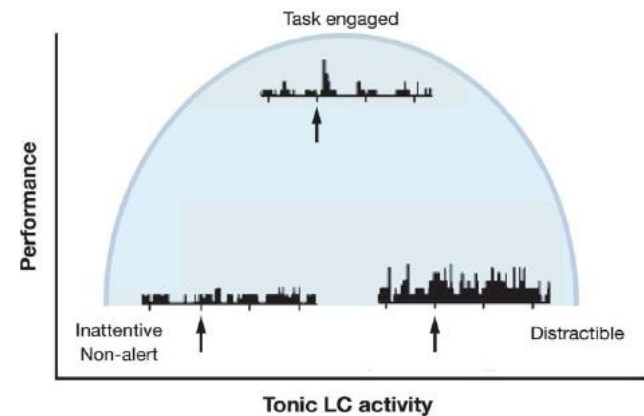
Norepinephrine / noradrenaline (NE / NA)

Involves in arousal, stress, “fight-or-flight” response, attention, etc

The locus coeruleus (LC) releases NE throughout the brain, modulating neural network.



Different LC/NE levels are correlated with different behaviours

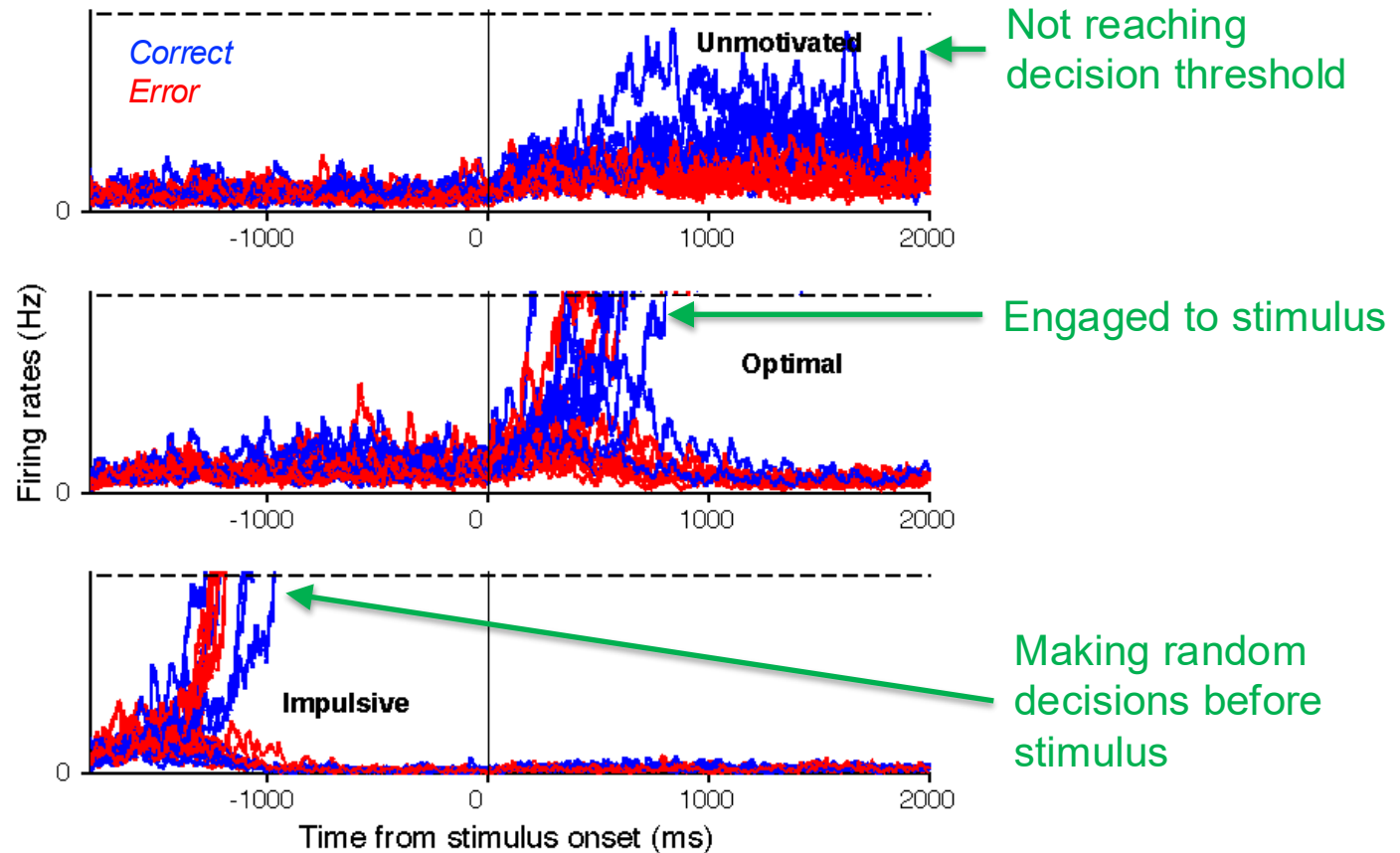


Aston-Jones et. al (1999); Aston-Jones & Cohen (2005)

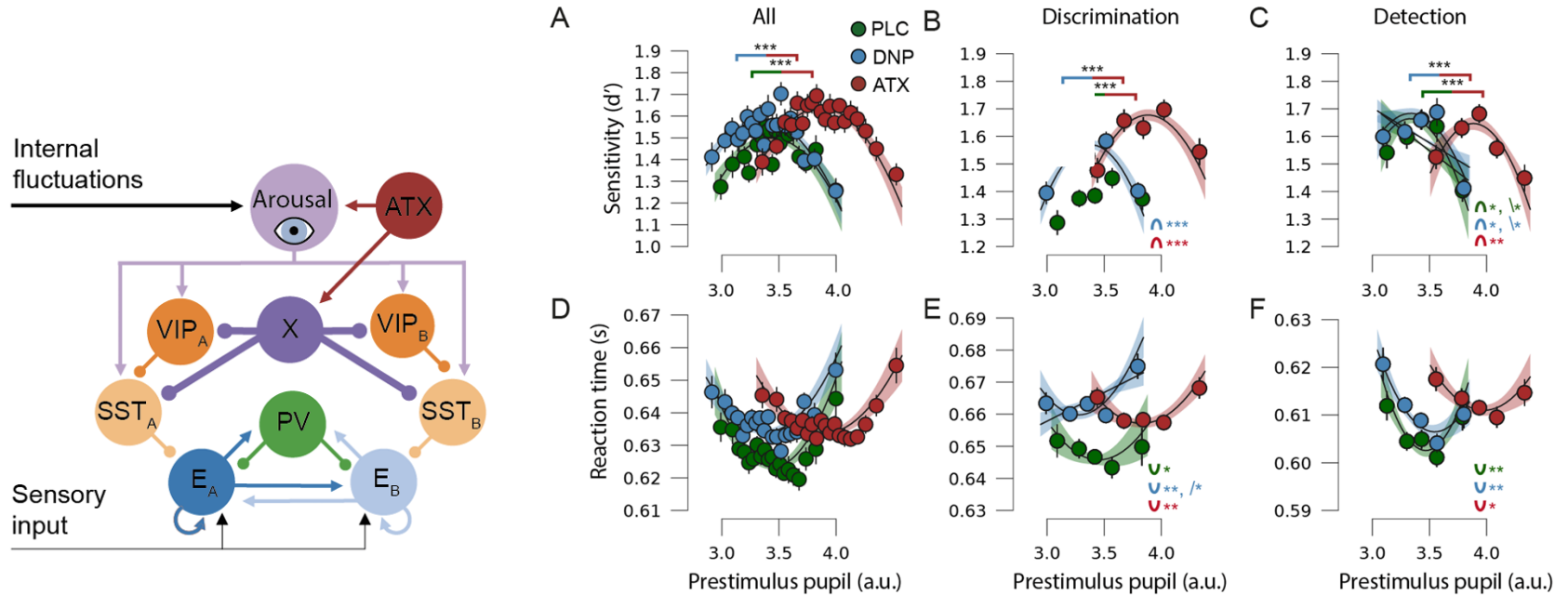
Change in model decision dynamics under “chemical” modulation (of internal state)

10 sample trials with the *same* external stimulus.

Increasing
LC / NE



Change in model decision dynamics under chemical neuromodulation due to internal state change



Atomoxetine (ATX), a selective norepinephrine reuptake inhibitors (SNRI), for treating attention-deficit hyperactivity disorder (ADHD)

Rule-based decisions and flexible task-switching

Classic Stroop task and task switching: Word reading or colour naming.

BLUE

GREEN

YELLOW

PINK

RED

ORANGE

GREY

BLACK

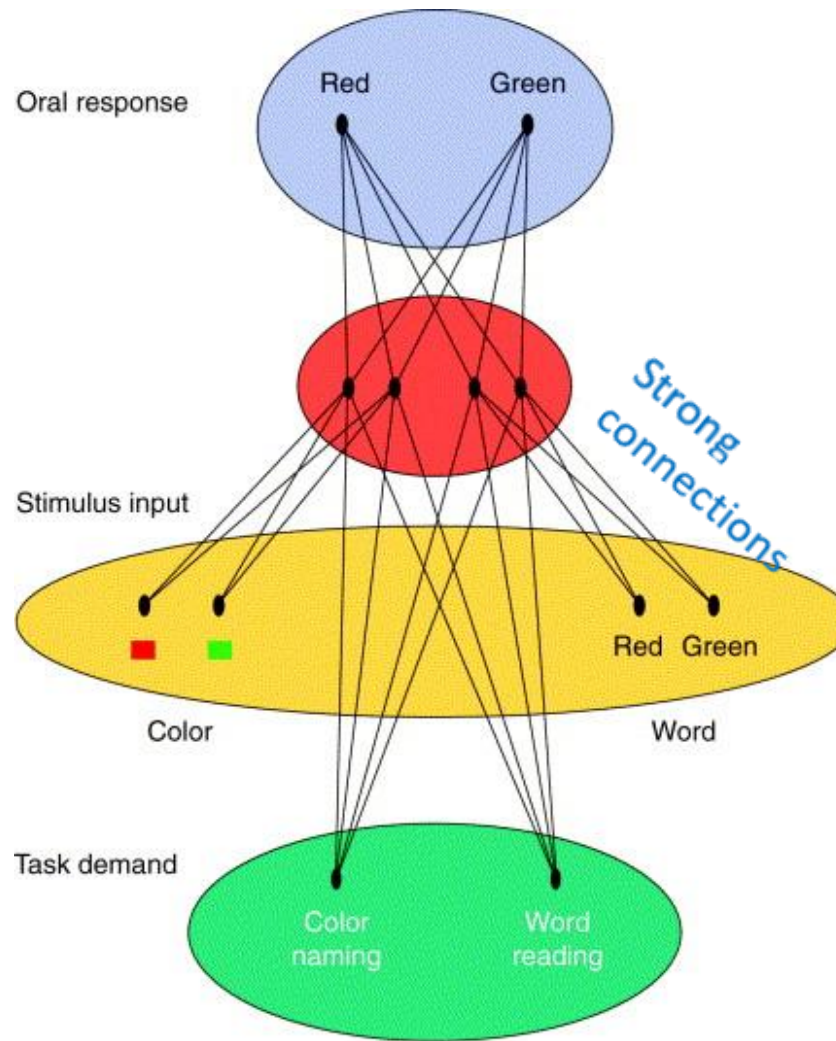
PURPLE

TAN

WHITE

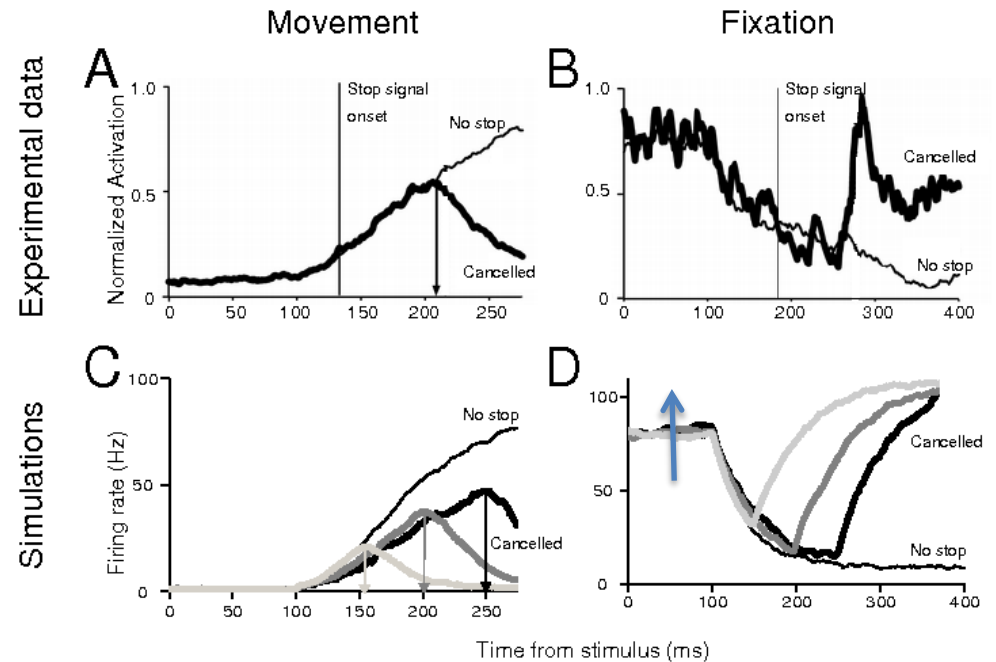
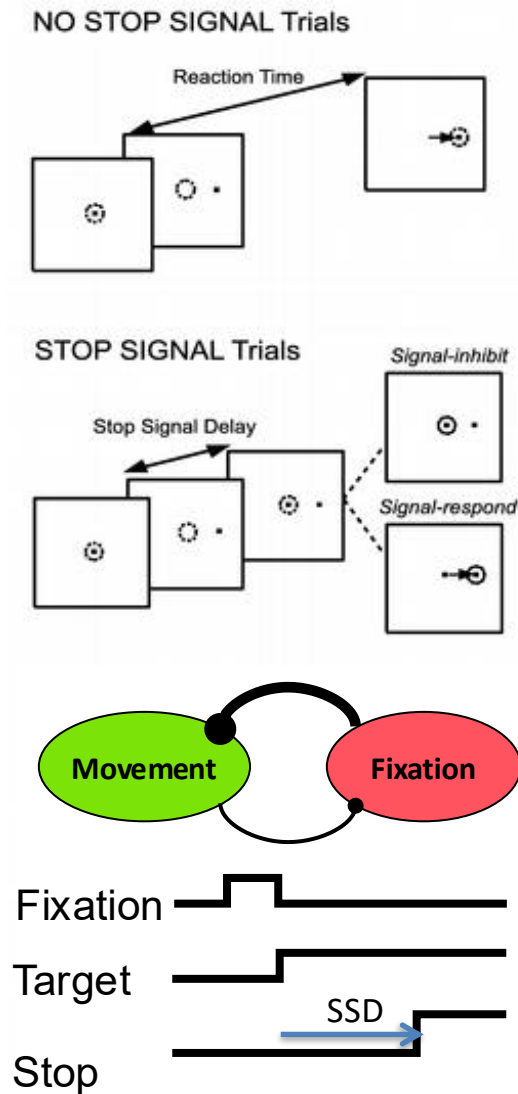
BROWN

Guided activation theory: Network model with cognitive control for Stroop task



Cohen & Huston (1994)
Colin & MacDonald (2000)
Ito et al. (2022)

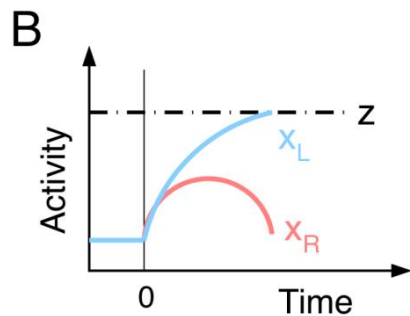
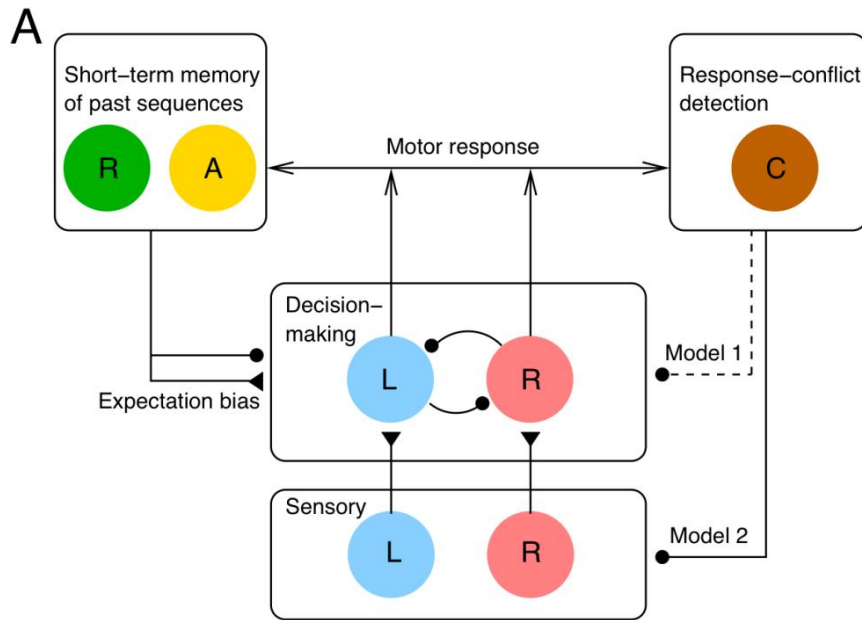
Proactive inhibition of an impending decision



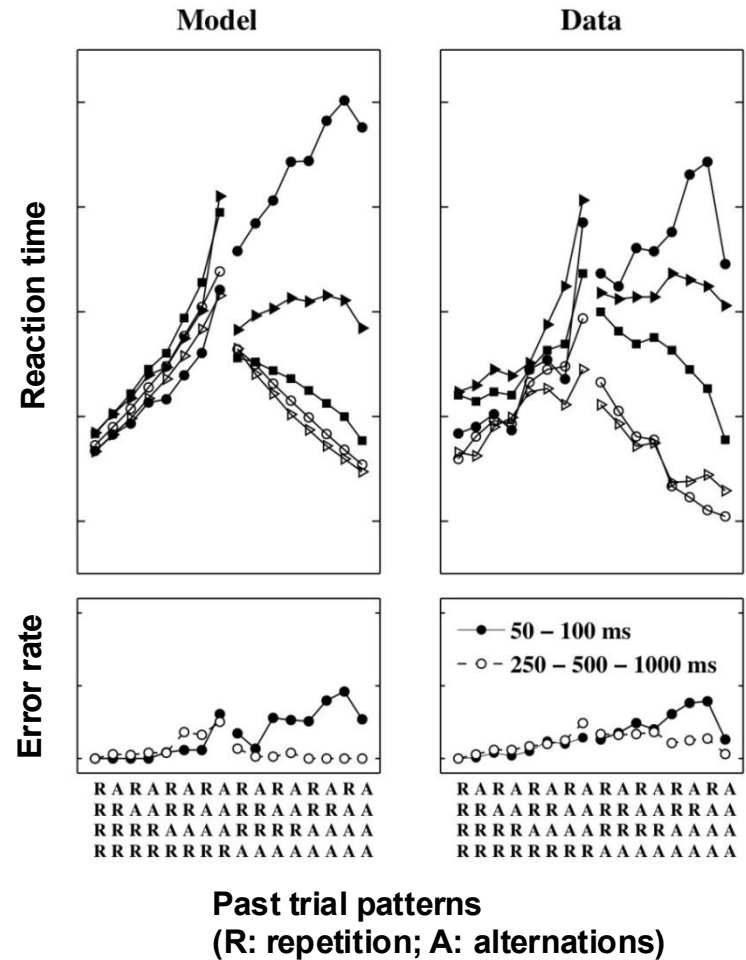
Wong-Lin, Eckhoff, Holmes & Cohen (2010)

Yang, McGinnity & Wong-Lin (2012)

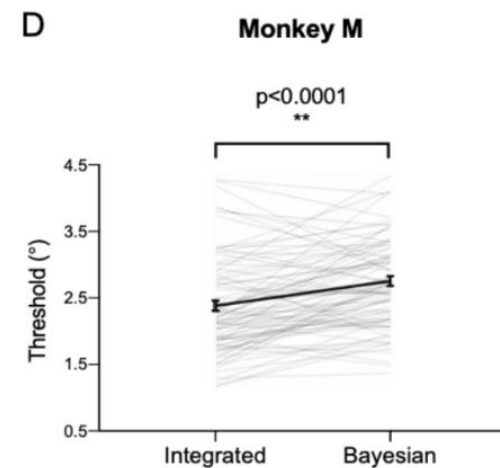
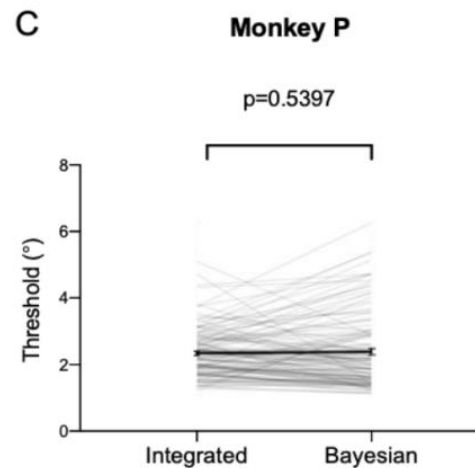
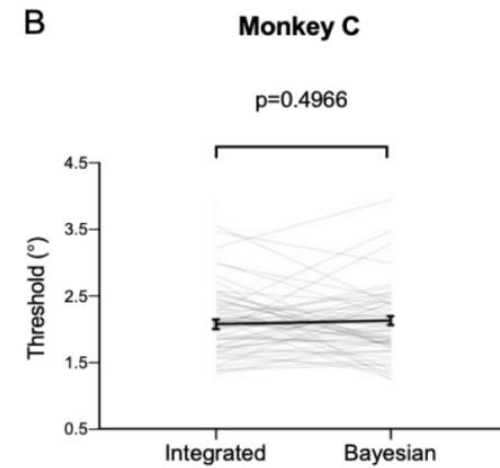
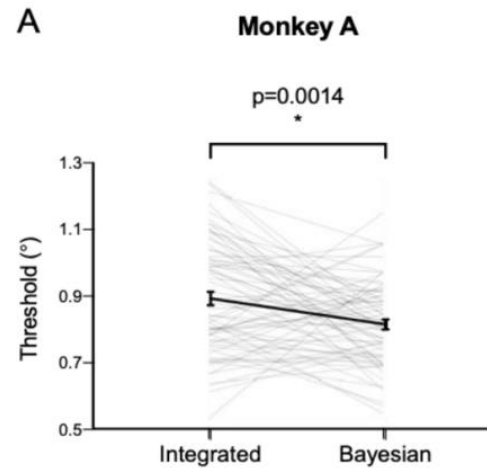
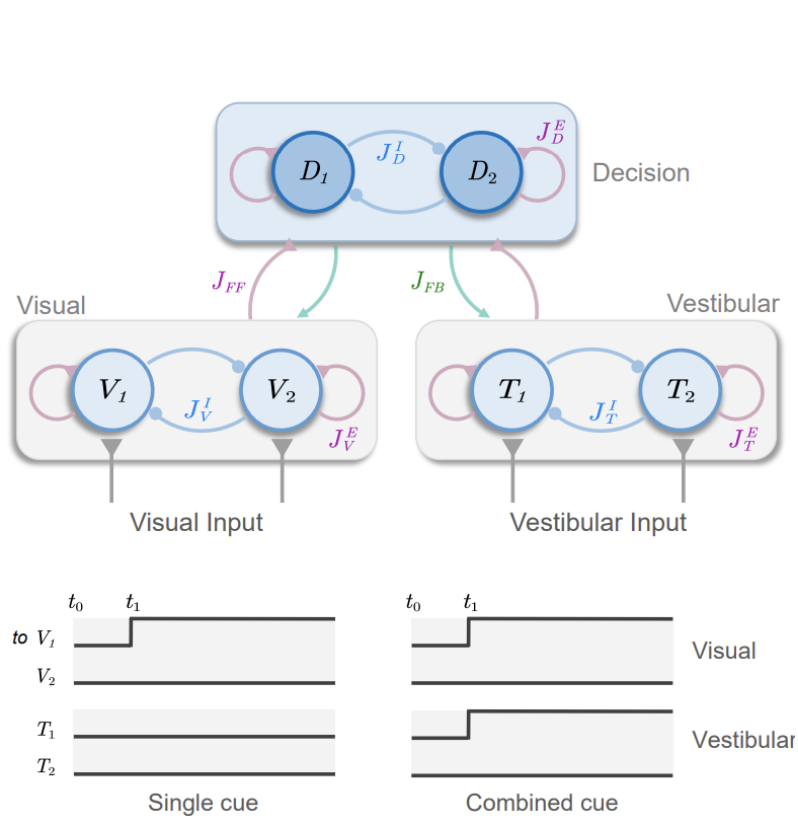
Sequential effects in simple decisions



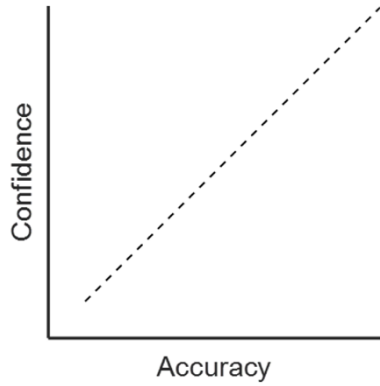
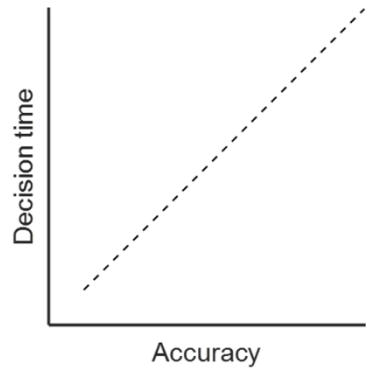
Modified LCA model
(2 biases in drift rate +
post-decision decay)



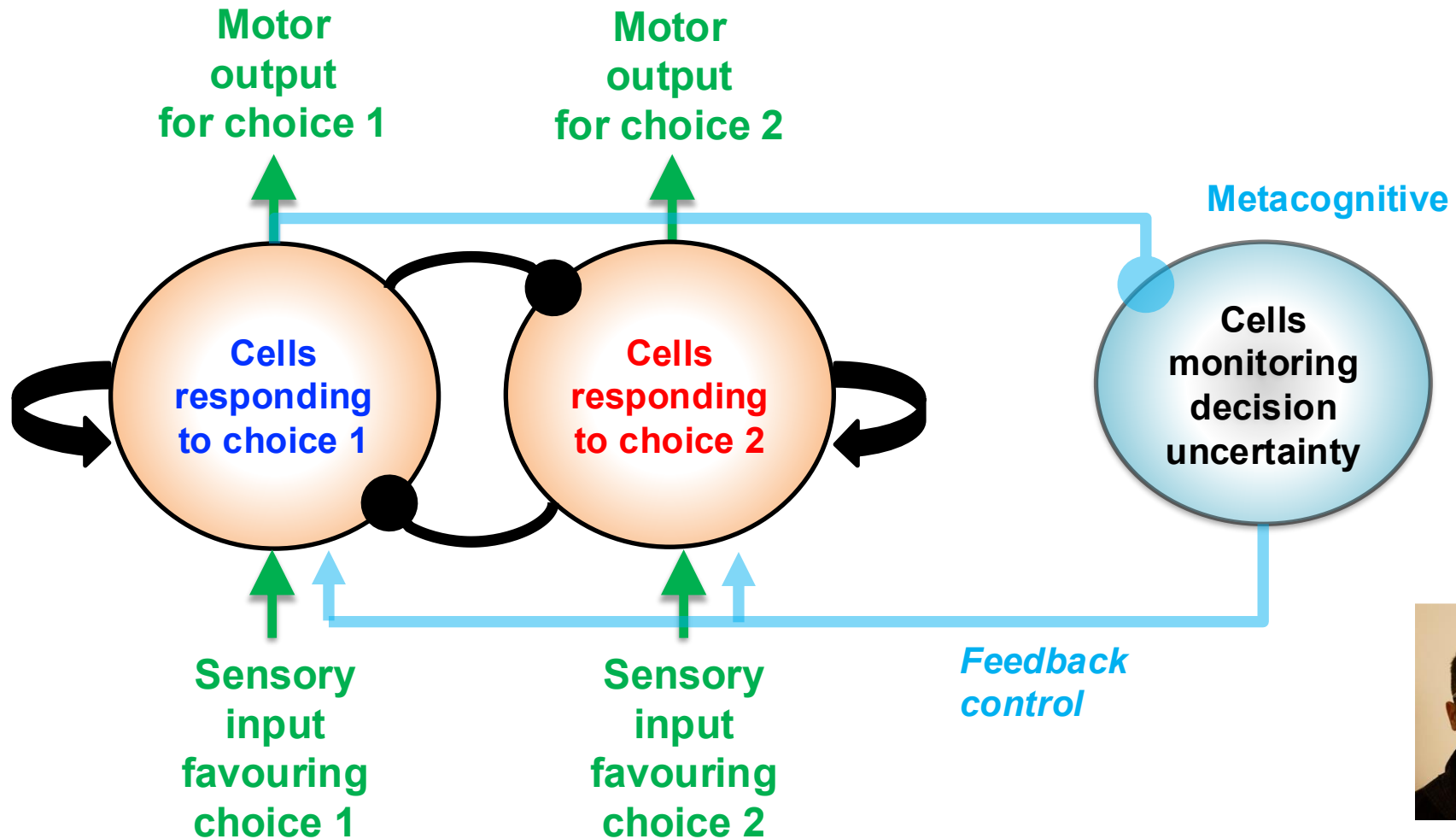
“Top-down” nonlinear feedback effects on multisensory decisions can vary away from optimal Bayesian integration



Decision confidence/uncertainty and change-of-mind



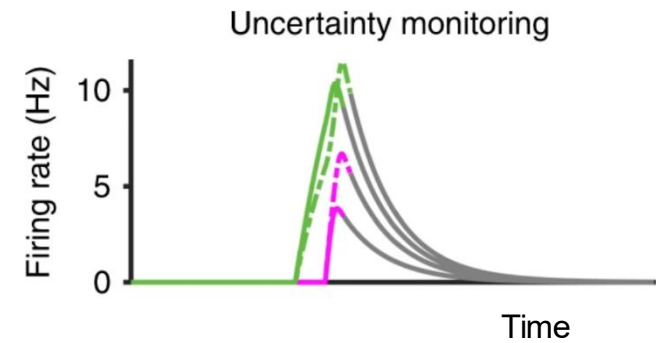
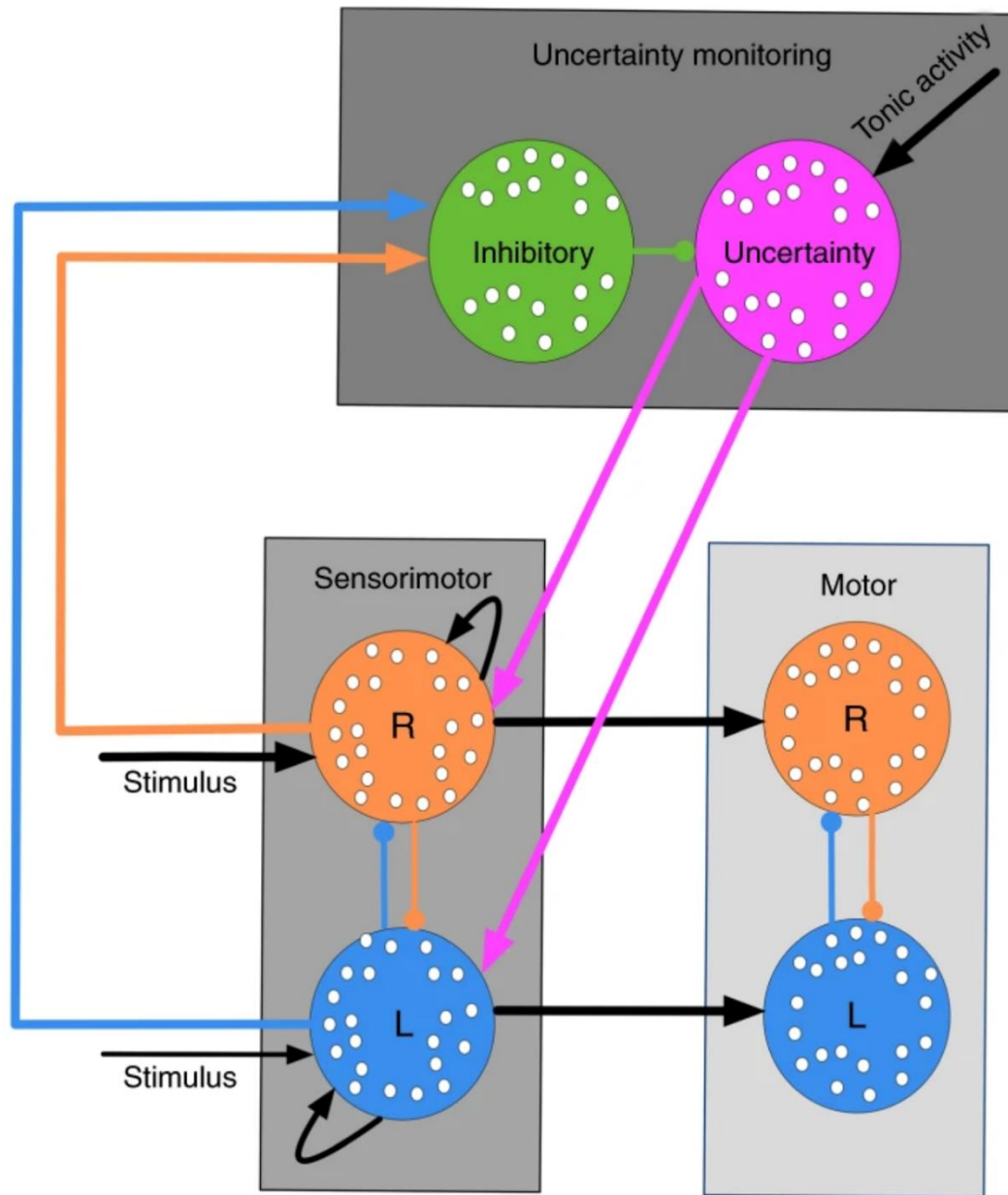
Real-time monitoring of decision uncertainty (“awareness”) and change-of-mind



Nadim Atiya

Atiya, Rano, Prasad & Wong-Lin (2019)

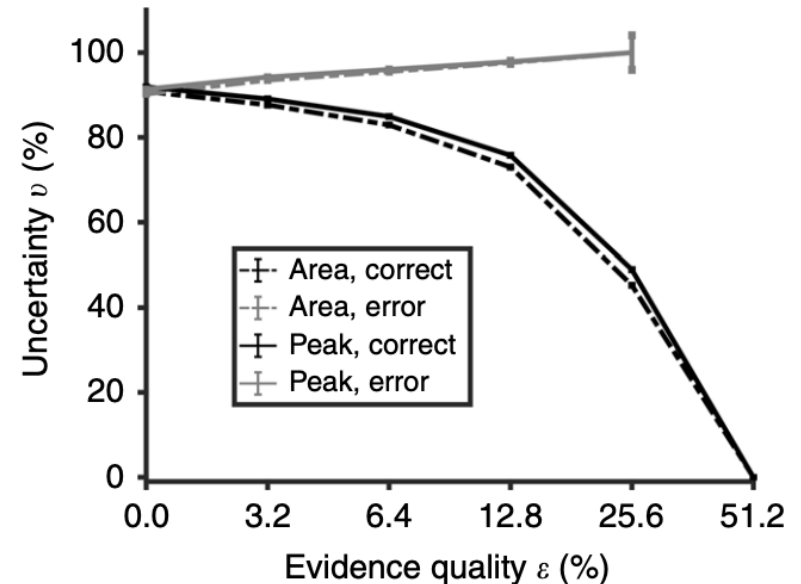
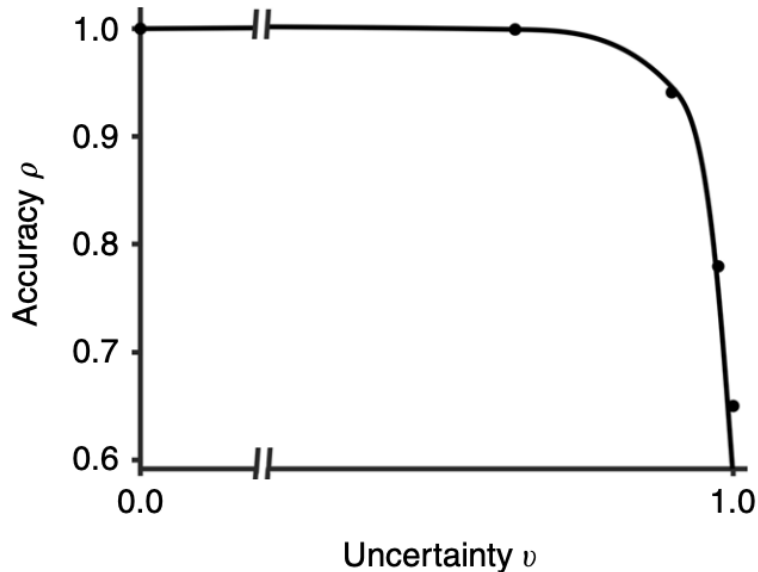
Atiya, Zgonnikov, O'Hora, Schoemann, Scherbaum & Wong-Lin (2020)



Dashed: Lower signal
Bold: Higher signal

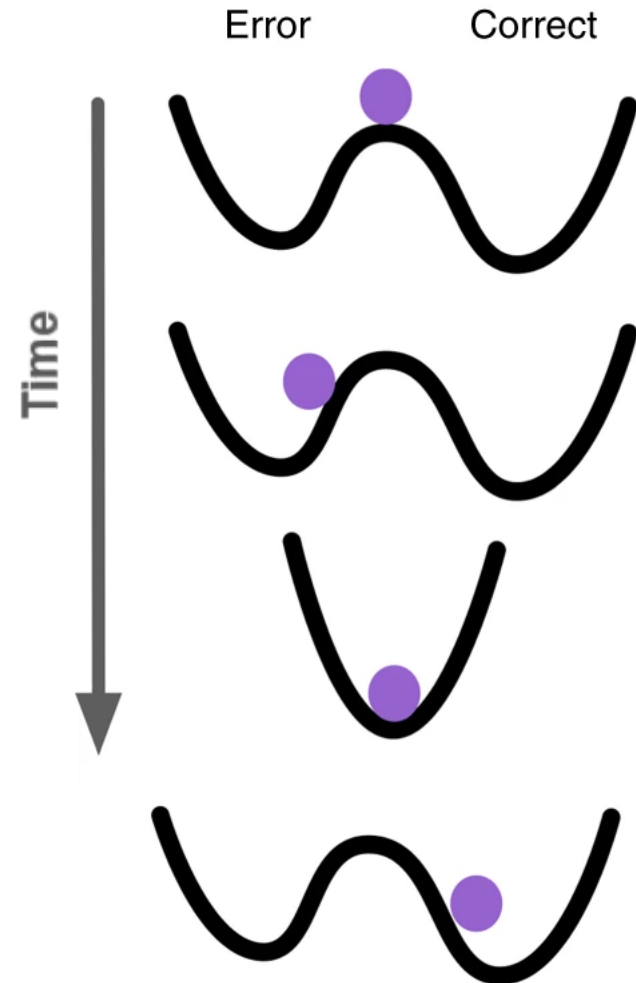
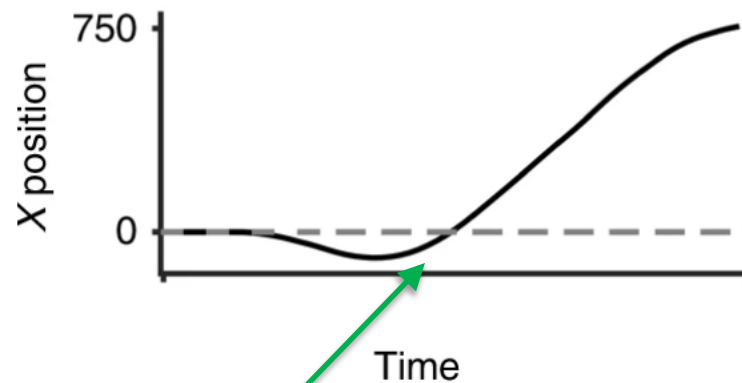
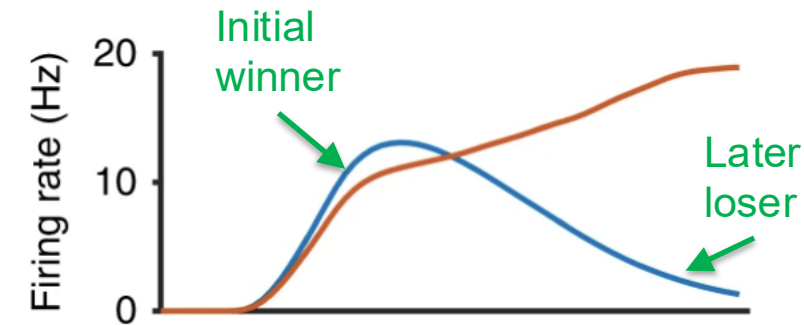
Model captures key characteristics of decision uncertainty

“<” or “X-shaped”



Atiya, Rano, Prasad & Wong-Lin (2019)

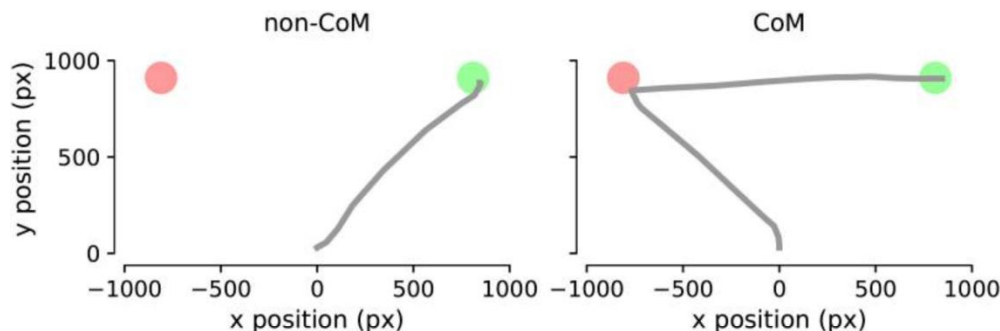
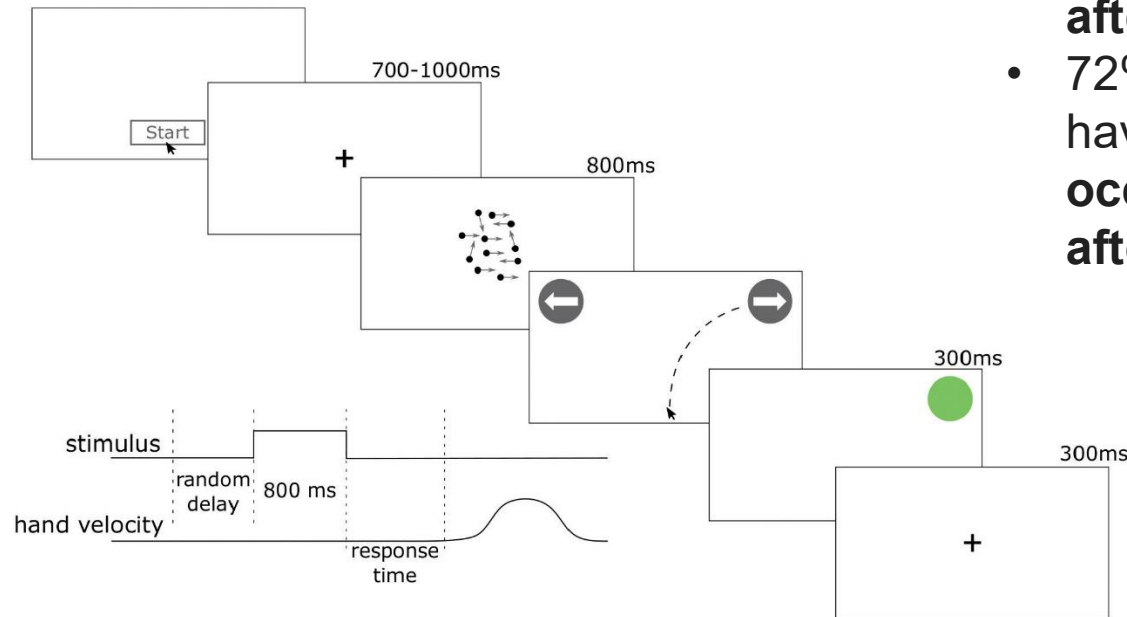
Model provides an explanation for change-of-mind



Change-of-mind
reflected in motor movement

Model prediction: Changes-of-mind can occur in the absence of additional evidence after initial decision

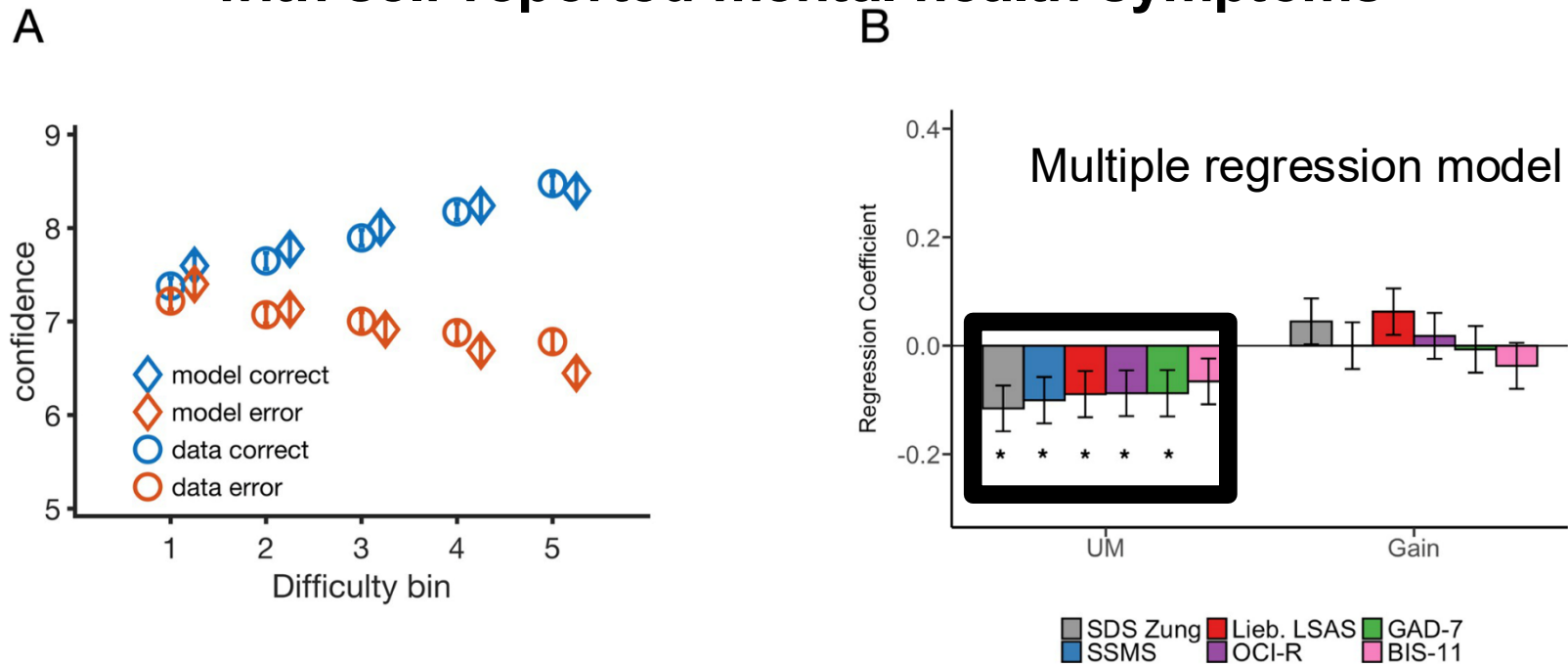
- Participants respond only **after** stimulus offset
- 72% change-of-mind trials have **decision reversal** occurring later than 450ms after stimulus offset



*Atiya, Zgonnikov et al.
(2020)*

Model accounts for data in separate study:

Weaker model uncertainty monitoring connectivity associated with self-reported mental health symptoms



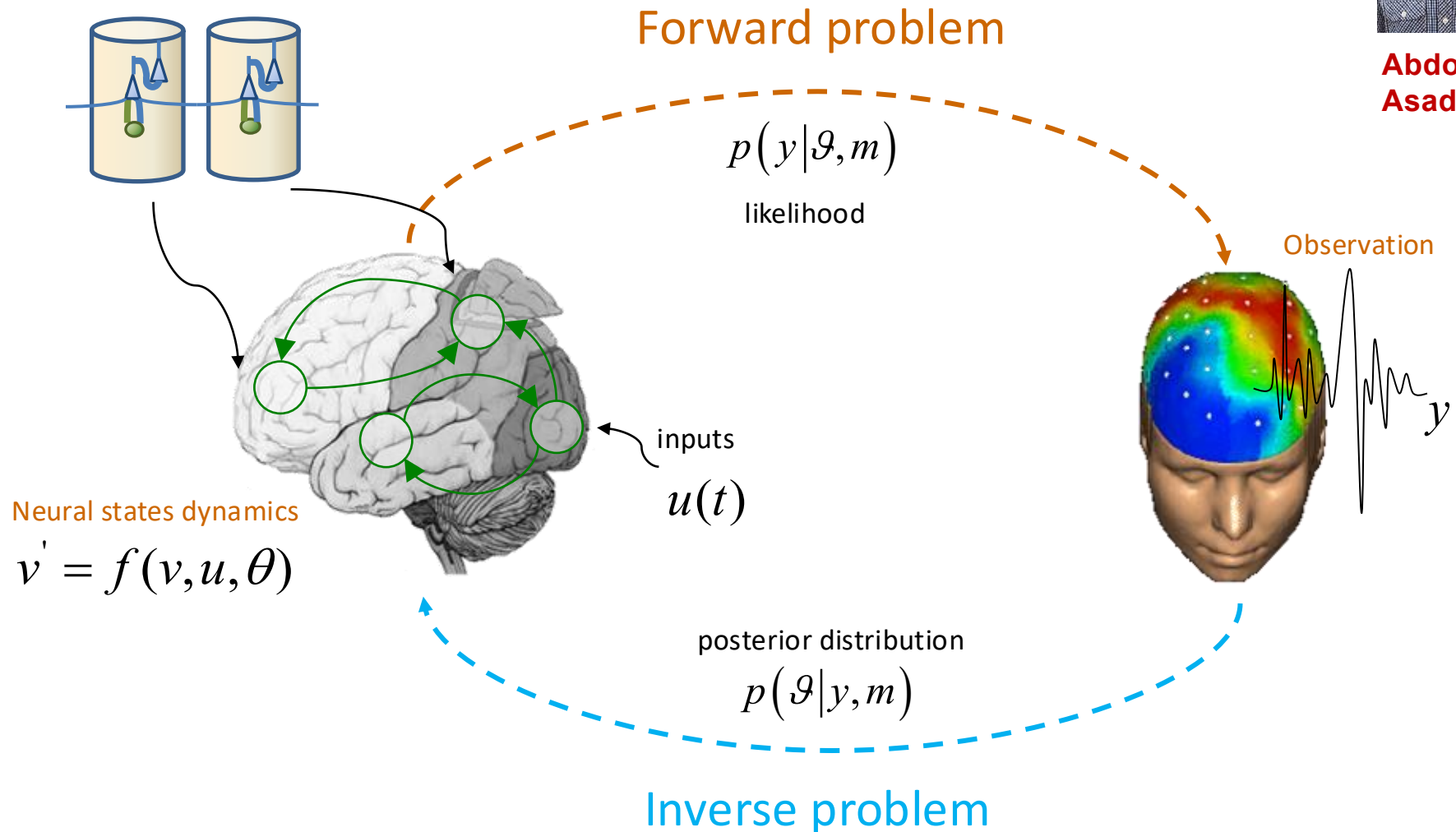
Atiya, Huys, Dolan & Fleming 2021)

Self-report measures of depression, **schizotypy**, **social anxiety**, **obsessive & compulsive symptoms** & **generalised anxiety** associated with weaker uncertainty modulation but not **impulsivity**. No association with stimulus gain parameter.

Dynamic causal modelling (DCM) to understand effective connectivity in human decision confidence

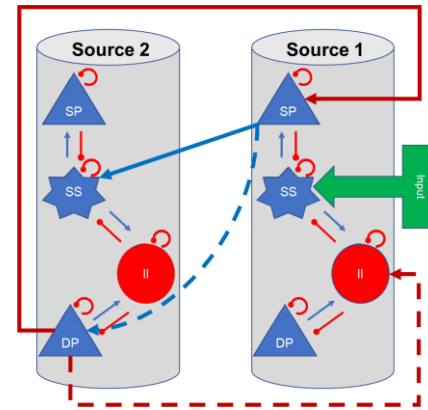


Abdoreza
Asadpour

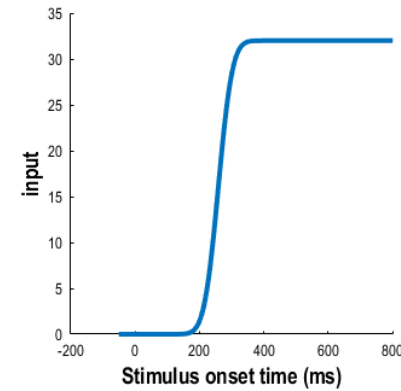


fMRI informed EEG-DCM identifies different networks for early perceptual decision confidence, uncertainty and speed (and across cortical layers)

Common regions: left Middle Frontal Gyrus & left Precuneus

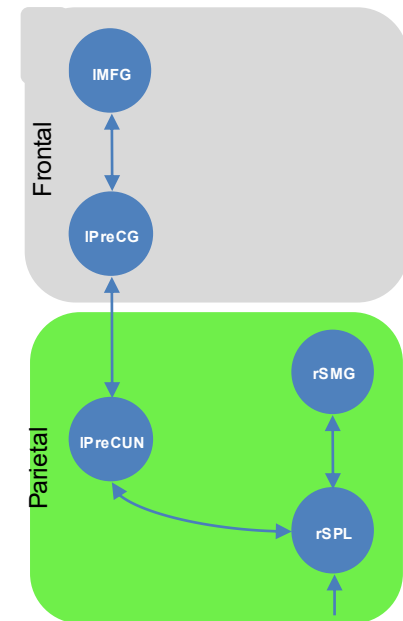
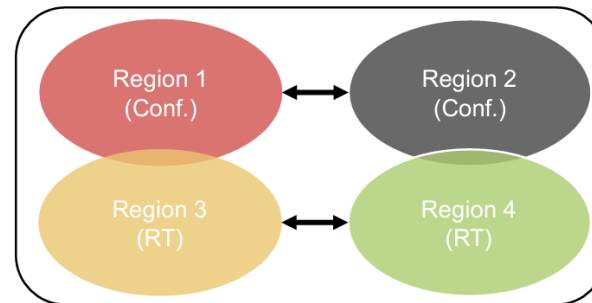
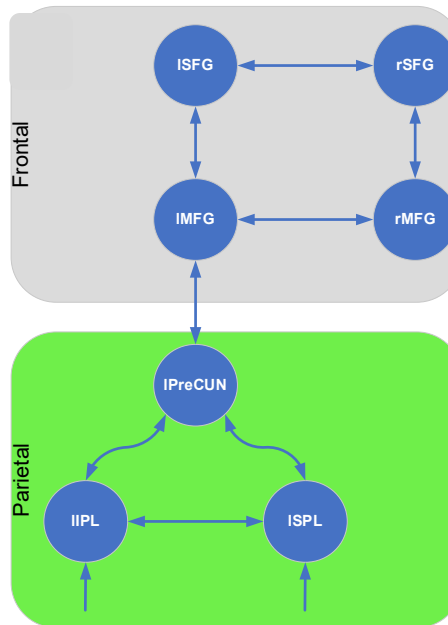


Canonical model



Model input

Low vs high confidence rating

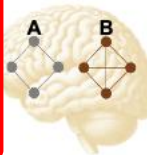


Neural oscillation selectively modulate decision accuracy, speed & confidence → violate SAT

Azimi & Wong-Lin (2025)

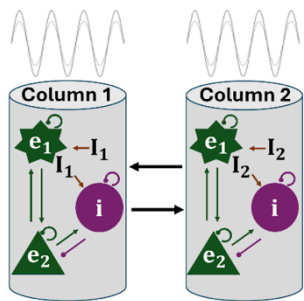
Faster E than I

Network A ($\tau_I > \tau_E$):
 $\tau_E = 10$ ms
 $\tau_I = 200$ ms

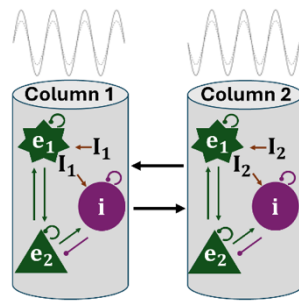


Slower E than I

Network B ($\tau_E > \tau_I$):
 $\tau_E = 100$ ms
 $\tau_I = 5$ ms



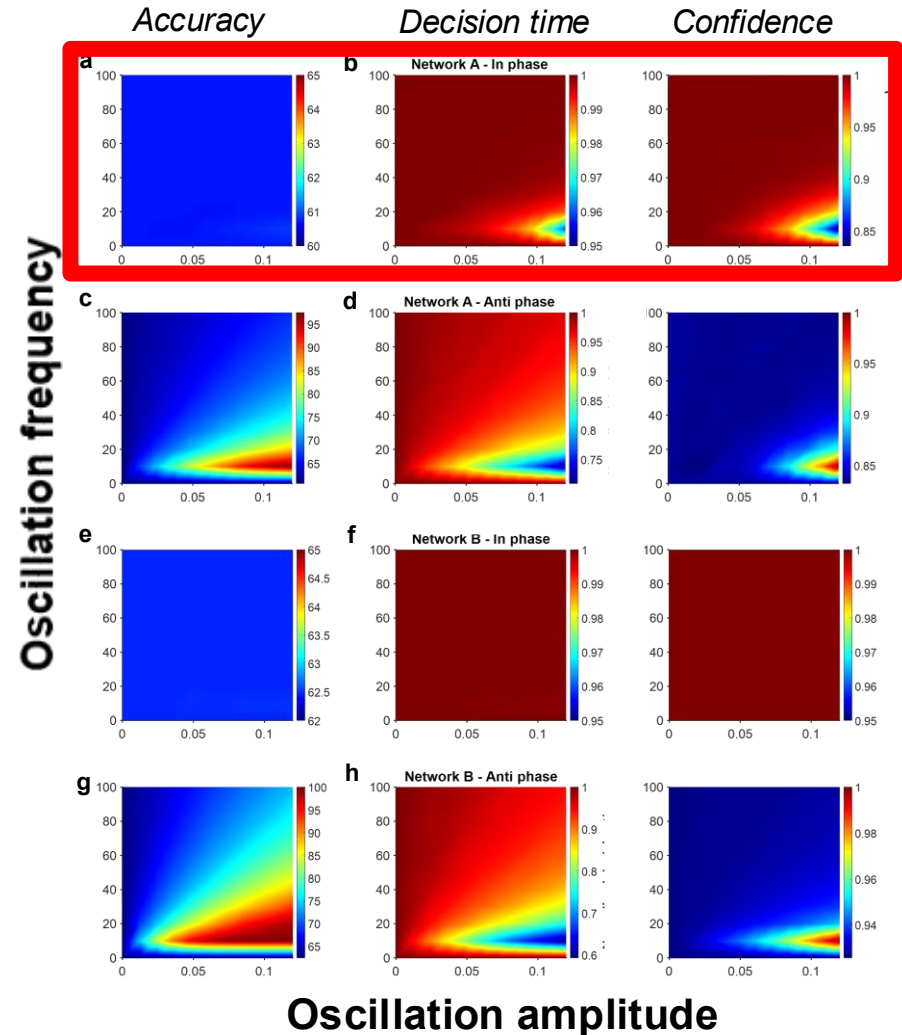
In-phase modulation



Anti-phase modulation



Amin Azimi

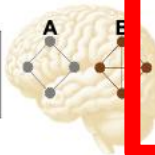


Neural oscillation selectively modulate decision accuracy, speed & confidence → violate SAT

Azimi & Wong-Lin (2025)

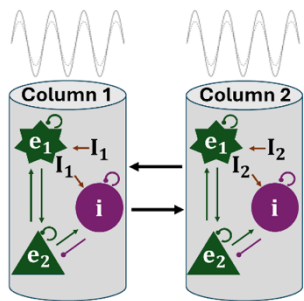
Faster E than I

Network A ($\tau_I > \tau_E$):
 $\tau_E = 10$ ms
 $\tau_I = 200$ ms

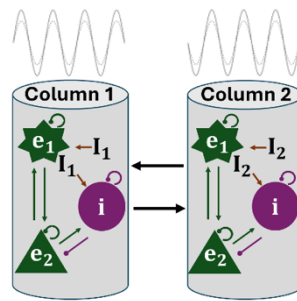


Slower E than I

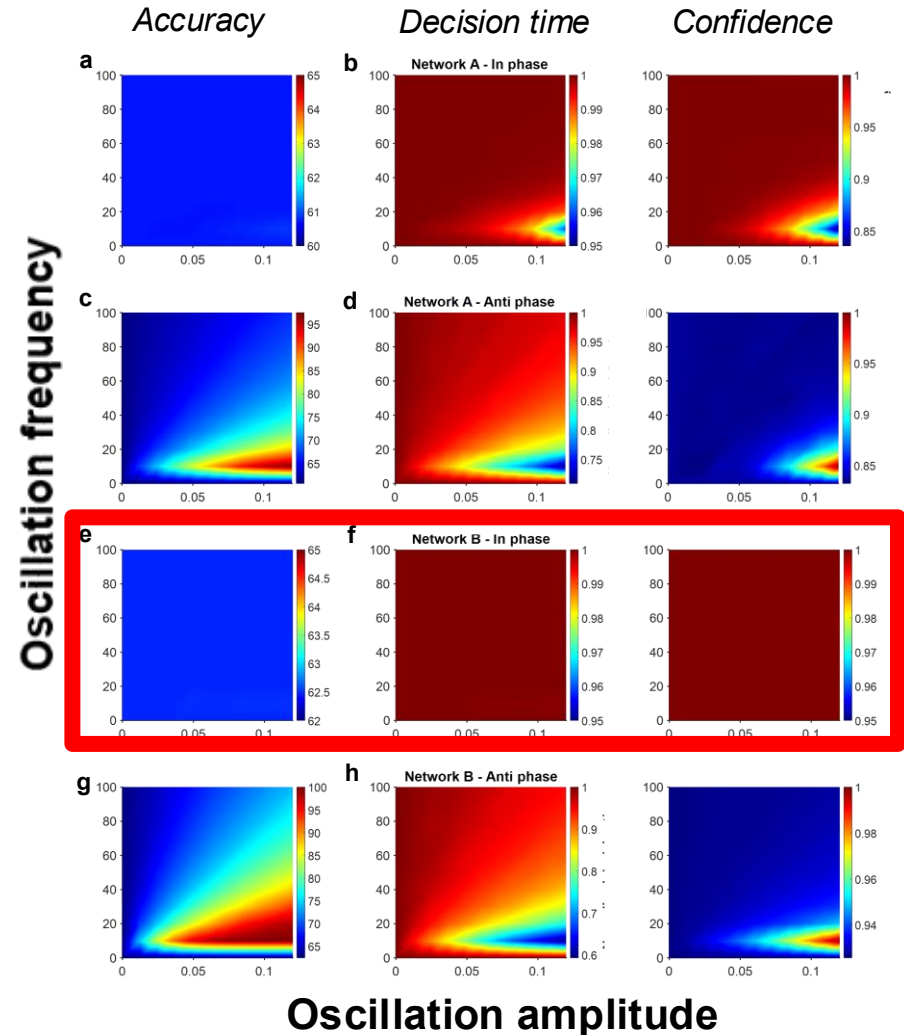
Network B ($\tau_E > \tau_I$):
 $\tau_E = 100$ ms
 $\tau_I = 5$ ms



In-phase modulation



Anti-phase modulation



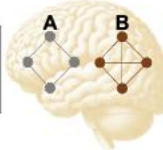
Amin Azimi

Neural oscillation selectively modulate decision accuracy, speed & confidence $\rightarrow$ violate SAT

Azimi & Wong-Lin (2025)

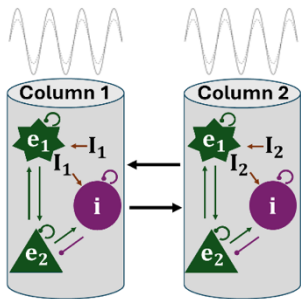
Faster E than I

Network A ($\tau_I > \tau_E$):
 $\tau_E = 10$ ms
 $\tau_I = 200$ ms

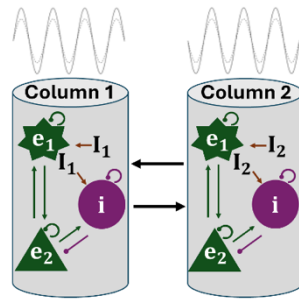


Slower E than I

Network B ($\tau_E > \tau_I$):
 $\tau_E = 100$ ms
 $\tau_I = 5$ ms

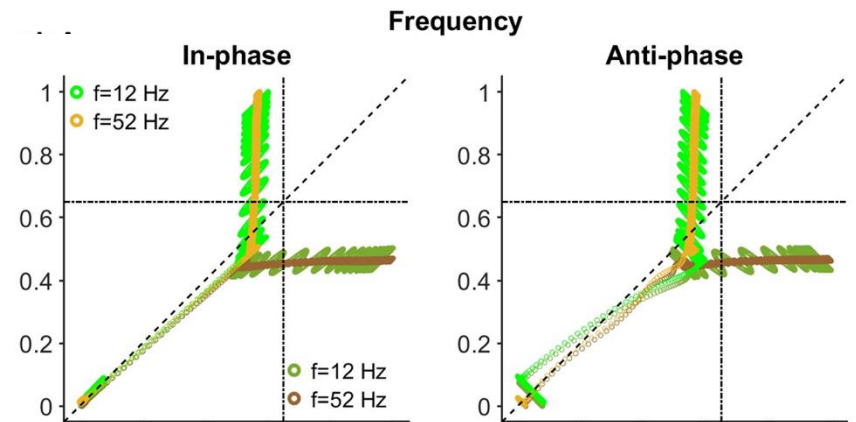
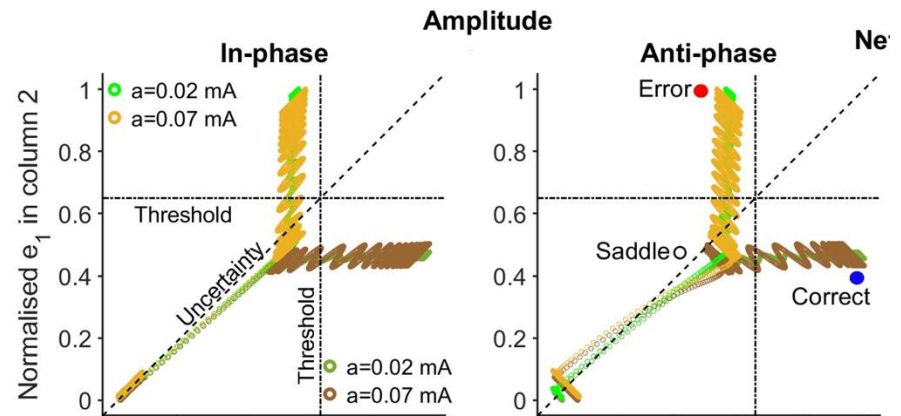


In-phase modulation



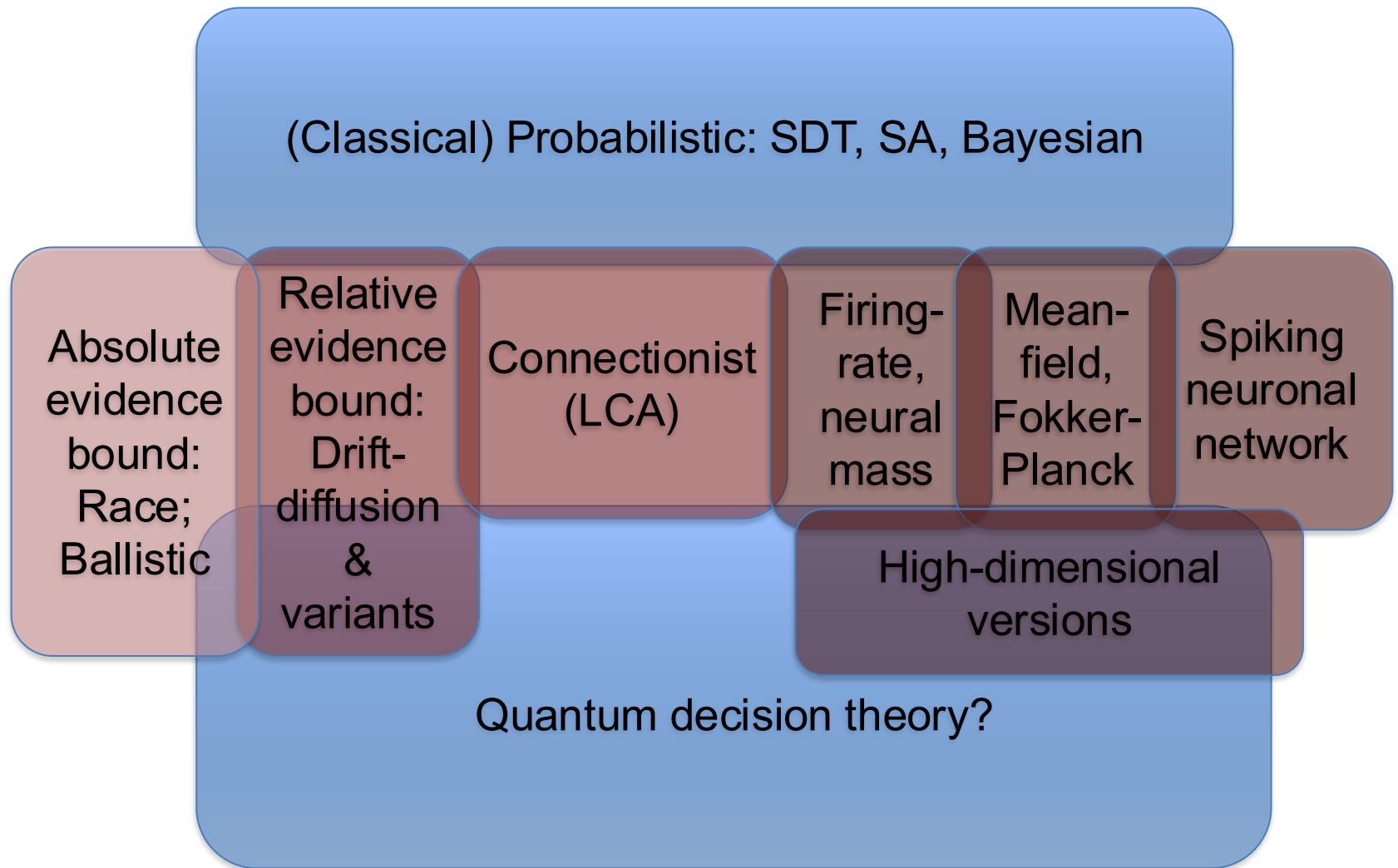
Anti-phase modulation

Momentum swinging with respect to network steady states and decision uncertainty manifold

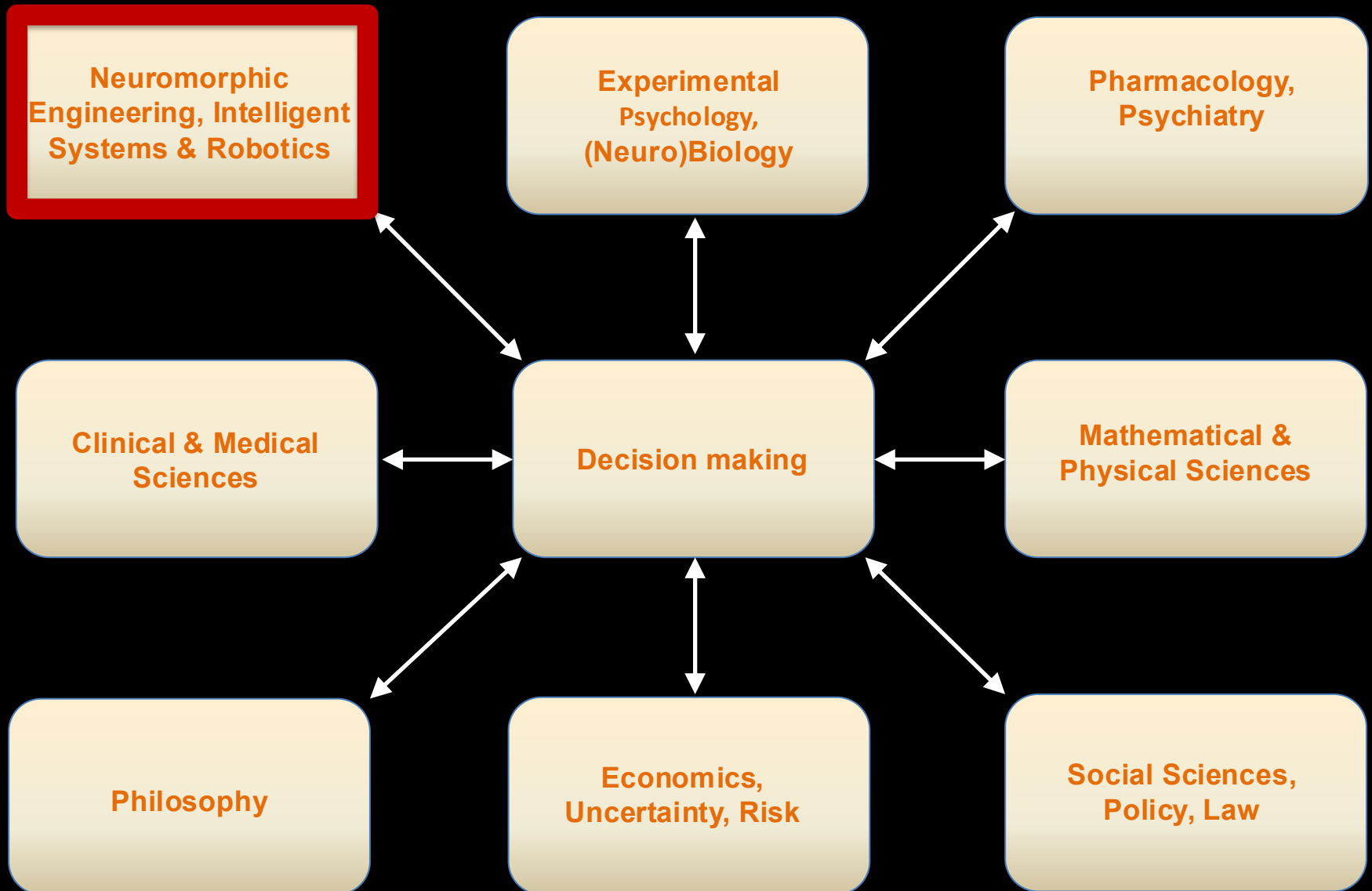


Amin Azimi

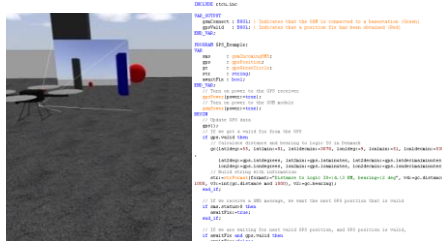
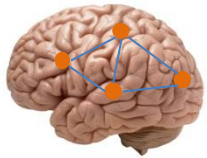
Taxonomy of decision models



Wide Applications and Links of Decision Making



Applications to intelligent technologies



Computational
cognitive
neuroscience

Computational
models, novel
algorithms, &
simulations

(Neuromorphic)
Implementations
in hardware for
real-time
applications

Implementation in
physical intelligent
machines & robots

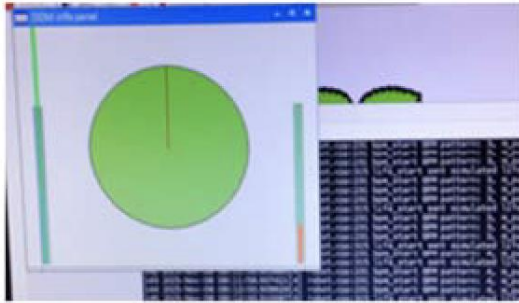
Decision making in AI / machines

A key component in AI and machine algorithms

E.g.

- Search
- Planning
- Probabilistic reasoning (uncertainty reasoning)
- Utility theory
- Probabilistic reasoning over time (e.g. HMM, Kalman filters)
- Utility theory
- Markov Decision Process (MDP)
- Game theory (multi-agent)
- Prediction (Classification, Regression, Reinforcement Learning)
- Association (Link analysis, Sequence analysis)
- Clustering

Sensor data fusion using DDM implemented in a single (FPGA) computing chip



(A)



(B)



(C)



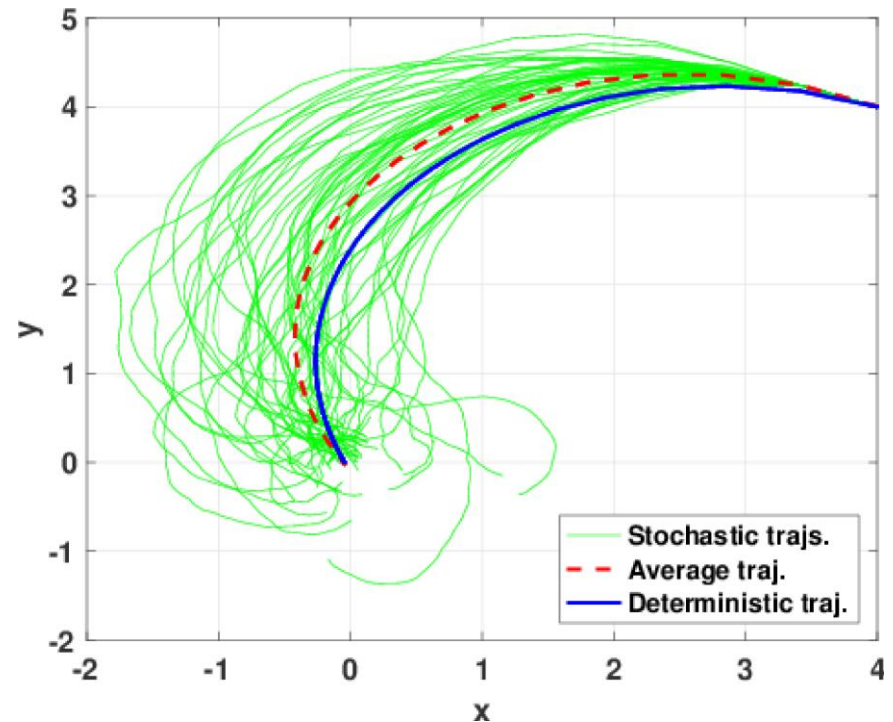
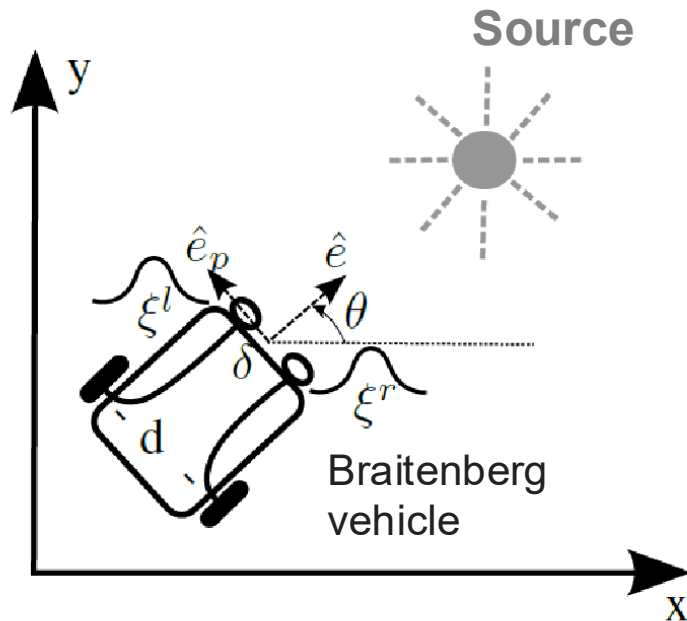
(D)



(F)

FPGA (Field-Programmable Gate Array): a type of integrated circuit that can configure and re-program after manufacturing

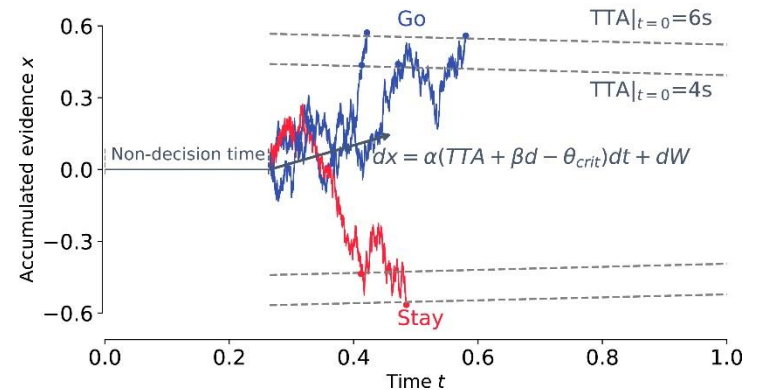
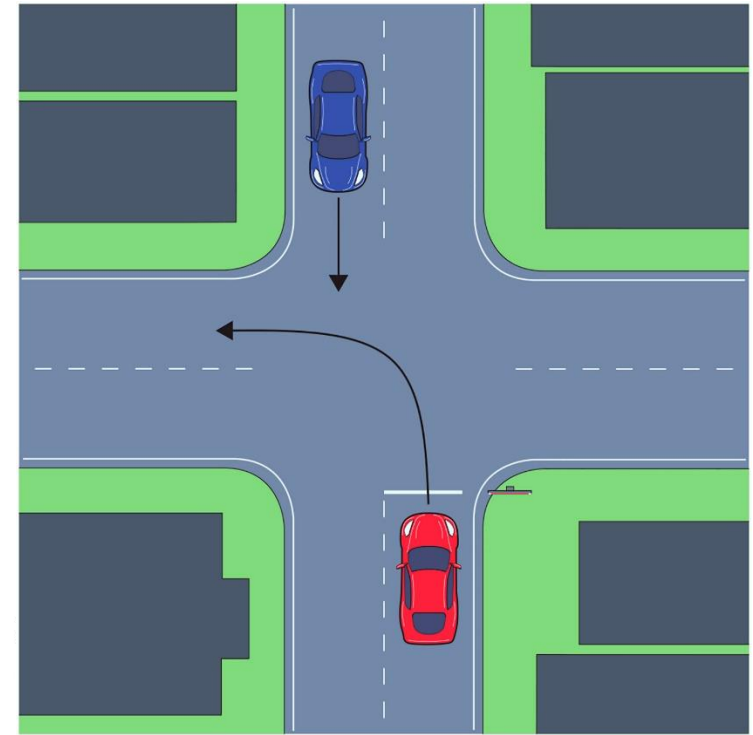
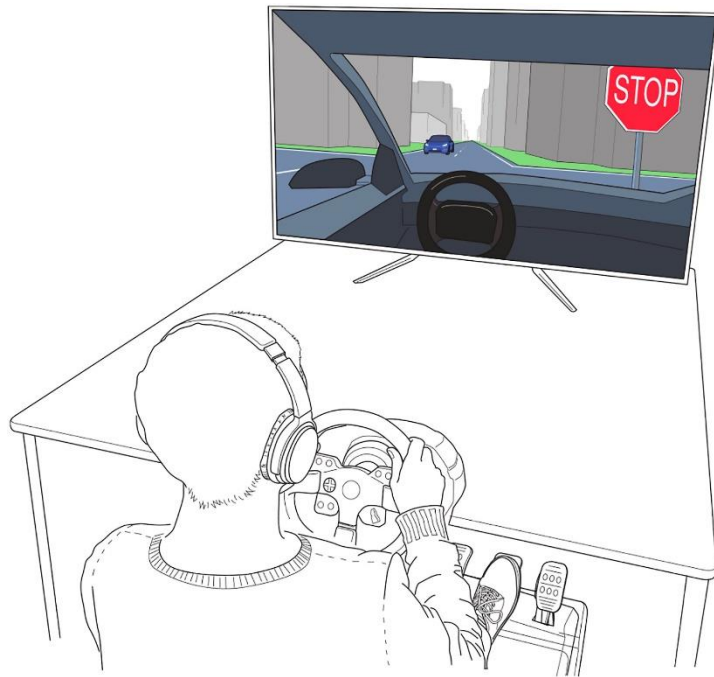
Source-seeking mobile robotic model (or heat avoidance of fruit fly) with noisy sensory information as a DDM process



Raño, Mehdi & Wong-Lin, ICRA (2017) A drift-diffusion model of biological source seeking for mobile robots.

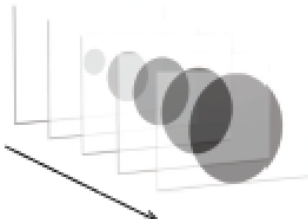
Simoes et al., Nat. Commun. (2021) Robustness and plasticity in Drosophila heat avoidance

Real-time prediction of human decisions by automated vehicles & simulating realistic human-like decisions in virtual environments for automated vehicles

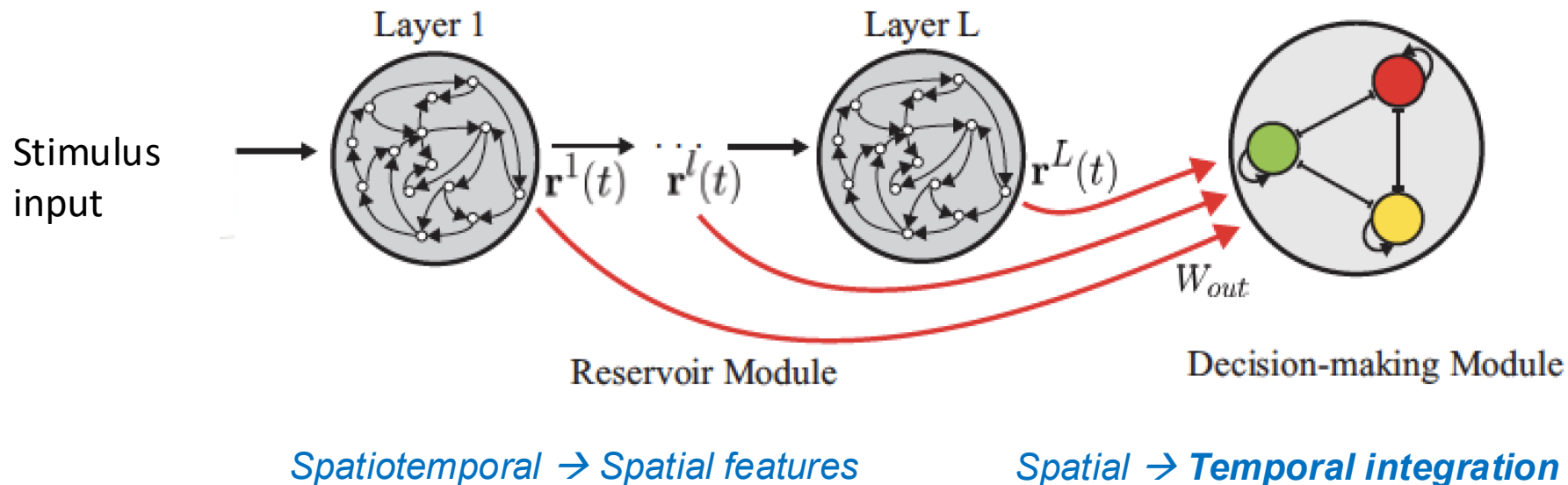
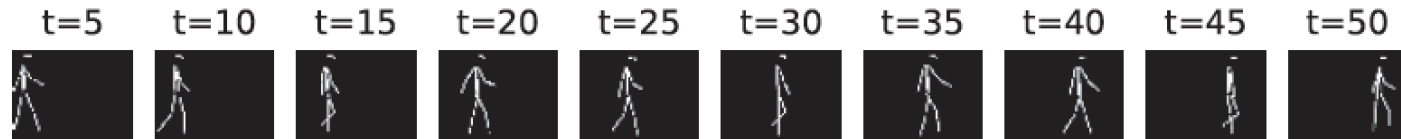


Few-shot machine learning over space & time outperforms state-of-the-art deep learning

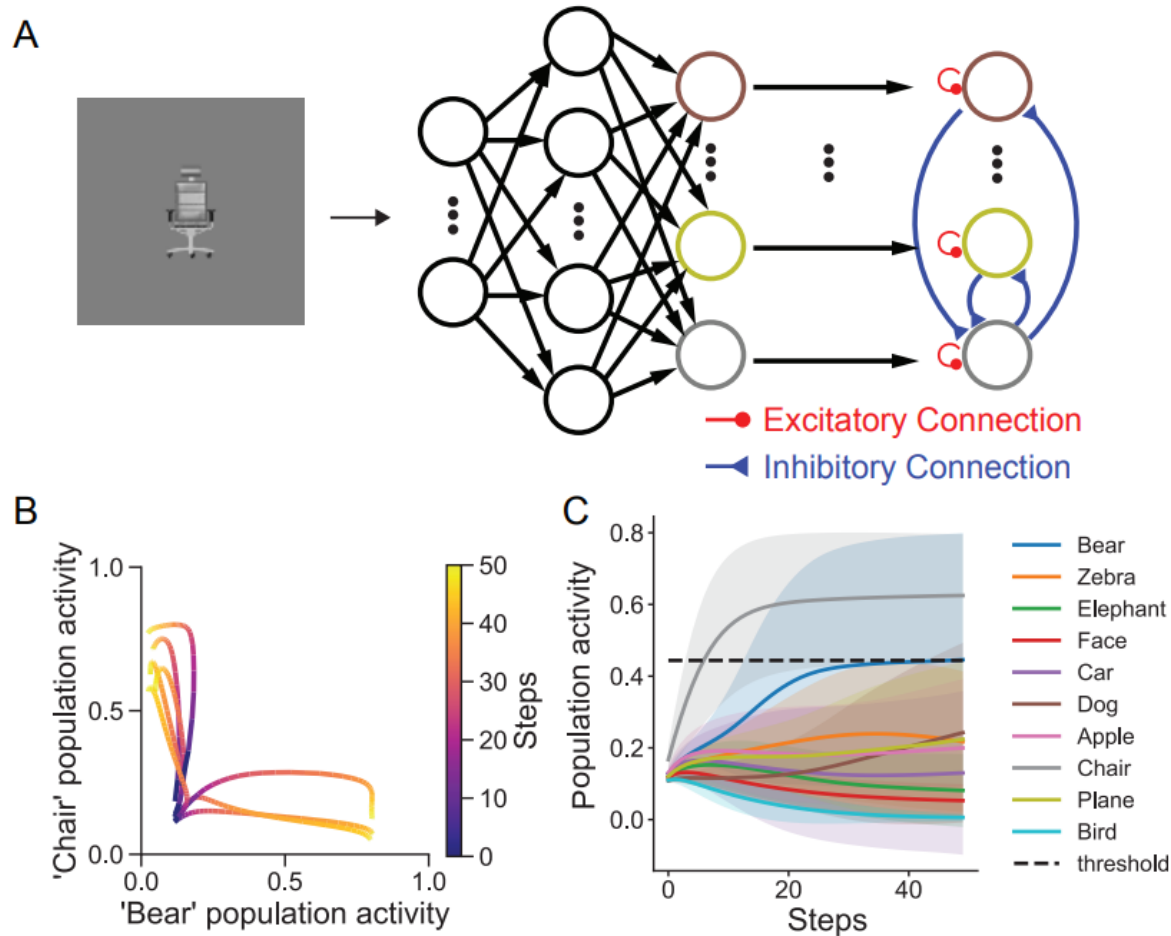
Looming pattern detection



Individual's gait identification



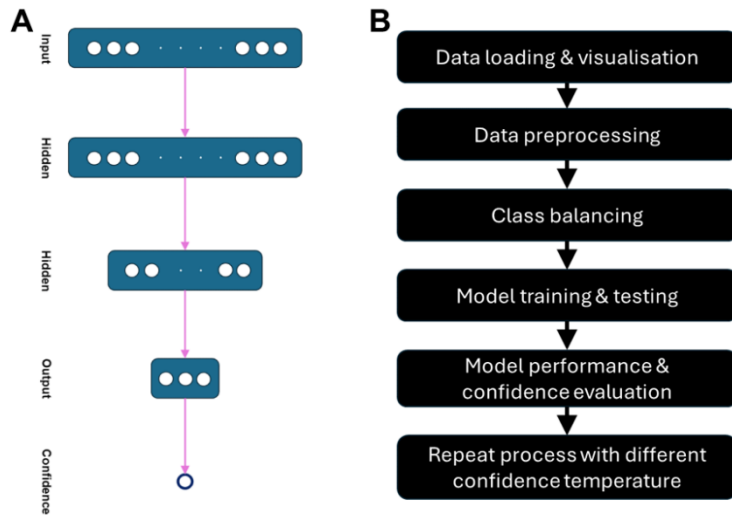
Neuro-inspired deep learning with evidence accumulation



Cheng, Rodriguez, Chen, Kar, Watanabe & Serre (2024)

Neuro-inspired deep learning with decision confidence for detecting individual with Alzheimer's disease

→ Understand reliability of model's decision for better trust



$$\text{softmax}, \sigma(\mathbf{z})_i = \frac{e^{\beta z_i}}{\sum_{j=1}^3 e^{\beta z_j}}$$

where $\beta = 1/(\text{Temperature})$

Confidence temperature	0.5	1.0	1.5	2.0	2.5
Average classification accuracy	0.8448	0.8421	0.8320	0.8466	0.8476
SD accuracy	0.3621	0.3647	0.3739	0.3603	0.3594
Average confidence	0.8152	0.8066	0.7888	0.8188	0.8163
SD confidence	0.1628	0.1681	0.1678	0.1578	0.1613
TP confidence	0.8024	0.6723	0.9401	0.8415	0.6889
TN confidence	0.8152	0.7511	0.5369	0.7839	0.6621
FP confidence	0.8084	0.6538	0.6690	0.6946	0.6871
FN confidence	0.7288	0.7796	0.7783	0.7238	0.6297

TP, TN > FP, FN

Summary

- Network dynamics as representation and tools for understanding decision making → links to cognitive models
- Low-dimensional decision models can account for (and explain) key neural and behavioural data
- More complex choice behaviour and dynamics can be built on simpler models, e.g. for cognitive control – cognitive building blocks
- Towards neurodynamical models of brain-wide activities for distributed encoding of decision processing
- AI (machine learning) tightly linked to decision-making → opportunities for brain-inspired intelligent technologies (NeuroAI)